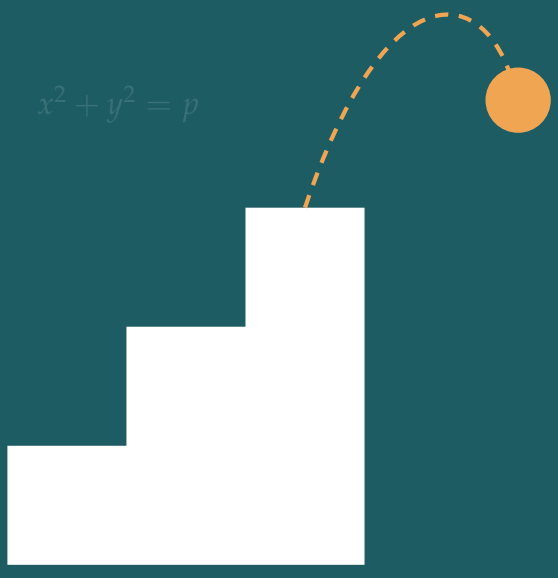
STEP PREPARATION · UNIVERSITY MATHEMATICS

Beyond STEP

STEP I, II AND III

1990 Worked Solutions



KEITH McNULTY

Beyond STEP

Past STEP questions as a doorway
to university mathematics

STEP I, II AND III

1990 Worked Solutions

Keith McNulty

Beyond STEP: Past STEP questions as a doorway to university mathematics

1990 Worked Solutions

Worked solutions to all 48 questions of the 1990 STEP papers (STEP I, II and III). Worked solutions to every STEP question since the specimen papers, the whole book *Beyond STEP* and every chapter on its own are free to read and download at `beyondstep.keithmcnulty.org`.

Have a proper go at a question before you read its solution. Every answer has been checked numerically, but with this many solutions some slips are inevitable: if something looks wrong, trust your own working and check it carefully.

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Corrections are welcome as issues at github.com/keithmcnulty/step_book_mcnulty.

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WORKED SOLUTION

STEP I 1990, Question 1

Pure • Trigonometric Optimisation

STEP I 1990, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q1

The key idea

Every vertex of the blind is at distance r from the centre D, so the radii DX and DY cut the quadrilateral into three isosceles triangles with apex D. Each has area $\frac{1}{2}r^2\sin(\text{apex angle})$, and the three apex angles add up to π. That turns a geometry problem into maximising a sum of sines, and a sum-to-product formula then does the first part without any calculus.

Solution

The area of the blind

Since A, D, C are collinear, the angles at D satisfy

$$\angle ADX + \angle XDY + \angle YDC = \pi, \qquad \text{so} \qquad \angle YDC = \pi - \theta - \alpha.$$

Here $\theta \geq 0$, $\alpha > 0$ and $\theta + \alpha \leq \pi$, so that X and Y lie on the arc in the order A, X, Y, C.

The radii DX and DY split the blind $AXYC$ into triangles ADX, XDY and YDC, each with two sides of length r meeting at D. Using area $= \frac{1}{2}ab\sin C$ for each,

$$\begin{aligned}\text{Area} &= \tfrac{1}{2}r^2\sin\theta + \tfrac{1}{2}r^2\sin\alpha + \tfrac{1}{2}r^2\sin(\pi - \theta - \alpha)\\ &= \tfrac{1}{2}r^2[\sin\alpha + \sin\theta + \sin(\theta + \alpha)],\end{aligned}$$

since $\sin(\pi - u) = \sin u$.

The best θ for fixed α

With α fixed, only $\sin\theta + \sin(\theta + \alpha)$ varies. By the sum-to-product formula $\sin P + \sin Q = 2\sin\frac{P+Q}{2}\cos\frac{P-Q}{2}$,

$$\sin\theta + \sin(\theta + \alpha) = 2\cos\tfrac{\alpha}{2}\,\sin\left(\theta + \tfrac{\alpha}{2}\right).$$

Ask yourself Why does this make the maximum obvious?

The factor $2\cos\frac{\alpha}{2}$ is a fixed positive constant, because $0 < \frac{\alpha}{2} < \frac{\pi}{2}$. So the area is largest exactly when $\sin(\theta + \frac{\alpha}{2})$ is largest. As θ runs over $[0, \pi - \alpha]$, the angle $\theta + \frac{\alpha}{2}$ runs over $[\frac{\alpha}{2}, \pi - \frac{\alpha}{2}]$, which contains $\frac{\pi}{2}$. So the sine reaches its greatest possible value 1, when $\theta + \frac{\alpha}{2} = \frac{\pi}{2}$, that is,

$$\boxed{\theta = \frac{\pi}{2} - \frac{\alpha}{2} = \frac{\pi - \alpha}{2}.}$$

Then $\angle YDC = \pi - \theta - \alpha = \frac{\pi-\alpha}{2} = \theta$ as well: the best blind is symmetric about the vertical radius, which is what intuition suggests. The maximum area for this α is

$$\tfrac{1}{2}r^2\left[\sin\alpha + 2\cos\tfrac{\alpha}{2}\right].$$

Letting α vary

Now we maximise $f(\alpha) = \sin\alpha + 2\cos\frac{\alpha}{2}$ for $0 < \alpha < \pi$. (For each α the largest area is $\frac{1}{2}r^2 f(\alpha)$, so the overall largest area is $\frac{1}{2}r^2$ times the largest value of f.) Differentiating,

$$f'(\alpha) = \cos\alpha - \sin\tfrac{\alpha}{2}.$$

Write $s = \sin\frac{\alpha}{2}$, so $0 < s < 1$, and use $\cos\alpha = 1 - 2\sin^2\frac{\alpha}{2}$:

$$f'(\alpha) = 1 - s - 2s^2 = (1 - 2s)(1 + s).$$

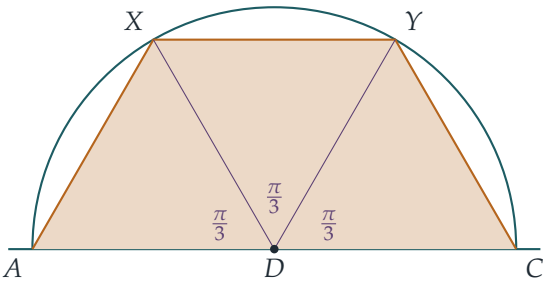
Since $1 + s > 0$, f' has the sign of $1 - 2s$. As s increases with α on $(0, \pi)$, f' is positive for $s < \frac{1}{2}$ and negative for $s > \frac{1}{2}$. So f increases and then decreases, with its greatest value where $\sin\frac{\alpha}{2} = \frac{1}{2}$, that is $\alpha = \frac{\pi}{3}$. There

$$f\left(\tfrac{\pi}{3}\right) = \sin\tfrac{\pi}{3} + 2\cos\tfrac{\pi}{6} = \tfrac{\sqrt{3}}{2} + \sqrt{3} = \tfrac{3\sqrt{3}}{2},$$

so the maximum possible area of the blind is

$$\boxed{\tfrac{1}{2}r^2 \cdot \tfrac{3\sqrt{3}}{2} = \frac{3\sqrt{3}}{4}r^2.}$$

At this point $\alpha = \theta = \angle YDC = \frac{\pi}{3}$, so the three triangles are equilateral and the blind is half of a regular hexagon inscribed in the full circle.



Checks

- Limiting cases: as $\alpha \to 0$ the blind becomes the triangle AXC with X at the top, area $\frac{1}{2} \cdot 2r \cdot r = r^2$, and indeed $\frac{1}{2}r^2 f(0) = r^2$. As $\alpha \to \pi$ the blind collapses onto the diameter, and $f(\pi) = 0$. The maximum $\frac{3\sqrt{3}}{4}r^2 \approx 1.299r^2$ beats both, and is less than the semicircle's area $\frac{\pi}{2}r^2 \approx 1.571r^2$, as it must be.
- Numerically: we checked the three-triangle formula against the shoelace formula for random positions of X and Y, located the best θ by a fine grid search for several values of α, and found that a joint search over θ and α with $r = 1$ gives largest area $1.299038\ldots = \frac{3\sqrt{3}}{4}$.

Common slips

- Forgetting the third triangle YDC, or giving its angle as $\pi - \alpha$ rather than $\pi - \theta - \alpha$.
- Using the shortcut $\theta = \alpha = \frac{\pi}{3}$ in the second part without justifying it. The calculus (or a symmetry argument) is needed to show that the best blind really is half of a regular hexagon.

Beyond STEP

The answer is a special case of a classical fact: among all n-gons inscribed in a circle, the regular one has the greatest area. One neat proof uses the concavity of sin on $[0, \pi]$ and Jensen's inequality: if the central angles $\phi_1, \ldots, \phi_n$ add to 2π, then $\sum \sin \phi_i \leq n \sin \frac{2\pi}{n}$. Our blind is half of the case $n = 6$. Convexity and inequalities of this kind are a running theme in first-year analysis; Abbott's *Understanding Analysis* is a good place to meet them properly, and Siklos's *Advanced Problems in Mathematics* has more STEP problems in the same spirit.

WORKED SOLUTION

STEP I 1990, Question 2

Pure • Roots of Unity Filter

STEP I 1990, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q2

This question has a full guided treatment in *Beyond STEP*, Section 5.2, *The Roots-of-Unity Filter*: https://beyondstep.keithmcnulty.org/ch05/roots-filter.html

The key idea

The cube roots of unity act as a filter. When we expand $(1+\omega^r)^n$ by the binomial theorem and add the results for $r = 1, 2, 3$, each binomial coefficient $\binom{n}{j}$ gets multiplied by $1+\omega^j+\omega^{2j}$, which is 3 when 3 divides j and 0 otherwise. So the sum keeps exactly the coefficients we want. The other way to evaluate each $(1+\omega^r)^n$ is to write $1+\omega^r$ in modulus–argument form and use de Moivre's theorem.

Solution

The cube roots of unity

Since $\omega = \mathrm{e}^{2\pi\mathrm{i}/3}$, we have $\omega^3 = \mathrm{e}^{2\pi\mathrm{i}} = 1$ but $\omega \neq 1$. Then

$$0 = \omega^3 - 1 = (\omega - 1)(1 + \omega + \omega^2),$$

and since $\omega - 1 \neq 0$ we may divide by it to get $1 + \omega + \omega^2 = 0$.

It follows that $1 + \omega^2 = -\omega$. Multiplying by $-1 = \mathrm{e}^{-\mathrm{i}\pi}$ rotates through $-\pi$, so

$$1 + \omega^2 = \mathrm{e}^{-\mathrm{i}\pi}\,\mathrm{e}^{2\pi\mathrm{i}/3} = \mathrm{e}^{-\mathrm{i}\pi/3}.$$

So $1 + \omega^2$ has modulus 1 and argument $-\frac{\pi}{3}$. (As a check, $1 + \omega^2 = 1 + \left(-\frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}\right) = \frac{1}{2} - \frac{\sqrt{3}}{2}\mathrm{i}$.) In the same way $1 + \omega = -\omega^2 = \mathrm{e}^{\mathrm{i}\pi/3}$, the complex conjugate.

The identity

First way: the binomial theorem. For $r = 1, 2, 3$,

$$(1+\omega^r)^n = \sum_{j=0}^{n} \binom{n}{j} \omega^{rj}.$$

Adding the three cases,

$$(1+\omega)^n + (1+\omega^2)^n + (1+\omega^3)^n = \sum_{j=0}^{n} \binom{n}{j} \left(\omega^j + \omega^{2j} + \omega^{3j}\right).$$

Ask yourself What is $\omega^j + \omega^{2j} + \omega^{3j}$ for a general integer j?

Since $\omega^{3j} = 1$, the bracket is $1 + \omega^j + \omega^{2j}$, and it depends only on j modulo 3, because $\omega^3 = 1$.

- If j is a multiple of 3, then $\omega^j = 1$ and the bracket is $1 + 1 + 1 = 3$.
- If $j \equiv 1 \pmod 3$, then $\omega^j = \omega$ and $\omega^{2j} = \omega^2$, so the bracket is $1 + \omega + \omega^2 = 0$.
- If $j \equiv 2 \pmod 3$, then $\omega^j = \omega^2$ and $\omega^{2j} = \omega^4 = \omega$, so again the bracket is 0.

So only the terms with $3 \mid j$ survive, and

$$(1+\omega)^n + (1+\omega^2)^n + (1+\omega^3)^n = 3\left[\binom{n}{0} + \binom{n}{3} + \cdots + \binom{n}{k}\right],$$

where k is the largest multiple of 3 that is at most n.

Second way: modulus and argument. From the first part, $1 + \omega = \mathrm{e}^{\mathrm{i}\pi/3}$ and $1 + \omega^2 = \mathrm{e}^{-\mathrm{i}\pi/3}$, and $1 + \omega^3 = 2$. By de Moivre's theorem,

$$\begin{aligned}(1+\omega)^n + (1+\omega^2)^n + (1+\omega^3)^n &= \mathrm{e}^{\mathrm{i}n\pi/3} + \mathrm{e}^{-\mathrm{i}n\pi/3} + 2^n \\ &= 2^n + 2\cos\left(\frac{n\pi}{3}\right).\end{aligned}$$

Comparing the two evaluations and dividing by 3,

$$\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \cdots + \binom{n}{k} = \frac{1}{3}\left(2^n + 2\cos\left(\frac{n\pi}{3}\right)\right),$$

which is the required result.

The case $n = 25$

The largest multiple of 3 not exceeding 25 is 24, so the first sum is exactly the identity with $n = 25$. Now $\frac{25\pi}{3} = 8\pi + \frac{\pi}{3}$, so $\cos\frac{25\pi}{3} = \cos\frac{\pi}{3} = \frac{1}{2}$, and

$$\binom{25}{0} + \binom{25}{3} + \cdots + \binom{25}{24} = \frac{1}{3}\left(2^{25} + 1\right) = \frac{33554433}{3} = 11184811.$$

The second sum

Here $n = 24$ and we want the coefficients $\binom{24}{j}$ with $j \equiv 2 \pmod 3$. Group all the coefficients $\binom{24}{j}$ by the remainder of j on division by 3:

$$S_0 + S_1 + S_2 = \sum_{j=0}^{24} \binom{24}{j} = 2^{24},$$

where S_m is the sum of $\binom{24}{j}$ over $j \equiv m \pmod 3$.

The symmetry $\binom{24}{j} = \binom{24}{24-j}$ pairs these up: if $j \equiv 2$ then $24 - j \equiv -2 \equiv 1 \pmod 3$, since 24 is a multiple of 3. So $j \mapsto 24 - j$ matches the terms of S_2 one-to-one with the terms of S_1, and $S_1 = S_2$.

The identity with $n = 24$ gives S_0: as $\cos 8\pi = 1$,

$$S_0 = \frac{1}{3}\left(2^{24} + 2\right).$$

Hence

$$S_2 = \frac{1}{2}\left(2^{24} - S_0\right) = \frac{1}{2} \cdot \frac{2 \cdot 2^{24} - 2}{3} = \frac{2^{24} - 1}{3}.$$

Halving the given value, $2^{24} = 16777216$, so

$$\binom{24}{2} + \binom{24}{5} + \cdots + \binom{24}{23} = \frac{16777215}{3} = 5592405.$$

Checks

- Small cases: for $n = 3$ the left side is $1 + 1 = 2$ and the right side is $\frac{1}{3}(8 + 2\cos\pi) = 2$; for $n = 4$ it is $1 + 4 = 5$ against $\frac{1}{3}\left(16 + 2\cos\frac{4\pi}{3}\right) = \frac{1}{3}(16 - 1) = 5$.
- Divisibility: $2^{25} + 1$ and $2^{24} - 1$ must both be multiples of 3, and they are, since $2 \equiv -1 \pmod 3$.
- We summed the binomial coefficients directly by computer: the identity holds for every n up to 199, the first sum is 11184811 and the second is 5592405.

Common slips

- Giving the argument of $1 + \omega^2$ as $\frac{\pi}{3}$: it lies in the fourth quadrant, so the principal argument is $-\frac{\pi}{3}$ (equivalently $\frac{5\pi}{3}$).
- Forgetting the case $r = 3$, which contributes the 2^n.
- In the second sum, trying to use the formula directly: it only gives multiples of 3, so another idea (here, symmetry) is needed for the remainder-2 terms.

Beyond STEP

This trick is called a *roots of unity filter*, and it is a first example of the discrete Fourier transform: averaging $f(\omega^r)$ over the mth roots of unity picks out the coefficients of a polynomial f in a chosen residue class modulo m. The same averaging over a group of symmetries reappears in the character theory of finite groups and in counting arguments such as Burnside's lemma. For more on why complex numbers make such calculations natural, see Needham, *Visual Complex Analysis*.

WORKED SOLUTION

STEP I 1990, Question 3

Pure • Normals and Parallel Curves

STEP I 1990, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q3

The key idea

Everything follows once we write down the downward unit normal explicitly and so get the coordinates of $\mathbf{s}$ in terms of x. Part (i) is then a matter of forcing the y-coordinate of $\mathbf{s}$ to vanish. In part (ii) the suggested substitution makes the coordinates of $\mathbf{s}$ very simple, and the gradient condition turns into a first-order differential equation that turns out to be an exact derivative.

Solution

Setting up. The tangent direction to $y = \mathrm{f}(x)$ is $(1, \mathrm{f}'(x))$, so the normals are $\pm(\mathrm{f}'(x), -1)$. The downward one has negative y-component, so

$$\mathbf{n} = \frac{1}{N}\begin{pmatrix} \mathrm{f}'(x) \\ -1 \end{pmatrix}, \qquad N = \sqrt{1 + \mathrm{f}'(x)^2},$$

and therefore

$$\mathbf{s} = \mathbf{r} + \mathrm{p}(x)\mathbf{n} = \begin{pmatrix} x + \dfrac{\mathrm{p}(x)\,\mathrm{f}'(x)}{N} \\ \mathrm{f}(x) - \dfrac{\mathrm{p}(x)}{N} \end{pmatrix}.$$

Note also that $|\mathbf{r} - \mathbf{s}| = |\mathrm{p}(x)|$, since $\mathbf{n}$ is a unit vector.

Part (i)

Here $\mathrm{f}(x) = x^k$, $\mathrm{f}'(x) = kx^{k-1}$ and $N = \sqrt{1 + k^2 x^{2k-2}}$. For $\mathbf{s}$ to lie on the x-axis we need its y-coordinate to be zero:

$$x^k - \frac{\mathrm{p}(x)}{N} = 0 \quad \Longrightarrow \quad \mathrm{p}(x) = x^k\sqrt{1 + k^2 x^{2k-2}}.$$

With this choice the x-coordinate of $\mathbf{s}$ is

$$x + x^k \cdot kx^{k-1} = x + kx^{2k-1},$$

which is continuous, increasing, tends to 0 as $x \to 0^+$ and tends to infinity as $x \to \infty$. So $\mathbf{s}$ really does sweep out the whole positive x-axis, not just part of it.

Now $|\mathbf{r}|^2 = x^2 + x^{2k}$, and $|\mathbf{r} - \mathbf{s}|^2 = \mathrm{p}(x)^2 = x^{2k} + k^2 x^{4k-2}$ (here $\mathrm{p} > 0$). So $|\mathbf{r}| = |\mathbf{r} - \mathbf{s}|$ for every $x > 0$ exactly when

$$x^2 = k^2 x^{4k-2} \quad \text{for all } x > 0.$$

Taking $x = 1$ forces $k^2 = 1$, so $k = 1$; and conversely $k = 1$ makes both sides equal to x^2. Hence $k = 1$.

Geometrically: on the line $y = x$, the downward normal from (x, x) meets the x-axis at $(2x, 0)$, and the triangle with vertices O, (x, x), $(2x, 0)$ is isosceles with both equal sides of length $\sqrt{2}\,x$.

Part (ii)

Now $\mathrm{f}(x) = x^2$, $\mathrm{f}'(x) = 2x$ and $N = \sqrt{1 + 4x^2}$. Writing $\mathrm{p} = \mathrm{q}N$ cancels the awkward square roots:

$$\mathbf{s} = \begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} x + 2x\,\mathrm{q}(x) \\ x^2 - \mathrm{q}(x) \end{pmatrix}.$$

The new curve is given parametrically by x, so its gradient at $\mathbf{s}$ is

$$\frac{\mathrm{d}Y}{\mathrm{d}X} = \frac{\mathrm{d}Y/\mathrm{d}x}{\mathrm{d}X/\mathrm{d}x} = \frac{2x - \mathrm{q}'}{1 + 2\mathrm{q} + 2x\mathrm{q}'}.$$

Setting this equal to $2x$, the gradient of $y = x^2$ at $\mathbf{r}$, and multiplying up:

$$\begin{aligned} & 2x - \mathrm{q}' = 2x + 4x\mathrm{q} + 4x^2\mathrm{q}' \\ \iff \quad & (1 + 4x^2)\,\mathrm{q}' + 4x\,\mathrm{q} = 0. \end{aligned}$$

> **Ask yourself** Can we solve this without dividing by q (which might be zero somewhere)?

Yes: divide instead by $N = \sqrt{1 + 4x^2}$, which is never zero. Since $N' = 4x/N$,

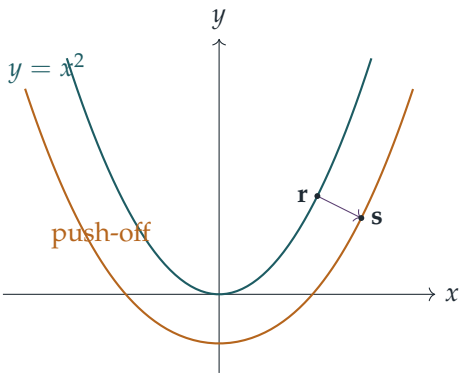
$$\frac{(1 + 4x^2)\,\mathrm{q}' + 4x\,\mathrm{q}}{N} = N\mathrm{q}' + \frac{4x}{N}\,\mathrm{q} = \frac{\mathrm{d}}{\mathrm{d}x}(N\mathrm{q}) = \mathrm{p}'(x).$$

So the condition is simply $\mathrm{p}'(x) = 0$ for all x, and hence

$$\mathrm{p}(x) = c, \qquad \text{a constant}$$

(equivalently $\mathrm{q}(x) = c/\sqrt{1 + 4x^2}$). Conversely, every constant c works, since each step above is reversible wherever the gradient of the new curve is defined.

The new curve is a *parallel curve*: every point of the parabola is pushed a fixed distance c along its normal. The picture shows $c = \frac{1}{2}$.



One subtlety: $\mathrm{d}X/\mathrm{d}x = 1 + 2\mathrm{q} + 2x\mathrm{q}'$, which with $\mathrm{q} = c/N$ simplifies to $1 + 2c/N^3$. For $c > -\frac{1}{2}$ this is always positive, so the push-off is a smooth curve. If $c \leq -\frac{1}{2}$ (pushing *upwards* by at least $\frac{1}{2}$, the radius of curvature at the vertex), it vanishes where $N^3 = -2c$ and the push-off develops cusps there.

Checks

- In (i) with $k = 2$ and $x = 1$: $\mathrm{p}(1) = \sqrt{5}$, and $\mathbf{s} = (1,1) + \sqrt{5} \cdot \frac{1}{\sqrt{5}}(2,-1) = (3,0)$, on the x-axis as required; but $|\mathbf{r}| = \sqrt{2} \neq \sqrt{5}$, so $k = 2$ fails.
- In (ii), at $x = 0$ the push-off with constant c passes through $(0,-c)$, and by symmetry its gradient there is 0, matching the parabola's vertex.
- Numerically, for $c = 1$ and $x = 0.8$, a finite-difference gradient of the push-off agrees with $2x = 1.6$; with the non-constant choice $\mathrm{p} = x^2$ it comes out as about 0.41 instead.

Common slips

- Using the upward normal $(-\mathrm{f}', 1)/N$: the signs in $\mathbf{s}$ all flip.
- Writing the gradient of the new curve as $\mathrm{d}Y/\mathrm{d}x$ rather than $(\mathrm{d}Y/\mathrm{d}x)/(\mathrm{d}X/\mathrm{d}x)$; the new curve's x-coordinate is X, not x.
- In (i), concluding $k = 1$ from matching exponents ($4k - 2 = 2$) without also checking the coefficient $k^2 = 1$.

Beyond STEP

Push-offs at constant distance are called *parallel* or *offset* curves, and they matter in practice: a milling machine's cutter centre follows the parallel curve of the shape being cut. The cusps that appear when the offset exceeds the radius of curvature lie on the *evolute*, the locus of centres of curvature, a central object in the differential geometry of

curves. A good introduction is Pressley, *Elementary Differential Geometry*.

WORKED SOLUTION

STEP I 1990, Question 4

Pure • Arithmetic Progressions and Differences

STEP I 1990, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q4

The key idea

Every term of a progression is the first term plus a running total of differences. For an arithmetic progression the differences are constant, so the sum is easy. For a double arithmetic progression the differences themselves form an arithmetic progression, so we get one more layer of summing, and the binomial coefficients $\binom{n}{2}$ and $\binom{n}{3}$ appear naturally. The last part then becomes two linear equations for the two unknown differences.

Solution

Arithmetic progressions

Write $d = a_2 - a_1$ for the common difference, so $a_k = a_1 + (k-1)d$. Then

$$\sum_{k=1}^{n} a_k = na_1 + d\sum_{k=1}^{n}(k-1) = na_1 + \frac{n(n-1)}{2}(a_2 - a_1),$$

using $0 + 1 + \cdots + (n-1) = \frac{1}{2}n(n-1)$. Collecting the terms in a_1 and a_2,

$$a_1 + a_2 + \cdots + a_n = \frac{n}{2}\big[(3-n)a_1 + (n-1)a_2\big].$$

Double arithmetic progressions

Let the differences be $\delta_k = b_{k+1} - b_k$. These form an arithmetic progression with first term $p = \delta_1 = b_2 - b_1$ and common difference

$$e = \delta_2 - \delta_1 = (b_3 - b_2) - (b_2 - b_1) = b_3 - 2b_2 + b_1.$$

So $\delta_j = p + (j-1)e$, and adding up the first $k-1$ differences (the formula from the first part, with n replaced by $k-1$) gives

$$b_k = b_1 + \sum_{j=1}^{k-1}\delta_j = b_1 + (k-1)p + \frac{(k-1)(k-2)}{2}e = b_1 + \binom{k-1}{1}p + \binom{k-1}{2}e.$$

Now sum over $k = 1, \ldots, n$. We need two sums. The first is $\sum_{k=1}^{n}(k-1) = \binom{n}{2}$, as before. For the second, with $j = k-1$ running from 0 to $n-1$,

$$\sum_{j=0}^{n-1} \frac{j(j-1)}{2} = \frac{1}{2}\left[\frac{(n-1)n(2n-1)}{6} - \frac{(n-1)n}{2}\right] = \frac{n(n-1)}{12}\left[(2n-1)-3\right]$$
$$= \frac{n(n-1)(n-2)}{6} = \binom{n}{3}.$$

Therefore

$$\sum_{k=1}^{n} b_k = nb_1 + \binom{n}{2}(b_2 - b_1) + \binom{n}{3}(b_3 - 2b_2 + b_1).$$

This is already a formula in n, b_1, b_2, b_3. If you prefer the coefficient of each b_i separately, expanding gives

$$b_1 + \cdots + b_n = \frac{n(n^2 - 6n + 11)}{6}\, b_1 + \frac{n(n-1)(7-2n)}{6}\, b_2 + \frac{n(n-1)(n-2)}{6}\, b_3.$$

(Quick test: $n = 1, 2, 3$ give b_1, $b_1 + b_2$ and $b_1 + b_2 + b_3$.)

? **Ask yourself** Is there a neater way to see where $\binom{n}{3}$ comes from?

Yes: $\binom{k-1}{2}$ is the number of ways to choose 2 things from $k-1$, and the hockey-stick identity $\sum_{k=1}^{n}\binom{k-1}{2} = \binom{n}{3}$ (choose 3 from n by first fixing the largest, k) does the summing in one step. Each layer of "differences of differences" raises the binomial coefficient by one, just as integrating raises a power by one.

The factorial progression

Now $b_1 = 1$, and with p and e as above, $b_k = 1 + (k-1)p + \frac{1}{2}(k-1)(k-2)e$. So

$$b_2 = 1 + p, \qquad b_4 = 1 + 3p + 3e, \qquad b_6 = 1 + 5p + 10e.$$

In a factorial progression with $c_1 = b_2$, $c_2 = b_4$, $c_3 = b_6$, $c_4 = 220$, the successive differences are $1!\,d$, $2!\,d$, $3!\,d$, that is, d, $2d$, $6d$:

$$b_4 - b_2 = 2p + 3e = d,$$
$$b_6 - b_4 = 2p + 7e = 2d,$$
$$220 - b_6 = 219 - 5p - 10e = 6d.$$

Substituting the first into the second, $2p + 7e = 4p + 6e$, so $e = 2p$ and then $d = 8p$. The third equation becomes

$$219 - 5p - 20p = 48p \quad \Longrightarrow \quad 73p = 219 \quad \Longrightarrow \quad p = 3,$$

so $e = 6$ and $d = 24$, which is non-zero as required. The solution is unique, since every step was forced.

So $b_k = 1 + 3(k-1) + 3(k-1)(k-2) = 3k^2 - 6k + 4$, giving the sequence $1, 4, 13, 28, 49, 76, \ldots$, and indeed $b_2, b_4, b_6, 220 = 4, 28, 76, 220$ have differences $24, 48, 144 = 1! \cdot 24,\ 2! \cdot 24,\ 3! \cdot 24$.

Finally, using the sum formula with $b_1 = 1$, $b_2 - b_1 = 3$ and $b_3 - 2b_2 + b_1 = 6$:

$$\begin{aligned} 1 + b_2 + \cdots + b_n &= n + 3 \cdot \frac{n(n-1)}{2} + 6 \cdot \frac{n(n-1)(n-2)}{6} \\ &= \frac{n}{2}\left[2 + 3(n-1) + 2(n-1)(n-2)\right] \\ &= \frac{n(2n^2 - 3n + 3)}{2}. \end{aligned}$$

Checks

For $n = 1, 2, 3$ the final formula gives 1, 5 and 18, matching 1, $1 + 4$ and $1 + 4 + 13$. For $n = 6$ it gives $3 \cdot 57 = 171$, and directly $1 + 4 + 13 + 28 + 49 + 76 = 171$. A computer also checked both general sum formulas exactly (with fractions) for hundreds of random progressions and all $n \leq 24$.

Common slips

- Taking the differences of the factorial progression as $d, 2d, 3d$: they are $1!\,d$, $2!\,d$, $3!\,d$, so the third is $6d$.
- Writing $b_k = b_1 + kp + \ldots$: there are only $k - 1$ differences between b_1 and b_k.

Beyond STEP

This is the calculus of finite differences: a sequence whose mth differences are constant is a polynomial of degree m in n, and its partial sums are a polynomial of degree $m + 1$, best written in the basis $\binom{n}{0}, \binom{n}{1}, \binom{n}{2}, \ldots$ (Newton's forward-difference formula). Summing $\binom{n}{r}$ to get $\binom{n+1}{r+1}$ is the discrete twin of $\int x^r \,\mathrm{d}x = \frac{x^{r+1}}{r+1}$. For more in this spirit, try the problems in Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP I 1990, Question 5

Pure • Partial Fractions and Curve Sketching

STEP I 1990, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q5

The key idea

In (i) the quadratic factorises, so partial fractions turn the integral into a difference of logarithms. In (ii) the huge powers look frightening, but $1760 = 8 \times 220$, so the function is really $u^8 - u + q$ with $u = x^{220} \geq 0$. The shape of the graph comes from the stationary points, and the number of roots depends only on where the horizontal axis sits relative to the local maximum and the two minima.

Solution

Part (i)

Since $6 \times 15 = 90 = 9 \times 10$ and $9 + 10 = 19$, the quadratic factorises:

$$6x^2 + 19x + 15 = (2x+3)(3x+5).$$

Write

$$\frac{1}{(2x+3)(3x+5)} = \frac{A}{2x+3} + \frac{B}{3x+5}, \qquad \text{so} \qquad 1 = A(3x+5) + B(2x+3).$$

Putting $x = -\frac{3}{2}$ gives $\frac{1}{2}A = 1$, so $A = 2$; putting $x = -\frac{5}{3}$ gives $-\frac{1}{3}B = 1$, so $B = -3$. Both $2x+3$ and $3x+5$ are positive on $[1,3]$, so no modulus signs are needed:

$$\int_1^3 \frac{\mathrm{d}x}{6x^2+19x+15} = \int_1^3 \left(\frac{2}{2x+3} - \frac{3}{3x+5}\right)\mathrm{d}x = \Big[\ln(2x+3) - \ln(3x+5)\Big]_1^3$$
$$= \ln\frac{9}{14} - \ln\frac{5}{8} = \ln\frac{72}{70} = \ln\frac{36}{35}.$$

Part (ii)

Symmetry and end behaviour. Both powers are even, so f is an even function and its graph is symmetric in the y-axis. For large $|x|$ the term x^{1760} dominates, so $\mathrm{f}(x) \to +\infty$ as $x \to \pm\infty$. Also $\mathrm{f}(0) = q$ and $\mathrm{f}(\pm 1) = q$.

Stationary points. Differentiating,

$$\mathrm{f}'(x) = 1760x^{1759} - 220x^{219} = 220\,x^{219}\left(8x^{1540} - 1\right).$$

This vanishes at $x = 0$ and where $x^{1540} = \frac{1}{8}$, that is at $x = \pm a$ with

$$a = 8^{-1/1540} \approx 0.9987.$$

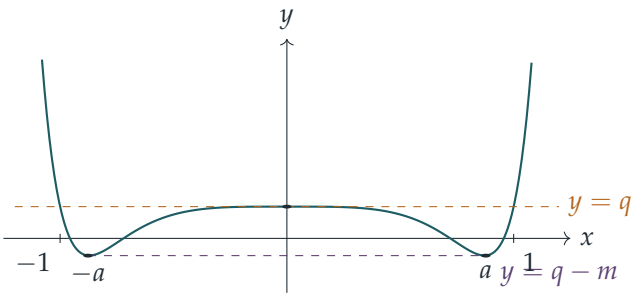
For $x > 0$, $\mathrm{f}'(x)$ has the sign of $8x^{1540} - 1$: negative on $(0, a)$ and positive for $x > a$. Since x^{219} is an odd power, the signs are reversed for $x < 0$. So f decreases on $(-\infty, -a)$, increases on $(-a, 0)$, decreases on $(0, a)$ and increases on (a, ∞): there is a local maximum q at $x = 0$ and two equal minima at $x = \pm a$.

Ask yourself How do we find the minimum value without wrestling with $8^{-1/1540}$?

Work with $u = x^{220}$. At $x = \pm a$ we have $x^{1540} = u^7 = \frac{1}{8}$, so $u = 8^{-1/7}$ and $u^8 = u \cdot u^7 = \frac{1}{8}u$. Hence

$$\mathrm{f}(\pm a) = u^8 - u + q = q - \tfrac{7}{8}u = q - m, \qquad \text{where} \qquad m = \tfrac{7}{8} \cdot 8^{-1/7} = 7 \cdot 2^{-24/7} \approx 0.650.$$

The sketch. The graph is a "W": it starts high, falls to the minimum $q - m$ at $x = -a$, rises to the local maximum q at the origin, falls to the minimum $q - m$ at $x = a$ and then rises steeply. Because the powers are so large, the real graph is extremely flat: x^{220} is tiny unless $|x|$ is very close to 1, so $\mathrm{f}(x) \approx q$ on most of $(-1, 1)$, with two narrow dips just inside $x = \pm 1$ (where the graph returns to height q) and near-vertical walls outside. The picture below exaggerates the dips so the features can be seen.



Counting roots. The roots of $\mathrm{f}(x) = 0$ are where the graph meets the x-axis, so we compare 0 with the local maximum q and the minimum value $q - m$.

- $q > m$: the minimum value $q - m$ is positive, so $\mathrm{f} > 0$ everywhere: **no roots**.
- $q = m$: the graph touches the axis only at the minima: **two roots**, $x = \pm a$.
- $0 < q < m$: now $q - m < 0 < q$. On each of the four intervals of monotonicity, $(-\infty, -a)$, $(-a, 0)$, $(0, a)$, (a, ∞), f is continuous and strictly monotonic and passes from a positive value to a negative one or vice versa, so it has exactly one root on each: **four roots**.

- $q = 0$: $f(x) = x^{220}(x^{1540} - 1)$, which vanishes exactly at $x = 0$ and $x = \pm 1$: **three roots**.
- $q < 0$: the local maximum $f(0) = q$ is negative, so $f < 0$ on the whole of $[-a, a]$, and f is strictly monotonic from a negative value to $+\infty$ on each of (a, ∞) and $(-\infty, -a)$: **two roots**.

So the possible numbers of distinct roots are 0, 2, 3 and 4, with

$$\begin{array}{ll} 0 \text{ roots}: & q > m, \\ 2 \text{ roots}: & q < 0 \text{ or } q = m, \\ 3 \text{ roots}: & q = 0, \\ 4 \text{ roots}: & 0 < q < m, \end{array} \qquad m = \tfrac{7}{8} \cdot 8^{-1/7} = 7 \cdot 2^{-24/7} \approx 0.650.$$

Checks

Simpson's rule on the integrand in (i) gives 0.0281709, matching $\ln \frac{36}{35} = 0.0281709$. For (ii), a fine numerical search for the minimum of $x^{1760} - x^{220}$ near $x = 1$ gives -0.650123 at $x \approx 0.99865$, in agreement with $-m$ and a. Counting sign changes of f on a fine grid for $q = 1,\ 0.5,\ 0.1,\ 0.001,\ 0,\ -0.01,\ -5$ gives $0, 4, 4, 4, 3, 2, 2$ roots, as the table predicts. The case $q = 0$ is a good sanity check on the sketch: the graph must pass through the origin and through $(\pm 1, 0)$.

Common slips

- Forgetting the boundary cases $q = 0$ (three roots, because the local maximum sits on the axis) and $q = m$ (two repeated roots at the minima).
- Claiming two roots for $q < 0$ "by symmetry" without checking that nothing happens between $-a$ and a: it is the fact that the local maximum q is negative which rules out extra roots there.
- Sketching a gently curved "W". With powers this large the graph is almost flat at height q for $|x| < 1$; it is worth saying so even if the sketch exaggerates the dips.

Beyond STEP

Counting roots by locating the stationary values is a first taste of the intermediate value theorem and monotonicity arguments made rigorous in a first analysis course; the substitution $u = x^{220}$ also shows how x^n behaves as $n \to \infty$, converging to 0 on $(-1, 1)$ but not uniformly. Abbott's *Understanding Analysis* treats both continuity and uniform convergence with care and readability.

WORKED SOLUTION

STEP I 1990, Question 6

Pure • Vectors and Parallelograms

STEP I 1990, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q6

The key idea

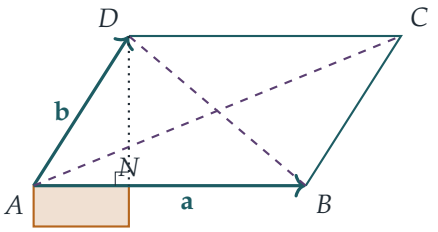
Put the origin at A and describe everything with the two side vectors $\mathbf{a} = \overrightarrow{AB}$ and $\mathbf{b} = \overrightarrow{AD}$. Then the diagonals are $\mathbf{a} + \mathbf{b}$ and $\mathbf{b} - \mathbf{a}$, and every squared length is a dot product of a vector with itself. Expanding those dot products does all the work. The only geometry needed is that $|\mathbf{u} \cdot \mathbf{v}|$ is the length of $\mathbf{u}$ times the length of the projection of $\mathbf{v}$ onto $\mathbf{u}$.

Solution

Take A as the origin, and let $\overrightarrow{AB} = \mathbf{a}$ and $\overrightarrow{AD} = \mathbf{b}$. Since $ABCD$ is a parallelogram, $\overrightarrow{BC} = \overrightarrow{AD} = \mathbf{b}$, so

$$\overrightarrow{AC} = \mathbf{a} + \mathbf{b}, \qquad \overrightarrow{BD} = \mathbf{b} - \mathbf{a}.$$

For any vector, $|\mathbf{v}|^2 = \mathbf{v} \cdot \mathbf{v}$, and the dot product is symmetric and distributes over addition. Write $a = |\mathbf{a}|$ and $b = |\mathbf{b}|$.



In the figure, N is the foot of the perpendicular from D to the line AB, so AN is the projection of AD on AB, with length $|\mathbf{a} \cdot \mathbf{b}|/a$.

Part (i)

Opposite sides are equal, so the left-hand side is $2a^2 + 2b^2$. For the diagonals,

$$AC^2 = (\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b}) = a^2 + 2\,\mathbf{a}\cdot\mathbf{b} + b^2,$$
$$BD^2 = (\mathbf{b}-\mathbf{a})\cdot(\mathbf{b}-\mathbf{a}) = a^2 - 2\,\mathbf{a}\cdot\mathbf{b} + b^2.$$

Adding, the cross terms cancel and $AC^2 + BD^2 = 2a^2 + 2b^2 = AB^2 + BC^2 + CD^2 + DA^2$, as required. (This is often called the parallelogram law.)

Part (ii)

Subtracting the same two lines instead,

$$AC^2 - BD^2 = 4\,\mathbf{a}\cdot\mathbf{b}, \qquad \text{so} \qquad \left|AC^2 - BD^2\right| = 4\,|\mathbf{a}\cdot\mathbf{b}|.$$

Now interpret $|\mathbf{a}\cdot\mathbf{b}|$. If θ is the angle between $\mathbf{a}$ and $\mathbf{b}$, then $|\mathbf{a}\cdot\mathbf{b}| = a\cdot b|\cos\theta|$, and $b|\cos\theta|$ is exactly the length AN of the projection of AD on AB. So $|\mathbf{a}\cdot\mathbf{b}|$ is the area of the rectangle (shaded in the figure) with sides AB and the projection of AD on AB.

> **Ask yourself** The result claims this works for *any* side and *either* adjacent side. Why don't we need eight separate calculations?

Every side of the parallelogram is a vector $\pm\mathbf{a}$ or $\pm\mathbf{b}$, and the two sides adjacent to it are the other vector, up to sign. For example, side BC is $\mathbf{b}$, and its neighbours AB and CD are $\mathbf{a}$ and $-\mathbf{a}$. Changing signs doesn't change the lengths of the vectors or of the projection, since the projection has length $|\mathbf{u}\cdot\mathbf{v}|/|\mathbf{u}|$. So every choice of side and adjacent side gives the same rectangle area $|\mathbf{a}\cdot\mathbf{b}|$: either $a\times$ (projection of $\mathbf{b}$ on $\mathbf{a}$) or $b\times$ (projection of $\mathbf{a}$ on $\mathbf{b}$), and both equal $|\mathbf{a}\cdot\mathbf{b}|$. Hence, in every case,

$$\left|AC^2 - BD^2\right| = 4\times(\text{area of that rectangle}). \qquad \blacksquare$$

The final part

Part (ii) came from the dot product of the two *sides*. Swap the roles of sides and diagonals and take the dot product of the two *diagonals*:

$$\overrightarrow{AC}\cdot\overrightarrow{BD} = (\mathbf{a}+\mathbf{b})\cdot(\mathbf{b}-\mathbf{a}) = b^2 - a^2,$$

because the cross terms $\mathbf{a}\cdot\mathbf{b}$ and $-\mathbf{b}\cdot\mathbf{a}$ cancel. Therefore

$$\left|AB^2 - AD^2\right| = \left|\overrightarrow{AC}\cdot\overrightarrow{BD}\right|.$$

Arguing as in part (ii), this gives the following.

Result. $\left|AB^2 - AD^2\right|$ equals the area of the rectangle whose sides are either diagonal and the projection of the other diagonal on it.

Proof. If φ is the angle between the diagonals, $\left|\overrightarrow{AC} \cdot \overrightarrow{BD}\right| = AC \cdot BD\,|\cos\varphi|$. Here $BD\,|\cos\varphi|$ is the length of the projection of BD on AC, and $AC\,|\cos\varphi|$ is the length of the projection of AC on BD. So the area is the same whichever diagonal we start from, and it equals $\left|AB^2 - AD^2\right|$ by the display above. ■

To make it look even more like (ii), let O be the point where the diagonals bisect each other. Then $AC = 2\,OA$ and the projection of BD on AC is twice the projection of OB on OA. So $\left|AB^2 - AD^2\right|$ is also 4 times the area of the rectangle whose sides are OA and the projection of OB on OA: half-diagonals now play the part of the sides.

Two pleasant corollaries fall out. The diagonals are equal exactly when $\mathbf{a} \cdot \mathbf{b} = 0$, so exactly when the parallelogram is a rectangle. They are perpendicular exactly when $a = b$, so exactly when it is a rhombus.

Checks

Take $A = (0,0)$, $B = (4,0)$, $D = (1,2)$, so $C = (5,2)$. Then $AC^2 = 29$ and $BD^2 = 13$, and $2(16+5) = 42 = 29 + 13$, which checks (i). For (ii), $AC^2 - BD^2 = 16$, while the projection of AD on AB has length 1, so the rectangle has area $4 \times 1 = 4$, and $16 = 4 \times 4$. For the final part, $AB^2 - AD^2 = 16 - 5 = 11$, and $\overrightarrow{AC} \cdot \overrightarrow{BD} = (5,2) \cdot (-3,2) = -11$. We also tested all three results, for every choice of side and adjacent side, on 10^5 random parallelograms; the largest discrepancy was about 10^{-13}, which is just rounding error.

Common slips

- Writing $\overrightarrow{BD} = \mathbf{a} - \mathbf{b}$ or $\mathbf{a} + \mathbf{b}$. The sign doesn't matter for BD^2, but getting the direction right avoids confusion later.
- Forgetting the modulus: $\mathbf{a} \cdot \mathbf{b}$ is negative when the angle at A is obtuse, but an area can't be, which is why the question has $\left|AC^2 - BD^2\right|$.
- Giving a factor of 4 in the final part while using whole diagonals. With whole diagonals the factor is 1; the factor 4 appears only with half-diagonals.

Beyond STEP

The identity $|\mathbf{a} + \mathbf{b}|^2 - |\mathbf{a} - \mathbf{b}|^2 = 4\,\mathbf{a} \cdot \mathbf{b}$ is the *polarisation identity*: it shows that the dot product can be recovered from lengths alone. Together with the parallelogram law of part (i), it is a key fact about inner product spaces. A norm comes from an inner product exactly when it satisfies the parallelogram law, which is a theorem of Jordan and von Neumann. Strang's *Introduction to Linear Algebra* is a good place to see dot products, projections and orthogonality developed in any number of dimensions.

WORKED SOLUTION

STEP I 1990, Question 7

Pure • First Order Linear ODEs

STEP I 1990, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q7

The key idea

Writing $y = uv$ gives us two unknown functions where we only need one, so we can afford to impose an extra condition on one of them. The natural choice is to make u kill the term containing P, which leaves a differential equation that integrates directly. This is the integrating-factor method in disguise: $1/u$ is exactly the integrating factor.

Solution

The substitution

With $y = uv$, the product rule gives $\dfrac{\mathrm{d}y}{\mathrm{d}x} = u\dfrac{\mathrm{d}v}{\mathrm{d}x} + v\dfrac{\mathrm{d}u}{\mathrm{d}x}$, so the equation becomes

$$u\frac{\mathrm{d}v}{\mathrm{d}x} + v\left(\frac{\mathrm{d}u}{\mathrm{d}x} + Pu\right) = Q.$$

If we choose u to satisfy the first order equation

$$\frac{\mathrm{d}u}{\mathrm{d}x} + Pu = 0,$$

the bracket vanishes and we are left with

$$\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{Q}{u},$$

an equation for $\frac{\mathrm{d}v}{\mathrm{d}x}$ in terms of x, Q and u only, as required. (We need $u \neq 0$ to divide; the non-zero solutions of $u' + Pu = 0$ are $u = A\mathrm{e}^{-\int P\,\mathrm{d}x}$ with $A \neq 0$, which never vanish.)

The particular equation

Here $P = -\dfrac{2}{x+1}$ and $Q = (x+1)^{5/2}$. Note that Q is only defined (as a real number) for $x \geq -1$ and P is undefined at $x = -1$, so we work on $x > -1$, where $x + 1 > 0$.

Ask yourself Do we need the general solution for u, or will any one solution do?

Any one non-zero solution will do: the arbitrary constant in the final answer appears when we integrate for v. So we solve

$$\frac{\mathrm{d}u}{\mathrm{d}x} = \frac{2u}{x+1}$$

by separating variables: $\ln|u| = 2\ln(x+1) + \text{const}$, and we take

$$u = (x+1)^2.$$

(Check: $u' = 2(x+1) = \frac{2u}{x+1}$.) Then

$$\frac{\mathrm{d}v}{\mathrm{d}x} = \frac{(x+1)^{5/2}}{(x+1)^2} = (x+1)^{1/2}, \qquad \text{so} \qquad v = \tfrac{2}{3}(x+1)^{3/2} + C.$$

Hence the general solution is

$$y = uv = \tfrac{2}{3}(x+1)^{7/2} + C(x+1)^2.$$

Applying $y(1) = 0$:

$$0 = \tfrac{2}{3}\cdot 2^{7/2} + 4C \quad \Longrightarrow \quad C = -\tfrac{1}{6}\cdot 2^{7/2} = -\tfrac{1}{6}\cdot 8\sqrt{2} = -\frac{4\sqrt{2}}{3}.$$

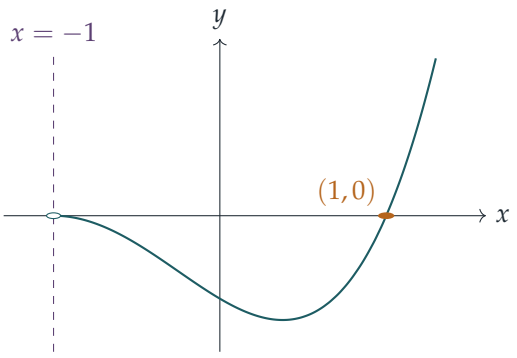
Therefore

$$\boxed{y = \tfrac{2}{3}(x+1)^{7/2} - \tfrac{4\sqrt{2}}{3}(x+1)^2 = \tfrac{2}{3}(x+1)^2\left[(x+1)^{3/2} - 2\sqrt{2}\right]}$$

Where the solution is valid

Ask yourself The formula for y makes sense at $x = -1$. Why isn't $x = -1$ included?

The solution is valid for $x > -1$. For $x < -1$ the right-hand side $(x+1)^{5/2}$ is not real, so the differential equation itself means nothing there (and nor does our $(x+1)^{7/2}$). At $x = -1$ the term $\frac{2y}{x+1}$ is undefined, so the equation doesn't hold there either, even though our formula gives $y \to 0$ as $x \to -1^+$. On the whole interval $x > -1$, with $x + 1 > 0$, every step above is legitimate, and the solution passes through the given point $x = 1$, which lies in this interval.



The open circle marks the excluded endpoint $x = -1$.

Checks

- Substitute back: $y' = \frac{7}{3}(x+1)^{5/2} - \frac{8\sqrt{2}}{3}(x+1)$ and $\frac{2y}{x+1} = \frac{4}{3}(x+1)^{5/2} - \frac{8\sqrt{2}}{3}(x+1)$; their difference is $(x+1)^{5/2}$, as required.
- $y(1) = \frac{2}{3} \cdot 4\left(2\sqrt{2} - 2\sqrt{2}\right) = 0$.
- Special values: $y(0) = \frac{2-4\sqrt{2}}{3} \approx -1.219$ and $y(3) = \frac{256-64\sqrt{2}}{3} \approx 55.16$. A numerical (Runge–Kutta) integration of the equation from $y(1) = 0$ agrees with both to nine figures.

Common slips

- Getting the sign of P wrong: here $P = -\frac{2}{x+1}$, so $u = (x+1)^2$, not $(x+1)^{-2}$.
- Forgetting the constant when integrating for v, which leaves no freedom to fit $y(1) = 0$.
- Giving $x \geq -1$ as the range: the equation is undefined at $x = -1$.

Beyond STEP

Choosing u to remove a troublesome term is the idea behind *variation of parameters*: solve the homogeneous equation first, then let the "constant" vary to absorb the forcing. The same trick works for second order linear equations, and the question of where a solution is valid grows into the existence and uniqueness theorem, which guarantees a unique solution on any interval where P and Q are continuous. Boyce and DiPrima, *Elementary Differential Equations*, covers both.

WORKED SOLUTION

STEP I 1990, Question 8

Pure • Infinite Trigonometric Products

STEP I 1990, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q8

The key idea

The double-angle formula $\sin 2\theta = 2\sin\theta\cos\theta$ turns each cosine into a ratio of two sines, $\cos\theta = \dfrac{\sin 2\theta}{2\sin\theta}$. Multiply such ratios together and almost everything cancels, like a telescoping sum. Once the finite product is a single neat fraction, letting $n \to \infty$ only needs the standard limit $\dfrac{\sin x}{x} \to 1$ as $x \to 0$.

Solution

A word about the condition on α

Every step below divides by $\sin(\alpha/2^m)$ for some $m \geq 0$, so we need these never to vanish. If $\sin(\alpha/2^m) = 0$ then $\alpha/2^m = j\pi$ for some integer j, so $\alpha = (2^m j)\pi$ is an integer multiple of π, which is excluded. So $\sin(\alpha/2^m) \neq 0$ for all $m \geq 0$.

The product of two cosines

Applying $\sin 2\theta = 2\sin\theta\cos\theta$ twice, first with $\theta = \alpha/2$ and then with $\theta = \alpha/4$,

$$\begin{aligned}\sin\alpha &= 2\sin\frac{\alpha}{2}\cos\frac{\alpha}{2}\\ &= 2\left(2\sin\frac{\alpha}{4}\cos\frac{\alpha}{4}\right)\cos\frac{\alpha}{2} = 4\sin\frac{\alpha}{4}\cos\frac{\alpha}{2}\cos\frac{\alpha}{4}.\end{aligned}$$

Since $\sin(\alpha/4) \neq 0$ we may divide by $4\sin(\alpha/4)$, which gives

$$\cos\frac{\alpha}{2}\cos\frac{\alpha}{4} = \frac{\sin\alpha}{4\sin(\alpha/4)},$$

as required.

The product of n cosines

Write $P_n = \cos\frac{\alpha}{2}\cos\frac{\alpha}{4}\cdots\cos\frac{\alpha}{2^n}$. We prove by induction on n that

$$P_n = \frac{\sin\alpha}{2^n\sin(\alpha/2^n)}.$$

For $n = 1$ this reads $\cos\frac{\alpha}{2} = \dfrac{\sin\alpha}{2\sin(\alpha/2)}$, which is the double-angle formula divided by $2\sin(\alpha/2) \neq 0$. (The case $n = 2$ is the result we have just shown.)

Now suppose the formula holds for some $n \geq 1$. The double-angle formula with $\theta = \alpha/2^{n+1}$ gives $\sin\frac{\alpha}{2^n} = 2\sin\frac{\alpha}{2^{n+1}}\cos\frac{\alpha}{2^{n+1}}$. Hence

$$\begin{aligned} P_{n+1} = P_n\cos\frac{\alpha}{2^{n+1}} &= \frac{\sin\alpha}{2^n\sin(\alpha/2^n)}\cos\frac{\alpha}{2^{n+1}} \\ &= \frac{\sin\alpha\,\cos(\alpha/2^{n+1})}{2^n\cdot 2\sin(\alpha/2^{n+1})\cos(\alpha/2^{n+1})} = \frac{\sin\alpha}{2^{n+1}\sin(\alpha/2^{n+1})}. \end{aligned}$$

Cancelling $\cos(\alpha/2^{n+1})$ is legitimate: it is a factor of $\sin(\alpha/2^n)$, which is non-zero, so it is non-zero too. This is the formula for $n+1$, so by induction it holds for every positive integer n.

Letting $n \to \infty$

The infinite product means the limit of the partial products P_n. Put $x_n = \alpha/2^n$, so $x_n \to 0$ as $n \to \infty$, and

$$2^n\sin\frac{\alpha}{2^n} = \alpha\cdot\frac{\sin x_n}{x_n} \longrightarrow \alpha\cdot 1 = \alpha.$$

? **Ask yourself** Why is it safe to turn $\lim P_n = \dfrac{\sin\alpha}{\alpha}$ upside down?

Because the limit is not zero: $\alpha \neq 0$ and $\sin\alpha \neq 0$ for our α. So $P_n \to \dfrac{\sin\alpha}{\alpha} \neq 0$, the infinite product exists, and dividing $\sin\alpha$ by it gives

$$\frac{\sin\alpha}{\cos\frac{\alpha}{2}\cos\frac{\alpha}{4}\cos\frac{\alpha}{8}\cdots} = \lim_{n\to\infty}\frac{\sin\alpha}{P_n} = \lim_{n\to\infty}2^n\sin\frac{\alpha}{2^n} = \alpha,$$

which is the required formula.

The formula for $\pi/2$

Take $\alpha = \pi/2$, which is not an integer multiple of π, and note $\sin\alpha = 1$. Then

$$\frac{\pi}{2} = \frac{1}{\cos\frac{\pi}{4}\cos\frac{\pi}{8}\cos\frac{\pi}{16}\cdots}.$$

It remains to show that the cosines are the nested square roots. Write $c_k = \cos(\pi/2^{k+1})$ for $k \geq 1$. Then $c_1 = \cos\frac{\pi}{4} = \sqrt{\frac{1}{2}}$. The half-angle formula $\cos^2\theta = \frac{1}{2} + \frac{1}{2}\cos 2\theta$ with $\theta = \pi/2^{k+2}$ gives

$$c_{k+1}^2 = \tfrac{1}{2} + \tfrac{1}{2}c_k, \qquad \text{so} \qquad c_{k+1} = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}c_k},$$

taking the positive root because $0 < \pi/2^{k+2} < \pi/2$, so $c_{k+1} > 0$. So

$$c_1 = \sqrt{\tfrac{1}{2}}, \quad c_2 = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{\tfrac{1}{2}}}, \quad c_3 = \sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{\tfrac{1}{2} + \tfrac{1}{2}\sqrt{\tfrac{1}{2}}}}, \quad \ldots,$$

and each factor is the previous one fed into $t \mapsto \sqrt{\frac{1}{2} + \frac{1}{2}t}$. These are exactly the factors in the required product, so

$$\frac{\pi}{2} = \frac{1}{\sqrt{\frac{1}{2}}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2} + \frac{1}{2}\sqrt{\frac{1}{2}}}}\cdots}.$$

Checks

- With $\alpha = \pi/2$ and $n = 1$ the finite formula gives $\cos\frac{\pi}{4} = \dfrac{1}{2\sin(\pi/4)} = \dfrac{1}{\sqrt{2}}$, which is correct.
- The reciprocals of the partial products of the nested radicals are 1.4142, 1.5307, 1.5607, 1.5683, 1.5702, …; after 60 factors we get 1.5707963267949, agreeing with $\pi/2$ to machine precision.
- The finite identity also checks numerically for random α and n up to 14, and $\sin\alpha / P_{60}$ reproduces α for values such as $\alpha = 2.5$ and $\alpha = -4$.

Common slips

- Dividing by $\sin(\alpha/2^n)$ without saying why it is non-zero. This is the reason for the condition $\alpha \neq k\pi$.
- Writing $1/\lim P_n = \lim(1/P_n)$ without noting that $\lim P_n \neq 0$.
- Forgetting to justify the positive square root in the half-angle step.

Beyond STEP

The final formula is Viète's product of 1593, the first exact infinite expression for π ever written down. At university you meet infinite products properly in analysis, where a product of non-zero factors is said to converge when its partial products

tend to a *non-zero* limit, which is exactly the point we checked. Euler's product $\sin\alpha = \alpha\prod_{k\geq 1}\left(1-\frac{\alpha^2}{k^2\pi^2}\right)$ is a deeper relative. Abbott's *Understanding Analysis* is a friendly place to see these limits handled rigorously.

WORKED SOLUTION

STEP I 1990, Question 9

Pure • Tangents and Area Bounds

STEP I 1990, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q9

The key idea

The curve $y = 1/x$ bends upwards (it is convex), so every tangent lies below it. The two trapezia under the tangents therefore fit inside the region under the curve between $x = 1$ and $x = b$, and that region has area $\ln b$. Comparing the two areas gives the lower bound for the logarithm. A rectangle that sits above the curve gives the upper bound in the same way.

Solution

The point C

For $y = 1/x$ we have $\frac{\mathrm{d}y}{\mathrm{d}x} = -1/x^2$. So the tangent at $(t, 1/t)$ is

$$y - \frac{1}{t} = -\frac{1}{t^2}(x - t), \qquad \text{that is} \qquad y = \frac{2}{t} - \frac{x}{t^2}.$$

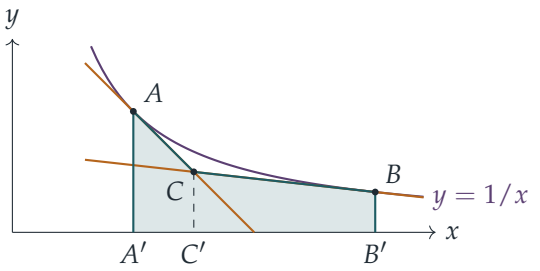
With $t = 1$ the tangent at A is $y = 2 - x$. With $t = b$ the tangent at B is $y = \frac{2}{b} - \frac{x}{b^2}$. Where they meet,

$$2 - x = \frac{2}{b} - \frac{x}{b^2} \quad \Longrightarrow \quad x\left(1 - \frac{1}{b^2}\right) = 2\left(1 - \frac{1}{b}\right).$$

Since $b > 1$ we may divide by $1 - \frac{1}{b} \neq 0$, which leaves $x\left(1 + \frac{1}{b}\right) = 2$. Hence

$$C = \left(\frac{2b}{b+1}, \frac{2}{b+1}\right),$$

where the y-coordinate comes from $y = 2 - x$. Note that $1 < \frac{2b}{b+1} < b$: the left inequality is $b + 1 < 2b$, and the right one is $2 < b + 1$. Both hold because $b > 1$, so C lies between A and B horizontally.



The figure is drawn for $b = 3$. The shaded region is made up of the two quadrilaterals.

The sum of the areas

Each quadrilateral has two vertical sides, so it is a trapezium, and its area is the width times the mean of the two vertical sides.

For $ACC'A'$ the width is $\frac{2b}{b+1} - 1 = \frac{b-1}{b+1}$, and the vertical sides are 1 and $\frac{2}{b+1}$:

$$\text{area}(ACC'A') = \frac{1}{2} \cdot \frac{b-1}{b+1}\left(1 + \frac{2}{b+1}\right) = \frac{(b-1)(b+3)}{2(b+1)^2}.$$

For $CBB'C'$ the width is $b - \frac{2b}{b+1} = \frac{b(b-1)}{b+1}$, and the vertical sides are $\frac{2}{b+1}$ and $\frac{1}{b}$:

$$\text{area}(CBB'C') = \frac{1}{2} \cdot \frac{b(b-1)}{b+1}\left(\frac{2}{b+1} + \frac{1}{b}\right) = \frac{(b-1)(3b+1)}{2(b+1)^2}.$$

Adding, the numerators combine to $(b-1)(4b+4) = 4(b-1)(b+1)$, so

$$\text{area}(ACC'A') + \text{area}(CBB'C') = \frac{2(b-1)}{b+1}.$$

The inequalities

? **Ask yourself** Why must the two trapezia fit under the curve?

For $x > 0$, the gap between the curve and the tangent at $(t, 1/t)$ is

$$\frac{1}{x} - \left(\frac{2}{t} - \frac{x}{t^2}\right) = \frac{t^2 - 2tx + x^2}{t^2 x} = \frac{(x-t)^2}{t^2 x} \geq 0.$$

So each tangent lies on or below the curve for all $x > 0$. The top edge of the shaded region is the tangent at A on $[1, \frac{2b}{b+1}]$ and the tangent at B on $[\frac{2b}{b+1}, b]$. Both pieces lie below $y = 1/x$, and the region sits above the x-axis. So the shaded region is contained in the region under the curve from $x = 1$ to $x = b$, and

$$\frac{2(b-1)}{b+1} \leq \int_1^b \frac{\mathrm{d}x}{x} = \ln b.$$

For the upper bound, $\frac{1}{x} \le 1$ for $x \ge 1$, so the region under the curve lies inside the rectangle with base $A'B'$ and height 1. Hence

$$\ln b = \int_1^b \frac{\mathrm{d}x}{x} \le \int_1^b 1\,\mathrm{d}x = b - 1.$$

Now let $z > 0$ and put $b = 1 + z$, so that $b > 1$ as required. Then $b - 1 = z$ and $b + 1 = 2 + z$, and the two inequalities together read

$$\frac{2z}{2+z} \le \ln(1+z) \le z,$$

which is what we had to prove. (Both inequalities are in fact strict for $z > 0$, because the curve and the bounding lines meet only at isolated points.)

Checks

- With $b = 1$, C would be $(1, 1) = A$ and the area sum would be 0, as it should.
- With $b = 2$: $C = (\frac{4}{3}, \frac{2}{3})$, and the area sum is $\frac{2}{3} \approx 0.667$, just below $\ln 2 \approx 0.693$, which is in turn below 1.
- For small z, $\frac{2z}{2+z} = z - \frac{z^2}{2} + \cdots$ agrees with $\ln(1+z) = z - \frac{z^2}{2} + \cdots$ up to the z^2 term, so the lower bound is very sharp near 0.

We checked the coordinates and both trapezium areas in exact arithmetic for several values of b, and the inequality for 10^5 random z between 10^{-6} and 10^6.

Common slips

- Forgetting to check that C lies between $x = 1$ and $x = b$. Without this, the "quadrilaterals" need not be the shapes you think.
- Saying that the tangents are below the curve "from the picture". The one-line calculation with $(x - t)^2$ is what makes it a proof.

Beyond STEP

Comparing a function with simple shapes above and below its graph is the idea behind Riemann sums, and it is how integrals are rigorously defined in a first analysis course. The fact that tangents of a convex function lie below its graph is the start of the theory of convex functions. It leads to Jensen's inequality and to many classical inequalities, including AM–GM. Abbott's *Understanding Analysis* develops the integral in exactly this spirit.

WORKED SOLUTION

STEP I 1990, Question 10

Mechanics • Power and Resistance

STEP I 1990, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q10

The key idea

"Steady speed" means no acceleration, so the runner's driving force exactly balances the resistances, and power is driving force times speed. The only subtlety is that air resistance depends on the speed *relative to the air*, which is different on the two legs. For the second half of the question, think about total energy rather than power: the energy spent on each leg is resistance times distance, and that turns the fixed energy budget into a linear condition on the two speeds.

Solution

Steady speeds and the time for the race

Model the runner as a particle moving along a straight line. Let v (in $\mathrm{m\,s^{-1}}$) be the runner's ground speed on a leg and D (in N) the forward driving force the runner produces. The resistances are $50\,\mathrm{N}$ and $2w\,\mathrm{N}$, both opposing motion relative to the ground and the air respectively.



Resolving along the track, the acceleration is zero at steady pace, so

$$D = 50 + 2w.$$

The rate at which the runner does work is the driving force times the speed of its point of application, so $300 = Dv$.

Outward leg. The wind blows at $3\,\mathrm{m\,s^{-1}}$ in the runner's direction of motion. If $v > 3$ the runner moves forward relative to the air at $w = v - 3$, and air resistance acts backwards. (We check $v > 3$ below.) Then $D = 50 + 2(v-3) = 44 + 2v$, and

$$v_1(44 + 2v_1) = 300 \iff v_1^2 + 22v_1 - 150 = 0 \iff v_1 = -11 + \sqrt{271} \approx 5.462,$$

taking the positive root. This is indeed greater than 3, so the assumption holds. (Alternatively: at $v = 3$ there is no air resistance and the power needed is only $3 \times 50 = 150 < 300$ W, so the runner must go faster than the wind.)

Return leg. Now the runner runs into the wind, so $w = v + 3$ and $D = 50 + 2(v + 3) = 56 + 2v$:

$$v_2(56 + 2v_2) = 300 \iff v_2^2 + 28v_2 - 150 = 0 \iff v_2 = -14 + \sqrt{346} \approx 4.601.$$

Time. Ignoring the acceleration at the start and turn,

$$T_0 = \frac{5000}{v_1} + \frac{5000}{v_2} \approx 915.4 + 1086.7 = 2002.1,$$

so the race takes 2002 s to the nearest second (about 33 minutes 22 seconds).

Alternative tactics

Suppose instead the runner works at x_1 W outward and x_2 W on the return, with steady speeds v_1 and v_2. The force balance is unchanged, so as long as $v_1 > 3$,

$$x_1 = v_1(44 + 2v_1), \qquad x_2 = v_2(56 + 2v_2).$$

The time on the outward leg is $5000/v_1$, so the energy used there is

$$x_1 \cdot \frac{5000}{v_1} = 5000(44 + 2v_1),$$

which is just the resistance times the distance, as it should be. Similarly the return leg uses $5000(56 + 2v_2)$. The total energy is therefore

$$E = 5000\big(100 + 2(v_1 + v_2)\big).$$

Ask yourself Where does the fixed total energy come in?

The runner must spend the same total energy as in the original race, so E is a fixed number. Since E depends on the speeds only through $v_1 + v_2$, that sum is forced:

$$v_1 + v_2 = \frac{E}{10\,000} - 50,$$

whatever the choice of x_1 and x_2. This proves the result. Its value is the one from the original race,

$$S = v_1 + v_2 = \sqrt{271} + \sqrt{346} - 25 \approx 10.063.$$

The best tactics. The time for the race is now

$$T = 5000\left(\frac{1}{v_1} + \frac{1}{v_2}\right) = \frac{5000\,(v_1 + v_2)}{v_1 v_2} = \frac{5000\,S}{v_1 v_2}.$$

With S fixed, this is smallest when $v_1 v_2$ is largest. By the AM–GM inequality, $v_1 v_2 \leq \left(\frac{S}{2}\right)^2$, with equality exactly when $v_1 = v_2 = \frac{S}{2}$. So the best plan is to run at the same speed both ways, $\frac{S}{2} \approx 5.032\,\mathrm{m\,s^{-1}}$. That is still faster than the wind, so the outward formula for x_1 remains valid. The minimum time is

$$T_{\min} = \frac{5000\,S}{S^2/4} = \frac{20\,000}{S} \approx 1987.4\ \mathrm{s}.$$

The runner saves $T_0 - T_{\min} \approx 2002.1 - 1987.4 = 14.7\,\mathrm{s}$, which is approximately 15 seconds, as required. The powers are about $x_1 \approx 272\,\mathrm{W}$ outward and $x_2 \approx 332\,\mathrm{W}$ back: the runner eases off with the wind behind and works harder into it.

Checks

- Energy consistency: $300 \times 2002.105 \approx 600\,632\,\mathrm{J}$, and $5000(100 + 2 \times 10.0632) \approx 600\,632\,\mathrm{J}$ also.
- With no wind, both legs would give $v^2 + 25v - 150 = 0$, so $v = 5$, and the two methods would coincide. The wind is what makes the saving possible.
- We checked numerically that $v_1 + v_2 = 10.0632$ for 1000 random choices of x_1 (with x_2 fixed by the energy budget), that no choice beats 1987.4 s, and that the saving is 14.66 s.

Common slips

- Using the ground speed in the air-resistance term instead of the speed relative to the air, or getting the sign of the wind the wrong way round on one leg.
- Thinking that equal *power* both ways is fair: it is equal *speed* both ways that minimises the time.
- Trying to prove $v_1 + v_2$ constant from the power equations alone. You need the fixed total energy.

Beyond STEP

The last part is a small constrained optimisation: minimise $1/v_1 + 1/v_2$ subject to a linear constraint. AM–GM settles it here, but the general tool, met in first-year courses, is the method of Lagrange multipliers, and the same ideas govern optimal pacing in cycling and rowing. For a careful treatment of work, power and drag forces, see Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP I 1990, Question 11

Mechanics • Projectiles and Momentum

STEP I 1990, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q11

The key idea

There are three stages: an ordinary projectile up to the top of its flight, an instantaneous collision (the catch), and an instantaneous "explosion" (the push). During the two instantaneous stages gravity has no time to act, so momentum of the shell–Superman system is conserved, separately in each direction. After that the shell is a projectile again, starting from a known height with a known downward speed.

Solution

We model the shell and Superman as particles, ignore air resistance, and take the shell to be fired from, and to land on, horizontal ground. Take x horizontal in the direction of motion and y vertically upwards.

Up to the top of the flight

The shell starts with components

$$u_x = v\cos\tfrac{\pi}{3} = \tfrac{1}{2}v, \qquad u_y = v\sin\tfrac{\pi}{3} = \tfrac{\sqrt{3}}{2}v.$$

At the top the vertical velocity is zero, so the time taken and the height reached are

$$t_1 = \frac{u_y}{g} = \frac{\sqrt{3}\,v}{2g}, \qquad H = \frac{u_y^2}{2g} = \frac{3v^2}{8g}.$$

At this moment the shell is moving horizontally with speed $\frac{1}{2}v$.

The catch

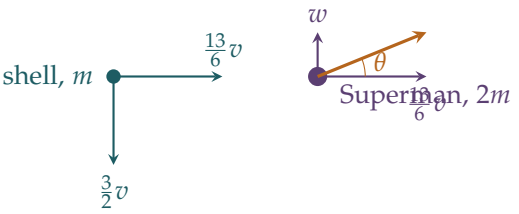
Superman, moving horizontally at $3v$, catches the shell from behind and they move together. Horizontal momentum is conserved during the (instantaneous) catch, so if V

is their common speed,

$$2m(3v) + m\left(\tfrac{1}{2}v\right) = 3mV \quad \Longrightarrow \quad V = \frac{13v}{6}.$$

Neither has any vertical velocity at this instant, so the combined vertical momentum is zero.

The push



Superman now pushes the shell straight down. The push is an internal impulse, so the total momentum of the pair is unchanged in both directions.

Horizontally: the shell keeps horizontal speed $\frac{13}{6}v$. The total horizontal momentum is still $3m \cdot \frac{13}{6}v$, so Superman keeps horizontal speed $\frac{13}{6}v$ as well. (This is no surprise: the impulse is vertical.)

Vertically: the total vertical momentum was zero, so if Superman moves upwards with speed w,

$$2m\,w - m\left(\tfrac{3}{2}v\right) = 0 \quad \Longrightarrow \quad w = \tfrac{3}{4}v.$$

Ask yourself Why can we use conservation of momentum when gravity is acting throughout?

The catch and the push are instantaneous. Gravity is a finite force, so over a time interval of length zero its impulse is zero; only the huge, short-lived internal forces between Superman and the shell matter, and those cancel in pairs.

The rest of the shell's flight

The shell now starts from height $H = \frac{3v^2}{8g}$ with downward vertical speed $\frac{3}{2}v$. Measuring s downwards, it reaches the ground after time t_2 where

$$\tfrac{3}{2}v\,t_2 + \tfrac{1}{2}gt_2^2 = \frac{3v^2}{8g}.$$

Multiplying by $8g$ gives a quadratic in gt_2:

$$4(gt_2)^2 + 12v(gt_2) - 3v^2 = 0 \quad \Longrightarrow \quad gt_2 = \frac{-12v + \sqrt{144v^2 + 48v^2}}{8} = \frac{-12 + 8\sqrt{3}}{8}\,v,$$

taking the positive root since $t_2 > 0$. So

$$t_2 = \left(\sqrt{3} - \tfrac{3}{2}\right)\frac{v}{g}.$$

(This is positive because $\sqrt{3} \approx 1.732 > 1.5$.) The total time of flight is

$$\begin{aligned} T = t_1 + t_2 &= \frac{\sqrt{3}}{2}\frac{v}{g} + \left(\sqrt{3} - \frac{3}{2}\right)\frac{v}{g} \\ &= \frac{3(\sqrt{3}-1)}{2}\frac{v}{g} \;\approx\; 1.10\,\frac{v}{g}. \end{aligned}$$

Superman's direction

Immediately after the push Superman has velocity components $\frac{13}{6}v$ horizontally and $\frac{3}{4}v$ upwards. His trajectory therefore starts at an angle θ above the horizontal with

$$\tan\theta = \frac{\frac{3}{4}v}{\frac{13}{6}v} = \frac{9}{26}, \qquad \theta = \arctan\frac{9}{26} \approx 19.09^\circ,$$

so the angle is 19° to the nearest degree.

Checks

- Without Superman the shell would land at time $2t_1 = \sqrt{3}\,v/g \approx 1.73\,v/g$. Pushing it down should shorten the flight, and indeed $T \approx 1.10\,v/g$ is smaller.
- Mass ratio: the shell is half Superman's mass, so it should get twice his vertical speed in the opposite direction. It does: $\frac{3}{2}v = 2 \times \frac{3}{4}v$.
- Numerically, with $v = 13.7$ and $g = 9.81$, stepping the shell's fall from height H in steps of 10^{-6} gives $t_2 \approx 0.32407\,\text{s}$, matching $(\sqrt{3} - \frac{3}{2})v/g$, and $T \approx 1.5335\,\text{s} = 1.0981\,v/g$.

Common slips

- Forgetting the catch: using the shell's own horizontal speed $\frac{1}{2}v$, or Superman's $3v$, after the push instead of the common speed $\frac{13}{6}v$.
- Giving Superman an upward speed of $\frac{3}{2}v$ by ignoring the masses.
- Taking the negative root of the quadratic, or adding t_2 to $2t_1$ instead of t_1.

Beyond STEP

The catch is a perfectly inelastic collision and the push is its reverse, an explosion: the same conservation law runs both ways, while kinetic energy is lost in one and

supplied in the other. In a first-year course this becomes the general theory of impulsive forces and of a system's centre of mass, whose motion depends only on the external forces, however violently the parts push on one another. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats momentum, impulse and the centre of mass in this spirit.

WORKED SOLUTION

STEP I 1990, Question 12

Mechanics • Elastic Springs and Energy

STEP I 1990, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

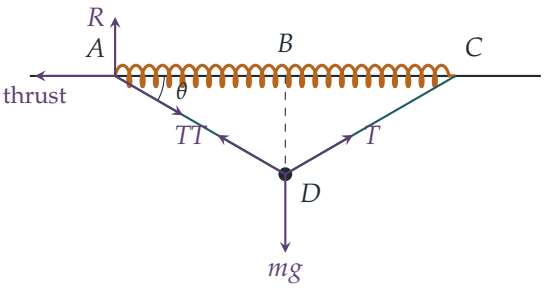
step.maths.org/questions/90-s1-q12

The key idea

Everything here is light except the particle, so the forces at the ends A and C must balance exactly: the horizontal pull of the string on each end equals the push of the compressed spring. That gives the equilibrium condition. For the motion, energy is the natural tool: the only kinetic energy is the particle's, so the particle is at rest precisely when the potential energy returns to its starting value.

Solution

Model. The spring and string are light, the string is inextensible, and the rod is smooth, so the rod can push on the spring only vertically. By symmetry about the vertical through B, the particle D stays directly below B, and $AD = DC = l$. Let T be the tension in each half of the string and θ the angle each half makes with the rod.



Equilibrium

With $\theta = \pi/6$, the half-length of AC is $l\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}l$, so

$$AC = \sqrt{3}\,l < 2l,$$

and the spring is *compressed* by $(2-\sqrt{3})l$. By Hooke's law it pushes outwards on A (and on C) with thrust

$$\frac{\lambda(2-\sqrt{3})l}{2l} = \lambda\Big(1-\frac{\sqrt{3}}{2}\Big).$$

Resolving vertically at D: $2T\sin\frac{\pi}{6} = mg$, so $T = mg$.

Resolving horizontally at A:

Ask yourself Why can we balance forces at A, when nothing is there?

The end of the spring is a light point: any net force on zero mass would give infinite acceleration, so the forces on it must balance at every instant (not just in equilibrium). The rod is smooth, so its reaction R is vertical, and horizontally only the string and the spring act:

$$T\cos\frac{\pi}{6} = \lambda\Big(1-\frac{\sqrt{3}}{2}\Big) \quad\Longrightarrow\quad mg\,\frac{\sqrt{3}}{2} = \lambda\,\frac{2-\sqrt{3}}{2},$$

so

$$mg = \lambda\,\frac{2-\sqrt{3}}{\sqrt{3}} = \lambda\Big(\frac{2}{\sqrt{3}}-1\Big),$$

as required.

The potential energy

Let $AC = 2xl$, so $0 < x \le 1$ while the string is taut. Half of AC is xl, and $AD = l$, so by Pythagoras D is a depth

$$BD = l\sqrt{1-x^2}$$

below the rod. The extension of the spring is $2xl - 2l$, so its elastic energy is $\dfrac{\lambda(2l-2xl)^2}{2\cdot 2l} = \lambda l(1-x)^2$. Taking the rod as the zero level of gravitational potential energy,

$$V(x) = \lambda l(1-x)^2 - mgl\sqrt{1-x^2}.$$

As a consistency check,

$$V'(x) = -2\lambda l(1-x) + \frac{mglx}{\sqrt{1-x^2}},$$

and at $x = \frac{\sqrt{3}}{2}$ (where $\cos\theta = x$, so $\theta = \frac{\pi}{6}$) this vanishes exactly when $mg\sqrt{3} = \lambda(2-\sqrt{3})$, which is the condition we found above.

The fall from B

Initially the particle is at B, so the string lies along the rod, $AC = 2l$ and $x = 1$. Then $V(1) = 0$ and the system starts from rest.

The rod is smooth and the string inextensible, so no energy is lost; and since the spring and string are light, the only kinetic energy is $\frac{1}{2}mv^2$ for the particle. Hence throughout the motion

$$\tfrac{1}{2}mv^2 + V(x) = V(1) = 0.$$

Ask yourself Why does "comes to rest" mean $V(x) = 0$, and not $V'(x) = 0$?

The particle is at rest when $v = 0$, which by energy conservation means $V(x) = 0$. The condition $V'(x) = 0$ describes equilibrium, which is where the particle moves *fastest*, not where it stops. So x satisfies

$$\lambda(1-x)^2 = mg\sqrt{1-x^2}, \qquad \text{that is,} \qquad (1-x)^2 = \left(\frac{2}{\sqrt{3}} - 1\right)\sqrt{1-x^2}.$$

The root $x = 1$ is the starting position. For $x < 1$ we may divide by $\sqrt{1-x} > 0$ and square, which gives a tidier cubic:

$$(1-x)^3 = k^2(1+x), \qquad k = \frac{2}{\sqrt{3}} - 1 \approx 0.15470, \quad k^2 \approx 0.023932.$$

Squaring is safe here because both sides of $(1-x)^{3/2} = k\sqrt{1+x}$ are positive. Let $f(x) = (1-x)^3 - k^2(1+x)$. Then

$$f(0.65) = 0.042875 - 0.039488 \approx +0.0034,$$
$$f(0.66) = 0.039304 - 0.039727 \approx -0.0004,$$

so there is a root between 0.65 and 0.66, very close to 0.66. Since $f'(x) = -3(1-x)^2 - k^2 < 0$, f is strictly decreasing, so this is the only root with $x < 1$. The particle next comes to rest when $x \approx 0.66$ (more precisely $x \approx 0.6589$), with D a depth of about $0.75l$ below the rod.

Checks

- At $\theta = \pi/6$ with $\lambda = l = m = 1$ and $g = \frac{2}{\sqrt{3}} - 1$, the horizontal pull at A and the spring thrust both equal 0.13397, and a numerical derivative of V at $x = \frac{\sqrt{3}}{2}$ is zero (with $V'' > 0$, so the equilibrium is stable).
- Bisection gives the root $x = 0.658861$, and stepping the equation of motion $m\ddot{y} = mg - 2\lambda(1-x)\,y/(lx)$ for the depth y of D (RK4) from rest at $y = 0$ shows the particle first stops at $x = 0.658861$, in agreement.
- The equilibrium value $x = \frac{\sqrt{3}}{2} \approx 0.866$ lies between 0.66 and 1, as it must: the particle oscillates about equilibrium.

Common slips

- Taking the spring to be stretched. Here $AC < 2l$, so it is compressed and pushes A and C apart.

- Setting $V'(x) = 0$ for the rest position. That gives equilibrium, where the speed is greatest.
- Forgetting that the gravitational potential energy is *negative* below the rod, which turns the energy equation into one with no solution in $(0, 1)$.

Beyond STEP

The idea that a massless body must have zero net force on it is a constraint, not a law of motion. Problems like this are handled most cleanly with Lagrangian mechanics, where you write down one energy function in a single coordinate (here x) and the constraint forces never appear. The potential $V(x)$ also gives the period of small oscillations about equilibrium through V''. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats energy methods and small oscillations in this spirit.

WORKED SOLUTION

STEP I 1990, Question 13

Mechanics • Friction and Rigid Body Equilibrium

STEP I 1990, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

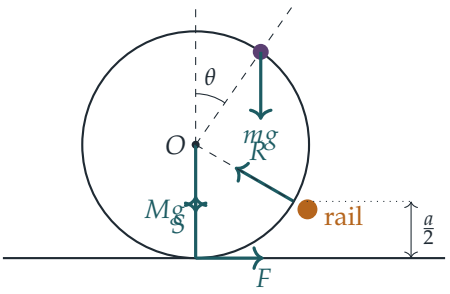
step.maths.org/questions/90-s1-q13

The key idea

Treat the particle first: it is an ordinary particle on a rough slope, the slope being the tangent plane to the cylinder at the point of contact. Then notice that, because the particle is in equilibrium, the total force it exerts on the cylinder is simply its weight, acting vertically down through the point of contact. Taking moments about the centre of the cylinder kills every force except that one and the friction from the plane, which is the quickest way in.

Solution

Model the particle as a point mass m and the cylinder as a uniform rigid cylinder of mass M whose centre of mass O is on its axis. The rail is smooth, so it exerts only a normal force; the plane and the cylinder's surface are rough with coefficient μ. Everything happens in the vertical cross-section through O, which we use from now on. Put the rail on the right.



The figure shows the forces on the cylinder (the mg at the particle is explained in the second part). Arrows are not to scale.

Equilibrium of the particle

At the point P of contact the outward normal makes angle θ with the upward vertical, so the tangent plane there is a slope inclined at θ to the horizontal. The forces on the particle are its weight mg, the normal reaction N along the outward normal and a friction force f along the tangent, up the slope. Resolving along and perpendicular to the normal,

$$N = mg\cos\theta, \qquad f = mg\sin\theta.$$

Friction can supply this provided $f \le \mu N$, that is,

$$\tan\theta \le \mu, \qquad \text{equivalently} \qquad \theta \le \arctan\mu.$$

(We need $0 \le \theta < 90^\circ$ for N to be positive, which holds as the particle is on top of the cylinder.)

Equilibrium of the cylinder once released

Ask yourself The particle presses on the cylinder with a normal force and a friction force. Do we need to handle them separately?

No. By Newton's third law the cylinder feels $-N$ and $-f$ at P. Since N and f together balance the particle's weight, their sum is mg vertically upwards, so the particle's total force on the cylinder is *mg vertically downwards, acting at P*. Its line of action is at horizontal distance $a\sin\theta$ from O, on the rail's side.

The other forces on the cylinder are:

- its weight Mg at O;
- the rail's reaction R, normal to the cylinder's surface and so directed along the radius towards O. The rail touches at height $a/2$, which is $a/2$ below O, so this radius makes 60° with the downward vertical. Hence R has a horizontal component $\frac{\sqrt{3}}{2}R$ away from the rail and an upward component $\frac{1}{2}R$;
- at the lowest point, a normal reaction S upwards and a friction force F horizontally (taken towards the rail).

Moments about O. The lines of action of Mg, R and S all pass through O. The particle's mg has moment $mg\,a\sin\theta$ (turning the cylinder towards the rail), and F acts at distance a below O, with the opposite sense. So

$$Fa = mg\,a\sin\theta \quad \Longrightarrow \quad F = mg\sin\theta.$$

This is positive, so friction does act towards the rail, as drawn. (Without the rail the cylinder would roll towards the rail's side; that is why the particle has to be on that side for the rail to be any use.)

Resolving horizontally: $\frac{\sqrt{3}}{2}R = F$, so

$$R = \frac{2mg\sin\theta}{\sqrt{3}} > 0,$$

consistent with the rail being able only to push.

Resolving vertically:

$$S = (M+m)g - \tfrac{1}{2}R = (M+m)g - \frac{mg\sin\theta}{\sqrt{3}},$$

which is positive since $\sin\theta < 1 < \sqrt{3}$ and $M > 0$, so the cylinder stays on the plane.

The plane can supply the friction provided $F \le \mu S$:

$$m\sin\theta \le \mu\left[(M+m) - \frac{m\sin\theta}{\sqrt{3}}\right] \iff m\sin\theta\left(\sqrt{3}+\mu\right) \le \sqrt{3}\,\mu\,(M+m).$$

So, once released, both bodies can remain in equilibrium provided $\tan\theta \le \mu$ and

$$\boxed{\sin\theta \le \frac{\sqrt{3}\,\mu\,(M+m)}{m\,(\sqrt{3}+\mu)}}$$

both hold. (No friction is needed at the smooth rail, and the friction between particle and cylinder was dealt with in the first part.)

The final part

With $m = 7M$ and $\mu = \frac{3}{4}$, the particle's condition is $\tan\theta \le \frac{3}{4}$, that is, $\sin\theta \le \frac{3}{5}$. The cylinder's condition is

$$\sin\theta \le \frac{\sqrt{3}\cdot\frac{3}{4}\cdot 8M}{7M\,(\sqrt{3}+\frac{3}{4})} = \frac{24\sqrt{3}}{7(4\sqrt{3}+3)} = \frac{24\sqrt{3}\,(4\sqrt{3}-3)}{7\cdot 39} = \frac{24(4-\sqrt{3})}{91}.$$

Which is the stricter? We have

$$\frac{24(4-\sqrt{3})}{91} < \frac{3}{5} \iff 120(4-\sqrt{3}) < 273 \iff 207 < 120\sqrt{3} \iff 42849 < 43200,$$

which is true. So the cylinder slipping on the plane is the binding constraint (only just), and the largest value of θ for equilibrium is

$$\theta_{\max} = \arcsin\frac{24(4-\sqrt{3})}{91} \approx 36.7^\circ,$$

just below $\arctan\frac{3}{4} \approx 36.9^\circ$.

Checks

- Limiting case $M \to \infty$: the cylinder's condition becomes vacuous, as a very heavy cylinder can always be held by friction; only $\tan\theta \le \mu$ remains.
- As $\theta \to 0$ the particle sits on top, its weight passes through O, and $F = R = 0$: no rail or friction is needed, as expected.
- Numerically, solving the three equilibrium equations from raw force vectors for 20 000 random choices of M, m, a, μ, θ agrees with the boxed condition every time; for $m = 7M$, $\mu = 0.75$ a fine scan of θ gives $\theta_{\max} \approx 36.737^\circ$, with $\sin\theta_{\max} \approx 0.59814$.

Common slips

- Taking the rail's force as horizontal. It is normal to the cylinder's surface, so it points along a radius, at 30° above the horizontal here.
- Forgetting the upward component $\frac{1}{2}R$ of the rail's force, which reduces S and so reduces the friction available from the plane.
- Stopping at $\theta = \arctan\frac{3}{4}$ without comparing with the cylinder's condition, which is (just) stricter.

Beyond STEP

The useful trick here, replacing the particle's contact forces by a single resultant equal to its weight, is a first glimpse of the idea that any system of forces on a rigid body reduces to a single force through a chosen point plus a couple. Statics of rigid bodies with several frictional contacts is a standard first-year topic, and the question of which contact slips first leads on to more subtle problems where the friction forces are not determined by the equations alone. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats forces, torques and rigid-body equilibrium carefully.

WORKED SOLUTION

STEP I 1990, Question 14

Probability and Statistics • Expectation and Optimal Stopping

STEP I 1990, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q14

The key idea

Each draw is independent, because the ball is replaced, so every draw is white, red or black with the same fixed probabilities. For Juggins we only need the chance of surviving three draws and the average score of a surviving draw. For Muggins the question is a trade-off: one more draw adds a little on average but risks losing everything already banked, and the size of that risk grows with N. Comparing "expected score if we draw" with "score if we stop" gives a threshold.

Solution

On any draw let W, R, B be the events that the ball is white, red or black. Since the ball is replaced, every draw is independent of the others, with

$$\mathrm{P}(W) = \tfrac{5}{10} = \tfrac{1}{2}, \qquad \mathrm{P}(R) = \tfrac{3}{10}, \qquad \mathrm{P}(B) = \tfrac{2}{10} = \tfrac{1}{5}.$$

Juggins' strategy

Juggins finishes on zero exactly when a black ball turns up among his first three draws. Otherwise all three draws are non-black and he stops. So

$$\mathrm{P}(\text{final score } 0) = 1 - \mathrm{P}(\text{three non-black draws}) = 1 - \left(\tfrac{4}{5}\right)^3 = 1 - \tfrac{64}{125} = \tfrac{61}{125}.$$

For the expected score, let S be his final score. If the first three draws are all non-black (probability $\frac{64}{125}$), then $S = X_1 + X_2 + X_3$, where X_i is the score of the ith draw given that it is not black. Given that a draw is not black, it is white with probability $\frac{1/2}{4/5} = \frac{5}{8}$ and red with probability $\frac{3/10}{4/5} = \frac{3}{8}$. Hence

$$\mathrm{E}(X_i) = 1 \cdot \tfrac{5}{8} + 2 \cdot \tfrac{3}{8} = \tfrac{11}{8}.$$

In every other case $S = 0$. Therefore

$$\mathrm{E}(S) = \tfrac{64}{125} \cdot 3 \cdot \tfrac{11}{8} = \tfrac{264}{125} = 2.112.$$

Ask yourself Why do we use $\frac{5}{8}$ and $\frac{3}{8}$ here, rather than $\frac{1}{2}$ and $\frac{3}{10}$?

We have already conditioned on surviving, so each of the three draws is known to be non-black, and the probabilities must be rescaled to add to 1. Equivalently, without conditioning: $\mathrm{E}(S)$ is the sum, over the $2^3 = 8$ white/red sequences, of (probability)×(score). Each draw contributes $1 \cdot \frac{1}{2} + 2 \cdot \frac{3}{10} = \frac{11}{10}$ in expectation, and it is kept only if the other two draws also survive (probability $\frac{16}{25}$). This gives $3 \cdot \frac{11}{10} \cdot \frac{16}{25} = \frac{264}{125}$ again.

Muggins' decision

Suppose Muggins has N points and draws once more (and then stops). His score becomes $N+1$, $N+2$ or 0, so

$$\mathrm{E}(\text{score after one more draw}) = \tfrac{1}{2}(N+1) + \tfrac{3}{10}(N+2) + \tfrac{1}{5} \cdot 0 = \tfrac{4}{5}N + \tfrac{11}{10}.$$

Drawing beats stopping on average exactly when

$$\tfrac{4}{5}N + \tfrac{11}{10} > N \iff \tfrac{11}{10} > \tfrac{1}{5}N \iff N < 5.5.$$

The average gain from a draw is $\frac{11}{10}$, and the average loss is $\frac{1}{5}N$, the black-ball chance times what is at stake. Since scores are whole numbers, this suggests the strategy:

keep drawing while the score is 5 *or less; stop as soon as it reaches* 6 *or more.*

Ask yourself Comparing just one more draw with stopping looks short-sighted. Could it pay to draw at $N \geq 6$ and plan to draw again later?

No. Once the score is 6 or more, any further draw leaves it at 6 or more (or ends the game at 0), so at every later decision the one-draw comparison again says "stop". Each extra draw from such a score lowers the expected final score, so the best plan from there is to stop at once. Below 6, one more draw already beats stopping, and the option of carrying on can only add to that. So the threshold rule is the best possible strategy, not just a sensible one.

Using this strategy, let $V(N)$ be the expected final score starting from N. Then $V(N) = N$ for $N \geq 6$, and $V(N) = \frac{1}{2}V(N+1) + \frac{3}{10}V(N+2)$ for $N \leq 5$. Working downwards:

N	6	5	4	3	2	1	0
$V(N)$	6	5.1	4.35	3.705	3.1575	2.690	2.292

So Muggins averages about 2.29 points a game, compared with Juggins' 2.112.

Checks

- A computer enumerated every sequence of up to three draws exactly. It confirmed $\mathrm{P}(0) = \frac{61}{125} = 0.488$ and $\mathrm{E}(S) = \frac{264}{125}$. A Monte Carlo run of 2×10^5 games gave 0.488 and 2.111.

- At $N = 5$ a draw gives $\frac{4}{5} \cdot 5 + 1.1 = 5.1 > 5$. At $N = 6$ it gives $5.9 < 6$. So the switch is between 5 and 6.
- Value iteration over all strategies gave the same rule. Stopping at ≥ 5 or at ≥ 7 instead gives a lower average, 2.261 and 2.265, against 2.292 for stopping at 6.

Common slips

- Calculating $\mathrm{E}(S)$ as $3 \times \frac{11}{10}$, which forgets that a later black ball wipes out the earlier points.
- Multiplying the conditional mean $\frac{11}{8}$ by the unconditional probabilities as well, which counts the survival factor twice.
- Writing the threshold as $N \leq 5.5$ without noting that N is an integer, so the rule is "draw at 5, stop at 6".

Beyond STEP

Muggins' problem is an *optimal stopping* problem. The rule "stop when stopping now beats one more step" is the *one-step look-ahead* rule, and it is optimal whenever, as here, the set of states where it says stop can never be left. The same ideas lead to Markov decision processes and dynamic programming. Grimmett and Welsh, *Probability: An Introduction*, covers the expectation and Markov chain background, and Blitzstein and Hwang, *Introduction to Probability*, has many similar first-step analyses.

WORKED SOLUTION

STEP I 1990, Question 15

Probability and Statistics • Negative Binomial Distribution

STEP I 1990, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q15

The key idea

For the first part, split the event into two independent pieces: what happens on the last toss, and what happens on all the tosses before it. For the rest, the important fact is that the coin has no memory. Once a head has just appeared, the wait for the next head starts afresh, so we only need to know how likely a geometric waiting time is to be even or odd.

Solution

Write $q = 1 - p$ throughout. Tosses are independent, each a head with probability p.

The distribution of the kth head

Let T_k be the number of the toss on which the kth head appears. The event $\{T_k = n\}$ happens exactly when

- toss n is a head, and
- among tosses $1, 2, \ldots, n-1$ there are exactly $k-1$ heads (and so $n-k$ tails).

Both conditions are needed. If toss n were a tail, the kth head would come before or after toss n, not on it. If the first $n-1$ tosses held a different number of heads, toss n would not give the kth head. Together, the two conditions are also enough.

These two events concern disjoint sets of tosses, so they are independent. The first has probability p. For the second, the $k-1$ heads can sit in any of $\binom{n-1}{k-1}$ sets of positions among the first $n-1$ tosses. Each such arrangement is a particular sequence of $k-1$ heads and $n-k$ tails, with probability $p^{k-1}q^{n-k}$ by independence, and different arrangements are mutually exclusive. Hence

$$\mathrm{P}(T_k = n) = p \cdot \binom{n-1}{k-1} p^{k-1} q^{n-k} = \binom{n-1}{k-1} p^k (1-p)^{n-k}, \qquad n \geq k,$$

as required.

Even to even

Let $G = T_k - T_{k-1}$ be the number of further tosses needed after the $(k-1)$th head (with $T_0 = 0$). The tosses after toss T_{k-1} are independent of everything up to that point. So G is independent of T_{k-1}, and it has the geometric distribution

$$\mathrm{P}(G = g) = q^{g-1}p, \qquad g = 1, 2, \ldots$$

(the case $k = 1$ of the first part).

Ask yourself If T_{k-1} is even, when is T_k even?

Since $T_k = T_{k-1} + G$, if T_{k-1} is even then T_k is even exactly when G is even. Because G is independent of T_{k-1}, conditioning on T_{k-1} being even doesn't change the distribution of G. Therefore

$$\mathrm{P}(T_k \text{ even} \mid T_{k-1} \text{ even}) = \mathrm{P}(G \text{ even}).$$

Summing the geometric series,

$$\mathrm{P}(G \text{ even}) = p\left(q + q^3 + q^5 + \cdots\right) = \frac{pq}{1-q^2} = \frac{pq}{(1-q)(1+q)} = \frac{q}{1+q},$$
$$\mathrm{P}(G \text{ odd}) = p\left(1 + q^2 + q^4 + \cdots\right) = \frac{p}{1-q^2} = \frac{1}{1+q}.$$

So the required probability is

$$\frac{q}{1+q} = \frac{1-p}{2-p}.$$

The answer doesn't depend on k.

Exactly two heads on even-numbered tosses

Now toss until the third head. Let $G_1 = T_1$, $G_2 = T_2 - T_1$ and $G_3 = T_3 - T_2$ be the three waiting times. By the same memoryless argument, they are independent, and each is geometric as above. Write

$$e = \mathrm{P}(G_i \text{ even}) = \frac{q}{1+q}, \qquad o = \mathrm{P}(G_i \text{ odd}) = \frac{1}{1+q}.$$

We start at toss 0, which is even. An even gap keeps the parity of the current position, and an odd gap switches it. The table lists the eight possible parity patterns of (G_1, G_2, G_3) and the resulting parities of (T_1, T_2, T_3):

G_1	G_2	G_3	T_1	T_2	T_3	even heads
E	E	E	E	E	E	3
E	E	O	E	E	O	**2**
E	O	E	E	O	O	1
E	O	O	E	O	E	**2**
O	E	E	O	O	O	0
O	E	O	O	O	E	1
O	O	E	O	E	E	**2**
O	O	O	O	E	O	1

Exactly two heads fall on even tosses in three of the cases, which have probabilities e^2o, eo^2 and o^2e. By independence of the gaps, the required probability is

$$\begin{aligned} e^2o + 2eo^2 = eo(e+2o) &= \frac{q}{(1+q)^2}\cdot\frac{q+2}{1+q} \\ &= \frac{q(2+q)}{(1+q)^3} = \frac{(1-p)(3-p)}{(2-p)^3}. \end{aligned}$$

Checks

- As a check on the first part, summing $\mathrm{P}(T_k = n)$ over $n \geq k$ should give 1. This is the negative binomial series $\sum_{n\geq k}\binom{n-1}{k-1}q^{n-k} = (1-q)^{-k}$.
- The two parity probabilities $\frac{q}{1+q}$ and $\frac{1}{1+q}$ add to 1. When p is close to 1, the first head almost always comes at once, on an odd toss, so $\mathrm{P}(G \text{ even}) \to 0$, as the formula gives.
- For a fair coin ($p = \frac{1}{2}$) the final answer is $\frac{(1/2)(5/2)}{(3/2)^3} = \frac{10}{27} \approx 0.370$. A simulation of 2×10^5 runs gave 0.369. Summing directly over the three waiting times agreed with the formula to 10^{-9} for $p = 0.3$, 0.5 and 0.8.

Common slips

- Using $\binom{n}{k}$ in the first part. The last toss is forced to be a head, so only the first $n-1$ tosses are free.
- Summing the negative binomial over even n in the second part. The memoryless gap avoids this entirely.
- In the last part, counting heads at even positions as if the positions were independent. They aren't, because each position is built from the one before.

Beyond STEP

The time of the kth head has a *negative binomial* distribution. Writing it as a sum of independent geometric gaps is the discrete version of how the arrival times of a Poisson

process are built from independent exponential waits. The parity bookkeeping in the last part is a two-state Markov chain, a topic met early in a first-year probability course. Grimmett and Welsh, *Probability: An Introduction*, covers all of this.

WORKED SOLUTION

STEP I 1990, Question 16

Probability and Statistics • Triangular Distribution and Testing

STEP I 1990, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s1-q16

The key idea

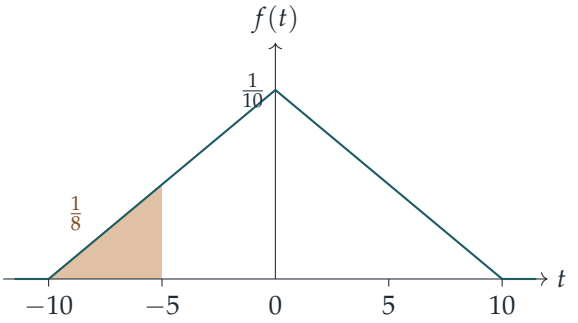
The density is a triangle, so every probability we need is the area of a small triangle, and the cumulative distribution function does all the work. The second thing to notice is that the 8 o'clock bus is the only bus that can turn up anywhere near 8 am: the 7 o'clock bus has gone by 7.10 and the 9 o'clock bus can't come before 8.50. So "catching a bus" in the morning really means "getting there before the 8 o'clock bus does". The final part then becomes a one-parameter binomial model and a significance test.

Solution

The density of T

The density rises linearly from 0 at $t = -10$ to a peak at $t = 0$, then falls at the same rate to 0 at $t = 10$, so it has the form $f(t) = k(10 - |t|)$ on $[-10, 10]$ and is zero elsewhere. The total area is that of a triangle with base 20 and height $10k$, so $\frac{1}{2} \cdot 20 \cdot 10k = 1$, giving $k = \frac{1}{100}$:

$$f(t) = \begin{cases} \dfrac{10+t}{100}, & -10 \le t \le 0, \\ \dfrac{10-t}{100}, & 0 \le t \le 10, \\ 0, & \text{otherwise.} \end{cases}$$



For later use, the cumulative distribution function on the left half is the area of a triangle with base $a+10$ and height $f(a)$:

$$\mathrm{P}(T<a)=\tfrac{1}{2}(a+10)\cdot\frac{a+10}{100}=\frac{(a+10)^2}{200},\qquad -10\le a\le 0,$$

and by symmetry about $t=0$, $\mathrm{P}(T>a)=\dfrac{(10-a)^2}{200}$ for $0\le a\le 10$. The shaded area in the figure is $\mathrm{P}(T<-5)=\frac{25}{200}=\frac{1}{8}$.

Catching a bus by 8.10 am

We assume that I wait at the stop until a bus comes. The bus due at 8 o'clock arrives at time T minutes after 8 am, with $-10\le T\le 10$; the previous bus has left by 7.10 and the next can't arrive before 8.50. So I catch a bus by 8.10 exactly when I reach the stop before the 8 o'clock bus arrives, and the 8.10 deadline is then automatically met.

If I reach the stop a minutes after 8 am, I catch it when $T>a$ (the chance of $T=a$ exactly is zero). The three cases give

$$\begin{aligned}
7{\cdot}55,\ a=-5:&\quad \mathrm{P}(T>-5)=1-\tfrac{25}{200}=\tfrac{7}{8},\\
7{\cdot}58,\ a=-2:&\quad \mathrm{P}(T>-2)=1-\tfrac{64}{200}=\tfrac{17}{25},\\
8{\cdot}02,\ a=2:&\quad \mathrm{P}(T>2)=\tfrac{64}{200}=\tfrac{8}{25}.
\end{aligned}$$

I reach the stop at 7.55 with probability 0.5, at 7.58 with probability $0.5\times 0.8=0.4$, and at 8.02 with probability $0.5\times 0.2=0.1$. My oversleeping has nothing to do with the bus, so by the law of total probability

$$\begin{aligned}
\mathrm{P}(\text{catch a bus by 8.10}) &= 0.5\cdot\tfrac{7}{8}+0.4\cdot\tfrac{17}{25}+0.1\cdot\tfrac{8}{25}\\
&= 0.4375+0.272+0.032=\frac{1483}{2000}=0.7415.
\end{aligned}$$

Is the new clock wrong?

Suppose first that the clock is correct, so I reach the stop at 7.55 every morning. Let T_8 and T_9 be the arrival times, in minutes after the hour, of the 8 o'clock and 9 o'clock buses; they are independent, each with density f. I am still waiting after 9 am exactly

when I miss the 8 o'clock bus ($T_8 < -5$) *and* the 9 o'clock bus is late ($T_9 > 0$). So on any one morning

$$p = \mathrm{P}(T_8 < -5)\,\mathrm{P}(T_9 > 0) = \tfrac{1}{8} \cdot \tfrac{1}{2} = \tfrac{1}{16}.$$

Ask yourself Which way should the test go: are 5 occasions "too many" or "too few"?

A wrongly set clock could push p either way. Say the clock is d minutes fast. Then I reach the stop at true time $7.55 - d$, and the clock shows 9.00 at true time $9.00 - d$, so the probability of waiting past "9 am" becomes

$$g(d) = \mathrm{P}(T_8 < -5 - d)\,\mathrm{P}(T_9 > -d).$$

A fast clock ($d > 0$) gets me there earlier, so I rarely miss the 8 o'clock bus and $g(d) < \frac{1}{16}$; a clock slightly slow does the opposite (for example $g(-2.5) \approx 0.079$). So the honest test is two-sided.

Let X be the number of the 100 mornings on which I wait past 9 am by the new clock. The arrival times of different buses are independent and I arrive at the same time every day, so the mornings are independent trials and

$$H_0:\ \text{clock correct}, \quad X \sim \mathrm{B}\big(100, \tfrac{1}{16}\big); \qquad H_1:\ \text{clock set wrongly, so } p \neq \tfrac{1}{16}.$$

Under H_0 we expect $\mathrm{E}(X) = 6.25$ occasions, with $\mathrm{Var}(X) = 100 \cdot \frac{1}{16} \cdot \frac{15}{16} \approx 5.86$, so the standard deviation is about 2.42. The observed 5 is just below the mean. Exactly,

$$\mathrm{P}(X \le 5) = \sum_{k=0}^{5} \binom{100}{k} p^k (1-p)^{100-k} \approx 0.40 \qquad \text{with } p = \tfrac{1}{16},$$

and the normal approximation with continuity correction gives $\Phi\big((5.5 - 6.25)/2.42\big) = \Phi(-0.31) \approx 0.38$. This is nowhere near 0.025 (the lower tail of a two-sided 5% test), or indeed any usual significance level. So we do not reject H_0: **there is no evidence that the new clock is set incorrectly.**

That is not the same as evidence that the clock is right. In fact $g(d) = g(-5 - d)$, because the density is symmetric (swap the roles of the two buses), so a clock that is 5 minutes slow gives exactly the same $p = \frac{1}{16}$ as a correct one. No number of mornings of this kind of data could tell those two apart.

Checks

- The density has area 1, and the triangle formula agrees with numerical integration of f, for example $\mathrm{P}(T < -2) = 0.32$.
- The answer 0.7415 sits between the best case $\frac{7}{8}$ (always arriving at 7.55) and the worst case 0.32 (always at 8.02), as it should. A simulation of 10^6 mornings gave 0.741.
- A simulation of 10^6 pairs (T_8, T_9) gave 0.0622 for $\frac{1}{16} = 0.0625$, and the exact binomial tail $\mathrm{P}(X \le 5)$ evaluates to 0.4002.

Common slips

- Forgetting that missing the 8 o'clock bus is necessary for waiting past 9: the answer $p = \frac{1}{2}$ (just "the 9 o'clock bus is late") ignores the first factor.
- Treating "catch a bus by 8.10" as needing $T \leq 10$ to be checked separately: the 8 o'clock bus always arrives by 8.10, so only $T > a$ matters.
- Running a one-tailed test for "too many" when the data show slightly too few; a wrong clock can move p in either direction.

Beyond STEP

The triangular density is exactly the distribution of the sum of two independent uniform random variables, a first example of a convolution, which is how densities of sums are found in a first-year probability course. The last part is a small taste of statistical identifiability: the symmetry $g(d) = g(-5 - d)$ means two different parameter values produce identical data, so no test can separate them. Grimmett and Welsh, *Probability: An Introduction*, covers convolutions and sums of random variables clearly.

WORKED SOLUTION

STEP II 1990, Question 1

Pure • Polynomial Inequalities

STEP II 1990, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q1

The key idea

A quartic with no real roots is just a quartic that stays positive, so we look for ways to split it into pieces we *know* are non-negative: perfect squares, or products whose sign is obvious on a particular interval. No single split works everywhere, but different splits work on different intervals, and together they cover the whole real line. The first part hands us the pieces; in the last part we have to find them ourselves.

Solution

The two non-negative expressions

Factorising the quartic and completing the square in the quadratic,

$$x^4 - 2x^3 + x^2 = x^2\left(x^2 - 2x + 1\right) = x^2(x-1)^2,$$
$$x^2 - 8x + 17 = (x-4)^2 + 1.$$

The first is a product of two squares, so it is ≥ 0 for every real x. The second is a square plus 1, so it is at least 1, and in particular non-negative.

The first equation

Write $p(x) = x^4 - 2x^3 + x^2 - 8x + 17$. Notice that p is *not* the sum of the two expressions above (that sum has $2x^2$, not x^2), so we can't simply add them. Instead we split p differently on each interval and show that $p(x) > 0$ throughout.

Ask yourself The question gives two non-negative expressions, but they don't add up to $p(x)$. How can we still use them?

Each of them can be "borrowed" on part of the line: on one interval we keep the quartic $x^2(x-1)^2$ and handle what's left, $17 - 8x$, directly; on another we keep the quadratic $(x-4)^2 + 1$ and handle the leftover $x^4 - 2x^3$ directly.

Case $x \leq 0$. Here $x^4 \geq 0$, $-2x^3 \geq 0$ (since $x^3 \leq 0$), $x^2 \geq 0$ and $-8x \geq 0$. So every term of $p(x)$ is non-negative, and the constant 17 makes the total at least 17. Hence $p(x) > 0$.

Case $0 < x \leq 2$. Write

$$p(x) = x^2(x-1)^2 + (17 - 8x).$$

The first bracket is ≥ 0, and since $x \leq 2$ we have $17 - 8x \geq 17 - 16 = 1$. So $p(x) \geq 1 > 0$.

Case $x > 2$. Write

$$p(x) = x^3(x-2) + \left(x^2 - 8x + 17\right) = x^3(x-2) + (x-4)^2 + 1.$$

Since $x > 2$, both x^3 and $x - 2$ are positive, so $x^3(x-2) > 0$; and $(x-4)^2 + 1 \geq 1$. So $p(x) > 1 > 0$.

The three intervals cover every real number and $p(x) > 0$ on each, so $p(x) = 0$ has no real roots.

The second equation

Let $q(x) = x^4 - x^3 + x^2 - 4x + 4$. We copy the method, now choosing our own split.

Ask yourself Where should we split the line, and which pairs of terms should we group?

We want each group to have an obvious sign on its own interval. The pairing $4 - 4x = 4(1-x)$ is non-negative for $x \leq 1$. For $x \geq 1$, the pairing $x^4 - x^3 = x^3(x-1)$ is non-negative, and what's left, $x^2 - 4x + 4 = (x-2)^2$, is a perfect square. So the natural boundary is $x = 1$.

Case $x \leq 1$. Write

$$q(x) = \left(x^4 - x^3 + x^2\right) + 4(1-x) = x^2\left[\left(x - \tfrac{1}{2}\right)^2 + \tfrac{3}{4}\right] + 4(1-x).$$

The first term is ≥ 0 and is zero only when $x = 0$ (the square bracket is at least $\frac{3}{4}$). The second term is ≥ 0 since $x \leq 1$, and is zero only when $x = 1$. Both terms cannot be zero at once, so $q(x) > 0$.

Case $x \geq 1$. Write

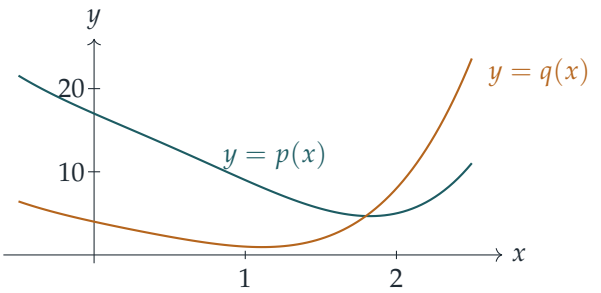
$$q(x) = x^3(x-1) + \left(x^2 - 4x + 4\right) = x^3(x-1) + (x-2)^2.$$

For $x \geq 1$ both x^3 and $x - 1$ are non-negative, so the first term is ≥ 0, and it is zero only at $x = 1$. The second term is a square, zero only at $x = 2$. Again they can't both vanish, so $q(x) > 0$.

So $q(x) > 0$ for all real x, and $q(x) = 0$ has no real roots.

Checks

- Expand each split to make sure it really reproduces p or q. For example, $x^3(x-1)+(x-2)^2 = x^4 - x^3 + x^2 - 4x + 4$.
- A numerical scan of $-5 \le x \le 8$ in steps of 10^{-4} gives a minimum of p of about 4.67 (near $x \approx 1.83$) and a minimum of q of about 0.94 (near $x \approx 1.11$). Both are comfortably positive, as the picture shows.



Common slips

- Adding the two given expressions and claiming the sum is $p(x)$: it isn't, because the x^2 terms don't match.
- Concluding $q(x) > 0$ from "a non-negative term plus a non-negative term". That only gives $q(x) \ge 0$; to rule out a root you must show the two terms can't vanish together.
- Thinking the sign of $-2x^3$ or $-8x$ is always negative. For $x \le 0$ they help rather than hurt.

Beyond STEP

Showing that a polynomial is positive by writing it as a sum of squares is a serious topic. Every non-negative polynomial in one variable is a sum of two squares of real polynomials, but Hilbert showed that this fails for several variables, and this led to his seventeenth problem. Today computers search for such decompositions in optimisation problems. For many more inequality tricks of this kind, Siklos's *Advanced Problems in Mathematics* is a good next step.

WORKED SOLUTION

STEP II 1990, Question 2

Pure • Ptolemy's Theorem

STEP II 1990, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q2

The key idea

The first part is a trigonometric identity that falls apart once every product of sines is turned into a difference of cosines; the condition $A + B + C + D = \pi$ is then used to rewrite $\cos(A + C)$ and $\cos(A - C)$. The second part is Ptolemy's theorem. The centre O is the hint: every chord of a circle of radius r has length $2r\sin(\text{half the angle it subtends at } O)$, and the four half-angles around O add up to π, which is exactly the condition in the first part.

Solution

The identity

We use the product-to-sum formula

$$\sin X \sin Y = \tfrac{1}{2}\big[\cos(X - Y) - \cos(X + Y)\big]$$

three times. On the left-hand side,

$$\sin(A + B)\sin(A + D) = \tfrac{1}{2}\big[\cos(B - D) - \cos(2A + B + D)\big],$$
$$\sin B \sin D = \tfrac{1}{2}\big[\cos(B - D) - \cos(B + D)\big],$$

so the $\cos(B - D)$ terms cancel and

$$\sin(A + B)\sin(A + D) - \sin B \sin D = \tfrac{1}{2}\big[\cos(B + D) - \cos(2A + B + D)\big].$$

On the right-hand side,

$$\sin A \sin C = \tfrac{1}{2}\big[\cos(A - C) - \cos(A + C)\big].$$

Now we use $C = \pi - A - B - D$. This gives

$$A + C = \pi - (B + D), \qquad \text{so} \quad \cos(A + C) = -\cos(B + D),$$
$$A - C = (2A + B + D) - \pi, \qquad \text{so} \quad \cos(A - C) = -\cos(2A + B + D),$$

using $\cos(\pi-\theta) = -\cos\theta$ and $\cos(\theta-\pi) = -\cos\theta$. Therefore

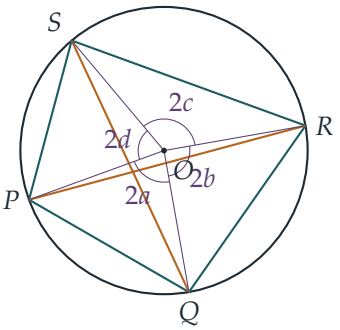
$$\sin A \sin C = \tfrac{1}{2}\big[-\cos(2A+B+D) + \cos(B+D)\big],$$

which is the same as the left-hand side. This proves the identity.

The circle

Let the circle have radius r. Going round the circle from P to Q to R to S and back to P, let the arcs PQ, QR, RS, SP subtend angles $2a$, $2b$, $2c$, $2d$ at O. Since the points are in that order, these four arcs make up the whole circle without overlapping, so

$$a, b, c, d > 0, \qquad 2a+2b+2c+2d = 2\pi, \qquad \text{i.e.} \quad a+b+c+d = \pi.$$



Ask yourself How long is a chord whose arc subtends an angle 2θ at the centre?

If $0 < 2\theta \le \pi$, the triangle formed by O and the two ends of the chord is isosceles with two sides r and apex angle 2θ. The perpendicular from O bisects both the chord and the apex angle, so the chord has length $2r\sin\theta$. If instead $\pi < 2\theta < 2\pi$, the angle at O inside the triangle is $2\pi - 2\theta$, and the chord has length $2r\sin(\pi-\theta) = 2r\sin\theta$. So in every case

$$\text{chord} = 2r\sin\theta.$$

Applying this to the four sides, and to the diagonals (the arc $P \to Q \to R$ subtends $2(a+b)$ and the arc $Q \to R \to S$ subtends $2(b+c)$), we get

$$PQ = 2r\sin a, \quad QR = 2r\sin b, \quad RS = 2r\sin c, \quad PS = 2r\sin d,$$

$$PR = 2r\sin(a+b), \qquad QS = 2r\sin(b+c).$$

Note that $a+b$ or $b+c$ may exceed $\frac{\pi}{2}$; this is why we needed the formula for arcs of more than a semicircle.

After dividing by $4r^2$, the result we want becomes

$$\sin a \sin c + \sin b \sin d = \sin(a+b)\sin(b+c),$$

or, equivalently,

$$\sin(b+a)\sin(b+c) - \sin a \sin c = \sin b \sin d.$$

This is exactly the identity from the first part with $A = b$, $B = a$, $D = c$ and $C = d$, which is allowed because $b + a + d + c = \pi$. Multiplying back by $4r^2$,

$$PQ \times RS + QR \times PS = PR \times QS,$$

as required.

Checks

- A square inscribed in a circle of radius $\frac{1}{\sqrt{2}}$ has sides 1 and diagonals $\sqrt{2}$: $1 \cdot 1 + 1 \cdot 1 = 2 = \sqrt{2} \cdot \sqrt{2}$.
- For any rectangle with sides p and q, Ptolemy's theorem reads $p^2 + q^2 = (\text{diagonal})^2$: Pythagoras's theorem is a special case. For a 3×4 rectangle, $9 + 16 = 25 = 5 \times 5$.
- A numerical check of the identity at 10^5 random values of A, B, D (with $C = \pi - A - B - D$) gave a largest error of about 5×10^{-15}, and 10^5 random cyclic quadrilaterals satisfied the final result to within a relative error of 10^{-15}. With the points taken out of order (for example P, R, Q, S in order round the circle) the equality fails, so the ordering really matters.

Common slips

- Using inscribed angles at a point on the circle without saying which arc they stand on. Working with angles at the centre, as the question suggests, avoids this.
- Forgetting that a diagonal can cut off an arc of more than a semicircle; the formula $2r \sin\theta$ still holds, but it needs the one-line justification above.
- Matching the wrong angles to A, B, C, D: A and C must be the two angles that appear only on the right-hand side, here b and d.

Beyond STEP

Ptolemy's theorem has a far-reaching sequel: for *any* four points in the plane, $PQ \cdot RS + QR \cdot PS \geq PR \cdot QS$, with equality exactly when the points lie in order on a circle (or a line). A slick proof treats the points as complex numbers p, q, r, s, notes the algebraic identity $(p - q)(r - s) + (q - r)(p - s) = (p - r)(q - s)$, and applies the triangle inequality; the equality case is where the geometry of the complex plane comes in. Needham's *Visual Complex Analysis* is a beautiful place to see this kind of argument, and inversion, which also proves the result neatly.

WORKED SOLUTION

STEP II 1990, Question 3

Pure • Curve Sketching Cubics

STEP II 1990, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q3

The key idea

Write the cubic as $f(x) = x(x^2 - 2bx + c^2)$. Two discriminants control everything: the one for $f'(x)$ tells us whether there are turning points, and the one for the quadratic factor tells us whether the curve meets the x-axis anywhere other than the origin. The seven cases are just the ways of placing b relative to the two thresholds $\frac{\sqrt{3}}{2}c$ and c. Once the cubics are drawn, $y^2 = f(x)$ comes almost for free: keep only the part where $f(x) \geq 0$, take square roots and reflect in the x-axis.

Solution

Facts that hold in every case

Let $f(x) = x^3 - 2bx^2 + c^2x = x(x^2 - 2bx + c^2)$. Then

$$f'(x) = 3x^2 - 4bx + c^2, \qquad f''(x) = 6x - 4b.$$

- *Turning points.* The discriminant of f' is $16b^2 - 12c^2 = 4(4b^2 - 3c^2)$. So f' has two distinct real zeros if $2b > c\sqrt{3}$, one repeated zero if $2b = c\sqrt{3}$, and none if $2b < c\sqrt{3}$. When there are zeros, their sum is $\frac{4}{3}b \geq 0$ and their product is $\frac{1}{3}c^2 \geq 0$, so neither is negative: all turning points lie in $x \geq 0$.
- *Roots.* $f(0) = 0$ always. The quadratic factor is $(x - b)^2 + c^2 - b^2$, so it has no real zeros if $b < c$, the double zero $x = b$ if $b = c$, and the two zeros $b \pm \sqrt{b^2 - c^2}$ if $b > c$.
- *Negative x.* For $x < 0$ each of x^3, $-2bx^2$ and c^2x is ≤ 0, and $x^3 < 0$, so $f(x) < 0$. The whole curve to the left of the y-axis lies below the x-axis.
- *Shape.* $f(x) \to \pm\infty$ as $x \to \pm\infty$; the gradient at the origin is $f'(0) = c^2$; and the only point of inflection is at $x = \frac{2}{3}b$, where f'' changes sign.

Since $c\sqrt{3} < 2c$ whenever $c > 0$, the cases run in order of increasing b for a fixed c: no turning points, then a stationary inflection, then turning points above the axis, then a touching root, then three roots.

The cubics

(i) $2b < c\sqrt{3}$. Here $f'(x) > 0$ for all x (its discriminant is negative and its leading coefficient is positive), so f is strictly increasing. Also $b < \frac{\sqrt{3}}{2}c < c$ (note $c > 0$ here), so the origin is the only root. The curve rises through the origin with gradient c^2 and has an inflection at $x = \frac{2}{3}b$.

(ii) $2b = c\sqrt{3} \neq 0$. Now $f'(x) = 3(x - \frac{2}{3}b)^2 \geq 0$, so f is still increasing, but it is momentarily level at $x = \frac{2}{3}b = c/\sqrt{3}$. This is also where $f'' = 0$, so it is a stationary point of inflection. Using $c^2 = \frac{4}{3}b^2$,

$$f\left(\tfrac{2}{3}b\right) = \tfrac{2}{3}b\left(\tfrac{4}{9}b^2 - \tfrac{4}{3}b^2 + \tfrac{4}{3}b^2\right) = \tfrac{8}{27}b^3 = \tfrac{\sqrt{3}}{9}c^3 > 0.$$

Again $b < c$, so the origin is the only root.

(iii) $c\sqrt{3} < 2b < 2c$. Now f' has two zeros $x_1 < x_2$, both positive, giving a maximum at $x_1 = \frac{1}{3}\left(2b - \sqrt{4b^2 - 3c^2}\right)$ and a minimum at $x_2 = \frac{1}{3}\left(2b + \sqrt{4b^2 - 3c^2}\right)$.

? **Ask yourself** Does the minimum dip below the x-axis?

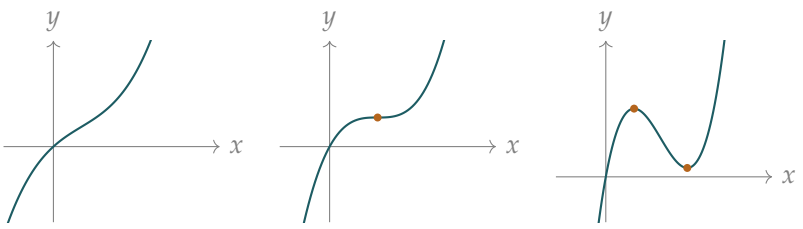
No. Since $b < c$, the factor $(x - b)^2 + c^2 - b^2$ is positive for every x, so $f(x) > 0$ for all $x > 0$. Both turning points therefore lie above the axis, and the curve still crosses the axis only at the origin. It climbs, falls back a little, then climbs for good.

(iv) $b = c \neq 0$. Now $f(x) = x(x - b)^2$. The double root at $x = b$ means the curve touches the axis there; this is the minimum, $(b, 0)$. The other zero of $f'(x) = (x - b)(3x - b)$ gives the maximum $\left(\frac{1}{3}b, \frac{4}{27}b^3\right)$.

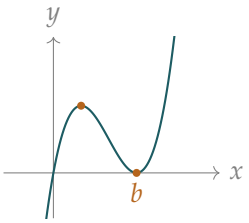
(v) $b > c > 0$. There are three roots, $0 < \alpha < \beta$, where $\alpha = b - \sqrt{b^2 - c^2}$ and $\beta = b + \sqrt{b^2 - c^2}$ (both positive because $\sqrt{b^2 - c^2} < b$). Between consecutive roots the curve must turn, so the maximum lies in $(0, \alpha)$ with $f > 0$ there and the minimum lies in (α, β) with $f < 0$ there. So $f \geq 0$ exactly on $[0, \alpha]$ and on $[\beta, \infty)$.

(vi) $c = 0$, $b \neq 0$. Now $f(x) = x^2(x - 2b)$ and $f'(x) = x(3x - 4b)$. The double root at the origin makes it a maximum where the curve touches the axis, and there is a minimum at $\left(\frac{4}{3}b, -\frac{32}{27}b^3\right)$. The curve crosses the axis at $x = 2b$.

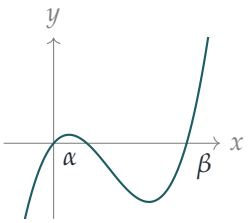
(vii) $b = c = 0$. This is $y = x^3$, with a stationary point of inflection at the origin.



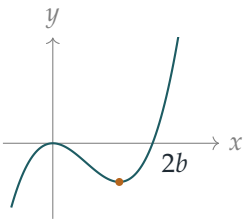
(i) (ii) (iii)



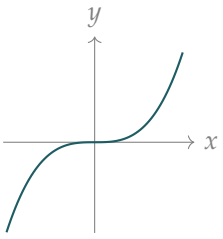
(iv)



(v)



(vi)



(vii)

The sketches use $c = 1$ with $b = \frac{1}{2}$, $\frac{\sqrt{3}}{2}$, 0.99, 1 in (i)–(iv), $(b, c) = (1, 0.8)$ in (v) and $b = 0.6$ in (vi). The vertical scales differ from panel to panel so that the small bumps in (iii) and (iv) can be seen.

The curves $y^2 = f(x)$

Three rules turn each cubic into the curve $y^2 = f(x)$:

- there are no points where $f(x) < 0$, and in particular none with $x < 0$;
- where $f(x) \geq 0$, $y = \pm\sqrt{f(x)}$, so the curve is symmetric in the x-axis and is tallest where f is largest;
- differentiating, $2y\dfrac{\mathrm{d}y}{\mathrm{d}x} = f'(x)$.

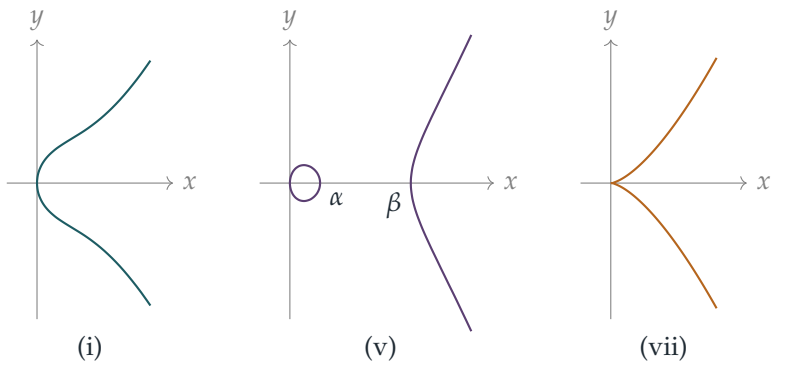
Ask yourself What does the curve look like where it meets the x-axis?

At a simple root x_0 of f we have $f'(x_0) \neq 0$, and $\dfrac{\mathrm{d}x}{\mathrm{d}y} = \dfrac{2y}{f'(x)} \to 0$ as $y \to 0$. So the tangent there is *vertical*, and the upper and lower halves join smoothly. Near such a root $y^2 \approx f'(x_0)(x - x_0)$, which is a parabola on its side. A repeated root behaves differently, as case (vii) shows.

Case (i). $f(x) \geq 0$ exactly when $x \geq 0$, and the root at 0 is simple ($f'(0) = c^2 > 0$). So the curve is a single unbounded branch. It leaves the origin with a vertical tangent, looking like $y^2 \approx c^2 x$ there, and $|y| = \sqrt{f(x)}$ increases steadily, since f is increasing, eventually growing like $x^{3/2}$. The upper half starts out concave, like $\sqrt{x}$, and ends up convex, like $x^{3/2}$, so it has a point of inflection (and, by symmetry, so does the lower half).

Case (v). $f(x) \geq 0$ on $[0, \alpha]$ and on $[\beta, \infty)$, and all three roots are simple. So the curve has two separate pieces: a closed oval through $(0,0)$ and $(\alpha,0)$, with vertical tangents at both, which is widest (in y) at the x-coordinate of the cubic's maximum; and an unbounded branch starting from $(\beta,0)$ with a vertical tangent and opening to the right like the curve in case (i). Nothing lies in the gap $\alpha < x < \beta$, where the cubic is negative.

Case (vii). $y^2 = x^3$, so $x \geq 0$ and $y = \pm x^{3/2}$. Here $\mathrm{d}y/\mathrm{d}x = \pm\frac{3}{2}x^{1/2} \to 0$ as $x \to 0$: both halves arrive at the origin *tangent to the x-axis*, from above and below, forming a sharp point called a cusp.



These use the same values of b and c as the corresponding cubics, with equal scales on the two axes.

Checks

- Put $b = c = 0$ into any general statement and it should match (vii); for example the stationary value $\frac{8}{27}b^3$ in (ii) and the minimum $-\frac{32}{27}b^3$ in (vi) both shrink to the origin.
- With $c = 1$ and $b = 0.99$ (case (iii)) the turning points are at about $(0.340, 0.150)$ and $(0.980, 0.020)$, both above the axis as claimed. With $(b, c) = (1, 0.8)$ the roots are 0, 0.4, 1.6 and the minimum is about -0.388 at $x \approx 1.147$, inside $(0.4, 1.6)$.
- A numerical test over thousands of random (b, c) in cases (i), (iii) and (v) confirmed the number and position of the turning points and roots, and the set where $f \geq 0$.

Common slips

- Drawing the minimum in case (iii) below the axis: with $b < c$ the cubic is positive for all $x > 0$.
- Drawing $y^2 = f(x)$ with a corner, or a horizontal tangent, where it crosses the x-axis at a simple root. The tangent there is vertical.
- Forgetting that $y^2 = f(x)$ has no points where the cubic is negative, so in case (v) the oval and the branch are disconnected.

Beyond STEP

A curve $y^2 = x^3 + Ax^2 + Bx + C$ whose cubic has distinct roots is an *elliptic curve*, one of the central objects of modern number theory. Case (v), with three real roots, gives the two-piece picture of an oval plus a branch; case (i), with one real root, gives one piece. When roots coincide the curve becomes singular: case (vii) has a cusp, and case (iv) would give a curve that crosses itself at $(b, 0)$. For an introduction to the remarkable way points on such curves can be "added", see Silverman and Tate, *Rational Points on Elliptic Curves*.

WORKED SOLUTION

STEP II 1990, Question 4

Pure • Line Arrangements and Induction

STEP II 1990, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q4

This question has a full guided treatment in *Beyond STEP*, Section 9.1, *Lines Cutting the Plane*: https://beyondstep.keithmcnulty.org/ch09/lazy-caterer.html

The key idea

Look at the picture one line at a time. When a new line is added, it crosses every old line exactly once, at different points, so it is chopped into a predictable number of pieces. Each of those pieces does two jobs: it is a new segment, and it cuts an existing region in two. Counting the effect of one new line gives a recurrence, and the recurrences are easy to sum.

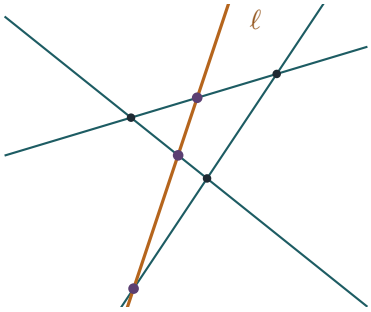
Solution

Small cases

Throughout, a "segment" means a piece of one of the lines between consecutive crossing points, including the two infinite pieces (rays) at the ends of each line, and with one line the whole line counts as one segment. Drawing the first few cases gives:

n	1	2	3	4
segments	1	4	9	16
regions	2	4	7	11

For instance, three lines form a triangle: each line is cut into a finite side and two rays, giving 9 segments, and the regions are the triangle, three regions touching its sides and three touching its corners, 7 in all. The segment counts suggest n^2.



The figure shows three lines (teal) and a fourth line ℓ (amber). The three plum points where ℓ crosses the old lines cut it into four pieces, and each piece splits one of the seven old regions, giving 11.

The number of segments

Fix one of the n lines. No two lines are parallel, so it meets each of the other $n-1$ lines, and since no three lines are concurrent, these $n-1$ crossing points are all different (if two coincided, three lines would meet there). So the line carries $n-1$ distinct points, which cut it into n pieces: $n-2$ finite segments and 2 rays. Doing this for each of the n lines,

$$\text{number of segments} = n \cdot n = n^2.$$

Each segment is counted once, because each lies on exactly one line.

If you prefer induction, let $S(n)$ be the number of segments. Adding an nth line to $n-1$ lines in general position, the new line is cut into n pieces, and each of the $n-1$ old lines it crosses has one of its segments split in two, adding one more segment each. So

$$S(n) = S(n-1) + n + (n-1) = S(n-1) + 2n - 1,$$

and with $S(1) = 1$, if $S(n-1) = (n-1)^2$ then $S(n) = (n-1)^2 + 2n - 1 = n^2$.

The number of regions

Let $R(n)$ be the number of regions made by n lines in general position, so $R(1) = 2$. Start from $n-1$ such lines and add the nth line ℓ. As above, ℓ crosses the old lines at $n-1$ distinct points and is cut into n pieces.

Take one of these pieces. Its interior doesn't meet any old line, so it lies inside a single old region, and it runs right across that region from boundary to boundary (or out to infinity). It therefore splits that region into two.

> **Ask yourself** Could two of the n pieces lie in the same old region, so that fewer than n new regions appear?

No. Each region is an intersection of half-planes (on a fixed side of each old line), so it is convex. A line meets a convex set in a single interval, so ℓ passes through each old

region at most once, and different pieces lie in different regions. Also, regions that ℓ misses are untouched.

So adding ℓ splits exactly n old regions in two and leaves the rest alone:

$$R(n) = R(n-1) + n.$$

Now we use induction. The formula $\frac{1}{2}(n^2+n+2)$ gives 2 when $n=1$, which is correct. If $R(n-1) = \frac{1}{2}\big((n-1)^2+(n-1)+2\big) = \frac{1}{2}(n^2-n+2)$, then

$$R(n) = \tfrac{1}{2}(n^2-n+2) + n = \tfrac{1}{2}(n^2+n+2),$$

so the formula holds for all $n \geq 1$, as required. (Equivalently, $R(n) = 1 + (1+2+\cdots+n) = 1 + \frac{1}{2}n(n+1)$, where the leading 1 is the whole plane with no lines.)

Checks

- The formula gives $2, 4, 7, 11$ for $n = 1, 2, 3, 4$, matching the drawings.
- Euler's formula gives an independent check. Join the infinite rays at a single "point at infinity". Then there are $V = \binom{n}{2} + 1$ vertices, $E = n^2$ edges and F faces with $V - E + F = 2$, so $F = n^2 - \binom{n}{2} + 1 = \frac{1}{2}(n^2+n+2)$.
- We drew random sets of up to 8 lines with integer coefficients on a computer, counted segments by locating the crossing points, and counted regions as the number of different patterns of "which side of each line" among sample points around every crossing and far away. For example, 8 lines gave 64 segments and 37 regions every time.

Common slips

- Forgetting the two infinite rays on each line, which gives $n(n-2)$ instead of n^2.
- Saying the new line adds $n-1$ regions (one per crossing) rather than n (one per piece).
- Not using the hypotheses: without "no three concurrent" the crossing points on a line need not be distinct, and the counts drop.

Beyond STEP

This is the simplest case of a *hyperplane arrangement*. The same argument one dimension up shows that n planes in general position cut space into $\frac{1}{6}(n^3+5n+6)$ pieces, because each new plane is itself cut into $R(n-1)$ regions by the old ones. The Euler-formula check is a first glimpse of the Euler characteristic, a central idea in topology. For a lovely chapter on Euler's formula and its uses, see Aigner and Ziegler, *Proofs from THE BOOK*.

WORKED SOLUTION

STEP II 1990, Question 5

Pure • Complex Numbers and Circles

STEP II 1990, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q5

The key idea

On a circle $|z| = r$ conjugation is division: $\bar{z} = r^2/z$. That turns "this ratio is purely imaginary" (the test for perpendicular directions) into a neat polynomial condition, $lm + pq = 0$. After that, the rest of the question is a chase: each perpendicular chord gives the new point by a formula of the form $z \mapsto c/z$, and you just compose these maps and see what comes out.

Solution

The perpendicularity criterion

Let S have radius r, so $|l| = |m| = |p| = |q| = r$ and therefore

$$\bar{l} = \frac{r^2}{l}, \qquad \bar{m} = \frac{r^2}{m}, \qquad \bar{p} = \frac{r^2}{p}, \qquad \bar{q} = \frac{r^2}{q}.$$

The chords LM and PQ have directions $l - m$ and $p - q$, both non-zero because the points are distinct. Two non-zero complex numbers represent perpendicular directions exactly when their quotient is purely imaginary, so put

$$w = \frac{l - m}{p - q}.$$

Then w is purely imaginary if and only if $w + \bar{w} = 0$. Now

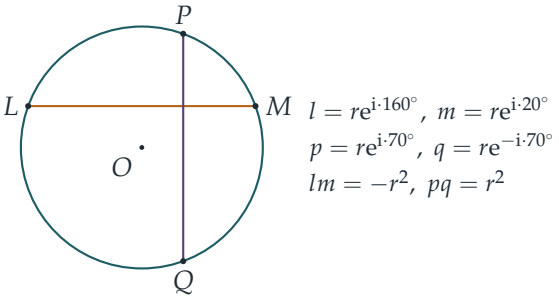
$$\bar{w} = \frac{r^2/l - r^2/m}{r^2/p - r^2/q} = \frac{(m - l)/(lm)}{(q - p)/(pq)} = \frac{l - m}{p - q} \cdot \frac{pq}{lm} = w\,\frac{pq}{lm},$$

so

$$w + \bar{w} = w\left(1 + \frac{pq}{lm}\right) = \frac{w}{lm}\,(lm + pq).$$

Since $w \neq 0$ and $lm \neq 0$, this vanishes if and only if $lm + pq = 0$. So $LM \perp PQ$ if and only if $lm + pq = 0$, as required.

(The perpendicular-bisector route is the same idea in geometric clothing: the perpendicular bisector of a chord LM passes through O, and the chord's direction is i times the direction of that bisector.)



Three points

Write a_1, a_2, a_3 for A_1, A_2, A_3, and z for A_1'. By the criterion, "$A_1'A_2' \perp A_1A_2$" means $z\,a_2' = -a_1a_2$, so

$$a_2' = -\frac{a_1a_2}{z}.$$

This point really is on S, because $|a_2'| = r \cdot r/r = r$; and it is the only point of S that works. In the same way

$$a_3' = -\frac{a_2a_3}{a_2'} = \frac{a_3}{a_1}\,z,$$
$$a_1'' = -\frac{a_3a_1}{a_3'} = -\frac{a_1^2}{z}.$$

Hence $A_1'' = A_1'$ if and only if $z = -a_1^2/z$, that is,

$$z^2 = -a_1^2 \iff z = \pm \mathrm{i}\,a_1.$$

Both values have modulus r, so lie on S, and they are different because $a_1 \neq 0$. So there are exactly two positions of A_1': the points obtained by rotating A_1 through $\pm 90^\circ$ about O.

Ask yourself What is the map $z \mapsto -a_1^2/z$ doing to the circle?

On S we have $1/z = \bar{z}/r^2$, so $-a_1^2/z = -(a_1^2/r^2)\,\bar{z}$, which is a conjugation followed by a rotation: a reflection of the circle in a diameter. A reflection fixes exactly the two ends of its mirror, which is why there are two solutions. Indeed every single step $z \mapsto -a_ka_{k+1}/z$ is such a reflection, and that explains everything that follows.

Four points

Continuing the same chase with a_4 in place of the last step,

$$a_2' = -\frac{a_1 a_2}{z}, \qquad a_3' = \frac{a_3}{a_1} z,$$
$$a_4' = -\frac{a_3 a_4}{a_3'} = -\frac{a_1 a_4}{z}, \qquad a_1'' = -\frac{a_4 a_1}{a_4'} = z.$$

So $A_1'' = A_1'$ for every choice of A_1' on S.

n points

By induction on k, the chase gives

$$a_k' = \frac{a_k}{a_1} z \quad (k \text{ odd}), \qquad a_k' = -\frac{a_1 a_k}{z} \quad (k \text{ even}),$$

since if one form holds for a_k', then $a_{k+1}' = -a_k a_{k+1}/a_k'$ has the other form. The last step is $a_1'' = -a_n a_1 / a_n'$:

- if n is even, $a_1'' = -a_n a_1 \cdot \dfrac{z}{-a_1 a_n} = z$, so A_1'' coincides with A_1' for every A_1';
- if n is odd, $a_1'' = -a_n a_1 \cdot \dfrac{a_1}{a_n z} = -\dfrac{a_1^2}{z}$, so, exactly as for $n = 3$, A_1'' coincides with A_1' for exactly two positions of A_1', namely $z = \pm \mathrm{i} a_1$.

In words: an even number of reflections makes a rotation, which here turns out to be the identity, while an odd number makes a reflection, which has two fixed points on the circle.

Checks

- The diagram: with $l = r\mathrm{e}^{\mathrm{i}\cdot 160^\circ}$, $m = r\mathrm{e}^{\mathrm{i}\cdot 20^\circ}$ the chord LM is horizontal, and with $p = r\mathrm{e}^{\pm \mathrm{i}\cdot 70^\circ}$ the chord PQ is vertical; indeed $lm = r^2 \mathrm{e}^{\mathrm{i}\pi} = -r^2 = -pq$.
- Numerically, for random points on circles of random radius, and n from 3 to 9, chasing the perpendicular chords gave a point back on S at every step, returned exactly to the start for even n, and for odd n a scan of 200 000 starting positions round the circle found exactly two fixed points, at $\pm \mathrm{i} a_1$.

Common slips

- Testing perpendicularity by "w is real". Parallel chords give a real ratio; perpendicular ones give a purely imaginary ratio.
- Forgetting that $\bar{z} = r^2/z$ needs $|z| = r$; it is the fact that all four points are on the same circle centred at O that makes the criterion so clean.
- In the n-point case, forgetting that the answer depends on the parity of n.

Beyond STEP

The maps $z \mapsto c/z$ restricted to the circle are reflections, and composing reflections is the beginning of the study of isometry groups: the symmetries of a circle form the orthogonal group $O(2)$, in which two reflections compose to a rotation. Maps such as $z \mapsto c/z$ and $z \mapsto (az+b)/(cz+d)$ are Möbius transformations, central to complex analysis and geometry. Needham's *Visual Complex Analysis* develops both pictures beautifully.

WORKED SOLUTION

STEP II 1990, Question 6

Pure • Integer Inequalities and Fractions

STEP II 1990, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q6

The key idea

Everything rests on one fact about integers: a strict inequality between integers can be sharpened by 1. If $a < b$ then $b \geq a + 1$, and if $d < c$ then $d \leq c - 1$. That turns (ii) into a one-line estimate. The later parts are then (ii) applied to cleverly chosen integers: clear the denominators in the fractional inequalities, and feed the resulting integer inequalities back into (ii).

Solution

Throughout, all letters stand for positive integers.

Part (i)

Since $b > a \geq 1$ and b is an integer, $b \geq 2$. Multiplying by $c > 0$ gives $bc \geq 2c$.

Since $c > 1$ and c is an integer, $c \geq 2$, so $2c = c + c \geq c + 2$. Together, $bc \geq 2c \geq 2 + c$, as required.

Part (ii)

Because the inequalities are strict and the numbers are integers, $b \geq a + 1$ and $d \leq c - 1$. Using $c > 0$ in the first and $a > 0$ in the second,

$$\begin{aligned} bc - ad &\geq (a+1)c - a(c-1) \\ &= ac + c - ac + a \\ &= a + c. \end{aligned}$$

So $bc - ad \geq a + c$, with equality exactly when $b = a + 1$ and $d = c - 1$.

(This is the same "sharpen by 1" trick as in (i), which is a warm-up; we won't need (i) again.)

Part (iii)

Multiply out the two inequalities, using $b, d > 0$:

$$\frac{a}{b} < p \iff a < bp, \qquad p < \frac{c}{d} \iff dp < c.$$

Now apply (ii) with b replaced by bp and d replaced by dp (both positive integers). The hypotheses $a < bp$ and $dp < c$ are exactly what we have, so

$$(bp)c - a(dp) \geq a + c, \qquad \text{that is,} \qquad (bc - ad)p \geq a + c.$$

Part (iv)

Clearing denominators (all positive) gives two integer inequalities:

$$aq < bp \qquad \text{and} \qquad dp < cq.$$

Also $\frac{a}{b} < \frac{c}{d}$, so $ad < bc$ and $bc - ad$ is a *positive* integer; we may therefore divide by it without reversing any inequality.

? **Ask yourself** Which four numbers should play the roles of a, b, c, d in (ii)?

We have two strict inequalities, and (ii) needs exactly two: "first $<$ second" and "fourth $<$ third". There are two natural ways to match them up, and each gives one of the bounds.

The bound on p. Take $aq < bp$ as the "$a < b$" inequality and $dp < cq$ as the "$d < c$" one. Part (ii), applied to aq, bp, cq, dp, gives

$$(bp)(cq) - (aq)(dp) \geq aq + cq, \qquad \text{so} \qquad pq(bc - ad) \geq q(a + c).$$

Dividing by $q > 0$ and then by $bc - ad > 0$,

$$p \geq \frac{a + c}{bc - ad}.$$

The bound on q. Now swap the roles: take $dp < cq$ as the "$a < b$" inequality and $aq < bp$ as the "$d < c$" one. Part (ii), applied to dp, cq, bp, aq, gives

$$(cq)(bp) - (dp)(aq) \geq dp + bp, \qquad \text{so} \qquad pq(bc - ad) \geq p(b + d).$$

Dividing by $p > 0$ and by $bc - ad > 0$,

$$q \geq \frac{b + d}{bc - ad}.$$

Both bounds are proved. (Part (iii) is the special case $q = 1$ of the first one.)

The fractions between $8/9$ and $9/10$

Take $a = 8$, $b = 9$, $c = 9$, $d = 10$. Then

$$bc - ad = 81 - 80 = 1,$$

so any fraction p/q (with p, q positive integers, in lowest terms or not) lying strictly between $8/9$ and $9/10$ satisfies

$$p \geq \frac{8+9}{1} = 17, \qquad q \geq \frac{9+10}{1} = 19.$$

A denominator less than 20 must therefore be exactly $q = 19$. Then we need

$$\frac{8}{9} < \frac{p}{19} < \frac{9}{10} \iff \frac{152}{9} < p < \frac{171}{10} \iff 16.88\ldots < p < 17.1,$$

so $p = 17$. And $17/19$ really does lie in the interval, since $8 \cdot 19 = 152 < 153 = 9 \cdot 17$ and $17 \cdot 10 = 170 < 171 = 9 \cdot 19$.

Hence the only fraction with denominator less than 20 between $8/9$ and $9/10$ is

$$\frac{17}{19}.$$

Because the bound $q \geq 19$ applies to every way of writing a fraction, not only to lowest terms, no non-reduced form like $34/38$ sneaks in with a smaller denominator either.

Checks

- $17/19$ is the *mediant* $\frac{8+9}{9+10}$, and both bounds in (iv) are attained by it, which is a good sign the bounds are sharp.
- Decimals: $8/9 \approx 0.8889$, $17/19 \approx 0.8947$, $9/10 = 0.9$.
- We tested (i) and (ii) for all values up to 29, tested (iii) and (iv) on about 60,000 random cases, and listed every p/q with $q \leq 19$ between $8/9$ and $9/10$ by brute force: only $17/19$ appears.

Common slips

- Forgetting to say that $bc - ad > 0$ before dividing by it in (iv).
- Using only the p bound in the final part. It allows q up to 19 but doesn't by itself rule out smaller denominators; the q bound is what does the work.
- Getting only one bound in (iv) from (ii): the second needs the roles of the two inequalities swapped.

Beyond STEP

Two fractions $a/b < c/d$ with $bc - ad = 1$ are called *Farey neighbours*, and this question proves the key fact about them: the fraction with the smallest denominator between them is their mediant $\frac{a+c}{b+d}$. Repeatedly inserting mediants builds the Farey sequences and the Stern–Brocot tree, which list every positive rational exactly once and are closely tied to continued fractions and the best rational approximations of real numbers. Davenport's *The Higher Arithmetic* is a lovely place to read about continued fractions and approximation.

WORKED SOLUTION

STEP II 1990, Question 7

Pure • Differential Equations

STEP II 1990, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q7

The key idea

An exponential trial solution turns a delay equation into an ordinary equation for p, because a time shift just multiplies e^{pt} by a constant. The equation $(*)$ mixes p with e^{-p}, so it can't be solved exactly, but multiplying through by e^{p} puts all the p-dependence on one side: the roots are where the single curve $y = (p+1)e^{p}$ meets the horizontal line $y = k$. Counting roots then comes down to sketching one curve.

Solution

The equation for p

Put $f(t) = Ae^{pt}$ with $A \neq 0$ (we want non-zero solutions). Then $f'(t) = pAe^{pt}$ and $f(t-1) = Ae^{p(t-1)} = e^{-p}\,Ae^{pt}$, so $(\dagger)$ becomes

$$Ae^{pt}\left(p + 1 - ke^{-p}\right) = 0 \quad \text{for all } t.$$

Since $A \neq 0$ and $e^{pt} > 0$, this holds for all t if and only if

$$p + 1 = ke^{-p},$$

which is $(*)$. So such solutions exist exactly when p is a real root of $(*)$, and then any A works.

How many real roots?

Multiplying $(*)$ by $e^{p} > 0$ gives the equivalent equation

$$g(p) = k, \qquad \text{where } g(p) = (p+1)e^{p}.$$

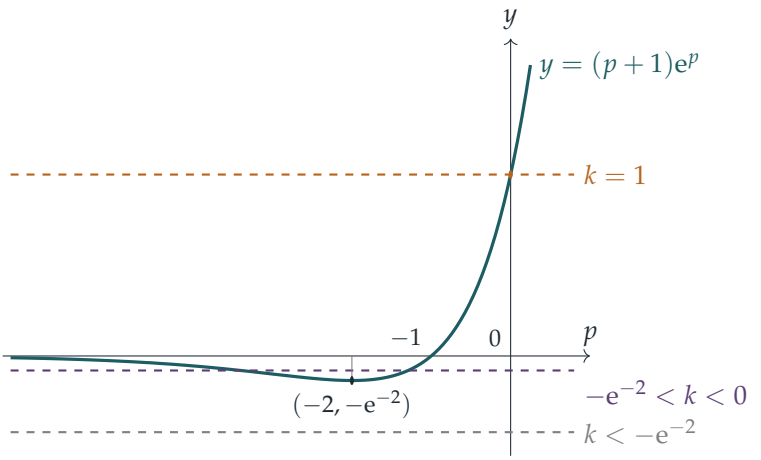
So we count the intersections of the curve $y = g(p)$ with the horizontal line $y = k$.

Ask yourself What does $y = (p+1)e^p$ look like? Where is it smallest, and what happens at each end?

Differentiating, $g'(p) = e^p + (p+1)e^p = (p+2)e^p$, which is negative for $p < -2$ and positive for $p > -2$. So g decreases on $(-\infty, -2]$, increases on $[-2, \infty)$, and has its global minimum

$$g(-2) = -e^{-2}.$$

At the ends: $g(p) \to \infty$ as $p \to \infty$, and $g(p) \to 0$ as $p \to -\infty$ (exponential decay beats linear growth); for $p < -1$ the factor $p+1$ is negative, so g approaches 0 from below. The curve crosses the axis only at $p = -1$, and $g(0) = 1$.



Reading off the picture (and the monotonicity above, which makes it rigorous):

- $k > 0$: on $(-\infty, -1]$ we have $g \le 0 < k$, and on $(-1, \infty)$ g increases strictly from 0 to ∞, so there is **exactly one** root, and it satisfies $p > -1$.
- $-e^{-2} < k < 0$: on $(-\infty, -2]$, g decreases strictly from values near 0 down to $-e^{-2}$, so it takes the value k once; on $[-2, -1]$ it increases strictly from $-e^{-2}$ to 0, so it takes the value k once more; for $p > -1$, $g > 0 > k$. Hence **two** roots, one in $(-\infty, -2)$ and one in $(-2, -1)$.
- $k = -e^{-2}$: the line touches the curve at its minimum, giving the **single** root $p = -2$.
- $k < -e^{-2}$: the line lies below the whole curve, so there are **no** roots.

So $(*)$ has 0, 1 or 2 real roots depending on k, as required, and solutions of the form Ae^{pt} exist precisely when

$$k \ge -e^{-2}, \quad k \ne 0.$$

(The value $k = 0$ is excluded by the question; it would give the single root $p = -1$.)

When do the solutions decay?

> **Ask yourself** Which p give $Ae^{pt} \to 0$, and how do we locate the roots relative to 0 without solving $(*)$?

With $A \neq 0$, $Ae^{pt} \to 0$ as $t \to +\infty$ if and only if $p < 0$. We never need the roots explicitly, only their position relative to 0.

- If $-e^{-2} \leq k < 0$, every root lies in $(-\infty, -1)$ (indeed $g \geq 0$ for $p \geq -1$, so $g(p) = k < 0$ forces $p < -1$). All such solutions decay.
- If $k > 0$, the unique root lies where g is strictly increasing, so $p < 0 \iff g(p) < g(0) \iff k < 1$. For $k = 1$ the root is $p = 0$, giving the constant solution $f = A$, which does not tend to zero; for $k > 1$ the root is positive and the solution grows.

Therefore the solutions tend to zero as $t \to +\infty$ exactly when

$$-e^{-2} \leq k < 1, \quad k \neq 0.$$

Checks

- $k = 1$: $p = 0$ satisfies $0 + 1 = 1 \cdot e^0$, and $f = A$ clearly solves $f' + f - f(t-1) = 0$. This is the borderline between decay and growth, as it should be.
- $k = -e^{-2} \approx -0.1353$: $p = -2$ gives $-1 = -e^{-2}e^2$. Numerically, $k = -0.13$ gives two roots, $p \approx -2.311$ and $p \approx -1.742$, either side of -2, while $k = -0.14$ gives none.
- Physical sense: with no feedback ($k \to 0$) the system is $f' = -f$, with decay rate $p = -1$, which is where the root for small $|k|$ sits. Strong positive feedback ($k > 1$) overcomes the damping and the solution grows.

Common slips

- Sketching $y = p + 1$ against $y = ke^{-p}$ and missing that for $k < 0$ the line can cut the curve twice or not at all; the tangency condition is easier to find with $g(p) = (p+1)e^p$.
- Forgetting the endpoint $k = -e^{-2}$, which does give a (repeated) root $p = -2$ and hence a decaying solution.
- Including $k = 1$ in the decay set: $p = 0$ gives a constant, not a decaying, solution.

Beyond STEP

Equation $(\dagger)$ is a *delay differential equation*, and $(*)$ is its characteristic equation. Unlike an ordinary linear ODE, it has infinitely many *complex* roots p, and the general solution

is built from all of them, so full stability needs every complex root to have negative real part; this is studied in control theory. The real roots of $(p+1)\mathrm{e}^p = k$ are values of the Lambert W function: $p+1 = W(k\mathrm{e})$, and $-\mathrm{e}^{-1}$ (here $k\mathrm{e} = -\mathrm{e}^{-1}$) is exactly the branch point of W. For linear ODEs and characteristic equations, see Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP II 1990, Question 8

Pure • Integration by Parts

STEP II 1990, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q8

The key idea

Every term of the differential equation is, or nearly is, an exact derivative. Since y and y' take the same values at $t = 0$ and $t = 1$, those exact derivatives integrate to zero. Each relation comes from multiplying the equation by a suitable function and then integrating by parts until every boundary term vanishes. Keep a list of the facts that kill boundary terms: $x(0) = 0$, $y(0) = y(1)$, $y'(0) = y'(1)$, and (once we have proved it) $x(1) = 0$.

Solution

The first identity

Since $x' = y$, the integrand $xy = xx'$ is the derivative of $\frac{1}{2}x^2$. So

$$\int_0^1 x(t)\, y(t)\, \mathrm{d}t = \Big[\tfrac{1}{2}x(t)^2\Big]_0^1 = \tfrac{1}{2}[x(1)]^2,$$

because $x(0) = \int_0^0 y(u)\, \mathrm{d}u = 0$. This is the required result.

Showing $x(1) = 0$

Write the middle term of $(*)$ as an exact derivative:

$$(y^2 - 1)\, y' = \frac{\mathrm{d}}{\mathrm{d}t}\Big(\tfrac{1}{3}y^3 - y\Big).$$

Now integrate $(*)$ from 0 to 1 term by term:

$$\Big[y'\Big]_0^1 + \Big[\tfrac{1}{3}y^3 - y\Big]_0^1 + \int_0^1 y\, \mathrm{d}t = 0.$$

The first bracket is $y'(1) - y'(0) = 0$, and the second is zero because $y(1) = y(0)$. The integral left over is $x(1)$, by the definition of x. Hence $x(1) = 0$.

The relation $\int y^2 = \frac{1}{3}\int y^4$

Multiply $(*)$ by x and integrate over $[0,1]$. There are three integrals, and we handle them one at a time.

The last term. By the first identity together with $x(1) = 0$,

$$\int_0^1 x\,y\,\mathrm{d}t = \tfrac{1}{2}[x(1)]^2 = 0.$$

The y'' term. We integrate by parts, differentiating x (so that $x' = y$):

$$\begin{aligned}\int_0^1 x\,y''\,\mathrm{d}t &= \Big[x\,y'\Big]_0^1 - \int_0^1 y\,y'\,\mathrm{d}t \\ &= x(1)y'(1) - x(0)y'(0) - \Big[\tfrac{1}{2}y^2\Big]_0^1 = 0,\end{aligned}$$

since $x(1) = x(0) = 0$ and $y(1) = y(0)$.

> **Ask yourself** How should we integrate $x\,(y^2-1)\,y'$ by parts? Which factor do we integrate?

The middle term. Integrate the factor $(y^2-1)y'$, which we already know is the derivative of $\frac{1}{3}y^3 - y$, and differentiate x:

$$\begin{aligned}\int_0^1 x\,(y^2-1)\,y'\,\mathrm{d}t &= \Big[x\big(\tfrac{1}{3}y^3 - y\big)\Big]_0^1 - \int_0^1 y\big(\tfrac{1}{3}y^3 - y\big)\,\mathrm{d}t \\ &= 0 - \tfrac{1}{3}\int_0^1 y^4\,\mathrm{d}t + \int_0^1 y^2\,\mathrm{d}t.\end{aligned}$$

The boundary term vanishes because x is zero at both ends.

Adding the three pieces gives zero:

$$\int_0^1 y^2\,\mathrm{d}t - \tfrac{1}{3}\int_0^1 y^4\,\mathrm{d}t = 0, \qquad \text{that is,} \qquad \int_0^1 [y(t)]^2\,\mathrm{d}t = \tfrac{1}{3}\int_0^1 [y(t)]^4\,\mathrm{d}t,$$

as required.

The relation $\int y'^2 = \int y^2$

> **Ask yourself** We want an integral of y'^2. Which multiplier will produce one?

Integrating $y\,y''$ by parts produces $-\int y'^2$, so this time we multiply $(*)$ by y and integrate. Taking the terms in turn:

$$\int_0^1 y\,y''\,\mathrm{d}t = \Big[y\,y'\Big]_0^1 - \int_0^1 y'^2\,\mathrm{d}t = -\int_0^1 y'^2\,\mathrm{d}t,$$

$$\int_0^1 (y^3 - y)\,y'\,\mathrm{d}t = \Big[\tfrac{1}{4}y^4 - \tfrac{1}{2}y^2\Big]_0^1 = 0.$$

The boundary term $y(1)y'(1) - y(0)y'(0)$ is zero because both y and y' repeat. The second line vanishes because $y(1) = y(0)$. What remains is

$$-\int_0^1 y'^2\,\mathrm{d}t + \int_0^1 y^2\,\mathrm{d}t = 0, \qquad \text{so} \qquad \int_0^1 [y'(t)]^2\,\mathrm{d}t = \int_0^1 [y(t)]^2\,\mathrm{d}t.$$

Checks

The first identity holds for any y. With $y = \cos 3t + t^2 - 0.7e^t$, quadrature gives $\int_0^1 xy\,\mathrm{d}t \approx 0.338191$, which equals $\frac{1}{2}x(1)^2$.

A non-trivial solution of $(*)$ that repeats after time 1 doesn't exist (see below). However, the same arguments work word for word on $[0, T]$ for any solution that repeats after time T. We integrated $(*)$ numerically onto its periodic orbit, which has $T \approx 6.6633$. Over one period we found $\int y \approx 10^{-9}$, $\int y^2 \approx 13.72222$, $\frac{1}{3}\int y^4 \approx 13.72222$ and $\int y'^2 \approx 13.72228$, which agree to within the step-size error.

Common slips

- Forgetting that $x(1) = 0$ is needed in the third part. Without it, neither $\int xy$ nor the boundary terms vanish.
- In the middle term, integrating x instead of $(y^2 - 1)y'$. Using $x' = y$ is what makes the integral collapse to powers of y.
- Saying a boundary term is zero "because $y(0) = 0$". We aren't told that. The boundary terms vanish because the values at 0 and 1 are equal.

Beyond STEP

Equation $(*)$ is the van der Pol oscillator, a classic model of a self-sustaining electrical circuit. It has a single periodic orbit, and the relations you proved are energy balances around one cycle. They also reveal a surprise. Since $\int_0^1 y = 0$, the Wirtinger (or Poincaré) inequality from Fourier series gives $\int_0^1 y'^2 \geq 4\pi^2 \int_0^1 y^2$. Together with the last relation, this forces $y \equiv 0$. So the only solution that repeats after time 1 is the trivial one, because the true period is about 6.66. For more on periodic orbits and limit cycles, see Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP II 1990, Question 9

Pure • Polar Curves and Arc Length

STEP II 1990, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q9

The key idea

Both curves are given as r in terms of θ, so compare them ray by ray: on each half-line from the origin the parabola has exactly one point, and it is either inside or outside the cardioid. That tells us at once where the parabola enters and leaves the cardioid. For the length, the cardioid is best left in polar form (its arc-length integrand collapses with a half-angle identity), while the parabola is much easier in Cartesian form.

Solution

The two curves

For the parabola, $r + r\cos\theta = 1$ says $r = 1 - x$. Squaring, $x^2 + y^2 = (1-x)^2$, so

$$y^2 = 1 - 2x.$$

This is a parabola with focus at the origin, vertex at $(\frac{1}{2}, 0)$, opening to the left. The origin lies on its concave side, where $y^2 < 1 - 2x$; call this the inside of the parabola.

The cardioid $r = 4(1+\cos\theta)$ has its cusp at the origin (when $\theta = \pi$) and reaches $x = 8$ when $\theta = 0$.

Where they meet

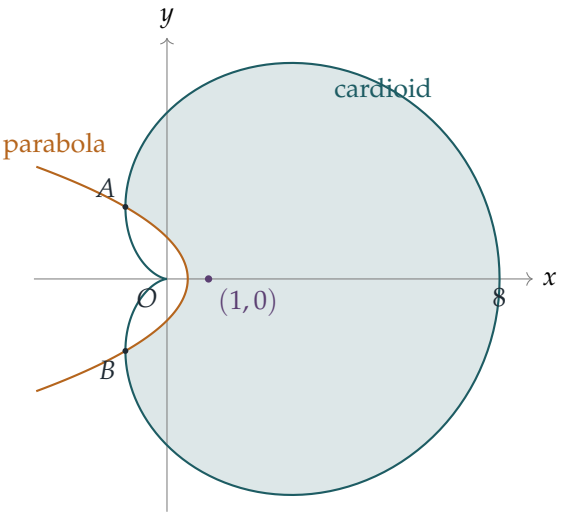
Fix a direction θ with $-\pi < \theta < \pi$, so that $1 + \cos\theta > 0$. On that ray the parabola is at distance $1/(1+\cos\theta)$ and the cardioid's boundary at distance $4(1+\cos\theta)$, and the interior of the cardioid is the segment of the ray before the boundary. So the parabola point is inside the cardioid exactly when

$$\frac{1}{1+\cos\theta} < 4(1+\cos\theta) \iff (1+\cos\theta)^2 > \tfrac{1}{4} \iff \cos\theta > -\tfrac{1}{2} \iff |\theta| < \tfrac{2\pi}{3},$$

where we multiplied by the positive quantity $1+\cos\theta$ and took the positive square root. Equality holds at $\theta = \pm\frac{2\pi}{3}$, where $r = 2$, giving the two crossing points

$$A = (-1, \sqrt{3}), \qquad B = (-1, -\sqrt{3}).$$

So the parabola is inside the cardioid along one unbroken arc, from B through the vertex $(\frac{1}{2}, 0)$ to A, and outside it everywhere else. This single arc joins two boundary points of the cardioid, so it cuts the interior into two parts: one inside the parabola (containing the origin, bounded by the small loop of the cardioid near the cusp) and one outside the parabola (bounded by the large outer arc of the cardioid).



(A neat detail visible in the sketch: $x = 4(1+\cos\theta)\cos\theta = (2\cos\theta+1)^2 - 1 \geq -1$, so A and B are also the leftmost points of the cardioid.)

The length of the boundary

The point $r = 1$, $\theta = 0$ is $(1, 0)$. It is inside the cardioid ($1 < 8$) and outside the parabola ($0^2 > 1 - 2$), so it lies in the shaded part. Its boundary consists of the cardioid arc $-\frac{2\pi}{3} \leq \theta \leq \frac{2\pi}{3}$ and the parabola arc from B to A.

Cardioid arc. In polar coordinates $\mathrm{d}s = \sqrt{r^2 + (\mathrm{d}r/\mathrm{d}\theta)^2}\,\mathrm{d}\theta$. Here $\mathrm{d}r/\mathrm{d}\theta = -4\sin\theta$, so

$$r^2 + \left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 = 16\left(1 + 2\cos\theta + \cos^2\theta + \sin^2\theta\right) = 32(1+\cos\theta) = 64\cos^2\tfrac{\theta}{2}.$$

Ask yourself Can we just write $\sqrt{64\cos^2(\theta/2)} = 8\cos(\theta/2)$?

Only because $\cos\frac{\theta}{2} \geq 0$ throughout: for $|\theta| \leq \frac{2\pi}{3}$ we have $|\frac{\theta}{2}| \leq \frac{\pi}{3}$. Using symmetry about the x-axis,

$$L_{\text{cardioid}} = 2\int_0^{2\pi/3} 8\cos\tfrac{\theta}{2}\,\mathrm{d}\theta = 32\left[\sin\tfrac{\theta}{2}\right]_0^{2\pi/3} = 32\cdot\tfrac{\sqrt{3}}{2} = 16\sqrt{3}.$$

Parabola arc. Use y as the parameter: $x = \frac{1}{2}(1 - y^2)$, so $\mathrm{d}x/\mathrm{d}y = -y$ and $\mathrm{d}s = \sqrt{1+y^2}\,\mathrm{d}y$, with y running from $-\sqrt{3}$ to $\sqrt{3}$. By symmetry,

$$L_{\text{parabola}} = 2\int_0^{\sqrt{3}} \sqrt{1+y^2}\,\mathrm{d}y.$$

With $y = \sinh u$, $\mathrm{d}y = \cosh u\,\mathrm{d}u$, the integrand becomes $\cosh^2 u = \frac{1}{2}(1 + \cosh 2u)$, so

$$\int \sqrt{1+y^2}\,\mathrm{d}y = \tfrac{1}{2}u + \tfrac{1}{4}\sinh 2u = \tfrac{1}{2}\left(\operatorname{arsinh} y + y\sqrt{1+y^2}\right) + C.$$

At $y = \sqrt{3}$, $\sqrt{1+y^2} = 2$ and $\operatorname{arsinh}\sqrt{3} = \ln\left(\sqrt{3}+2\right)$, so

$$L_{\text{parabola}} = \left[\operatorname{arsinh} y + y\sqrt{1+y^2}\right]_0^{\sqrt{3}} = \ln(2+\sqrt{3}) + 2\sqrt{3}.$$

Total. Adding,

$$L = 16\sqrt{3} + 2\sqrt{3} + \ln(2+\sqrt{3}) = 18\sqrt{3} + \ln(2+\sqrt{3}),$$

as required.

Checks

- The whole cardioid has length $\int_{-\pi}^{\pi} 8\cos\frac{\theta}{2}\,\mathrm{d}\theta = 32$, and the outer arc we used, $16\sqrt{3} \approx 27.71$, is most of it, which matches the sketch: the small loop near the cusp has length $32 - 16\sqrt{3} \approx 4.29$.
- The parabola arc, about 4.78, is a little longer than the two straight segments from B to the vertex $(\frac{1}{2}, 0)$ to A, of total length $2\sqrt{1.5^2 + 3} \approx 4.58$, as it should be.
- Numerically, midpoint-rule quadrature of the polar arc-length integrands for both curves gave a total of 32.49387, agreeing with $18\sqrt{3} + \ln(2+\sqrt{3}) \approx 32.49387$.

Common slips

- Taking the wrong part: $(1, 0)$ is outside the parabola (to the right of its vertex at $x = \frac{1}{2}$), so its boundary uses the large outer arc of the cardioid, not the loop near the cusp.
- Integrating the cardioid from 0 to 2π in one go with $8\cos\frac{\theta}{2}$, which goes negative for $\theta > \pi$; the integrand is $8\left|\cos\frac{\theta}{2}\right|$.

Beyond STEP

Both curves are conics or limaçons written with a focus or cusp at the pole, and $r = \ell/(1 + e\cos\theta)$ is the form in which conics appear in orbital mechanics, with the

Sun at the focus. The arc-length formula $\int \sqrt{r^2 + r'^2}\,\mathrm{d}\theta$ is a special case of the length of a parametrised curve, $\int |\mathbf{r}'(t)|\,\mathrm{d}t$, which in first-year vector calculus leads on to curvature and line integrals. Kleppner and Kolenkow's *An Introduction to Mechanics* uses the polar form of conics to derive Kepler's orbits.

WORKED SOLUTION

STEP II 1990, Question 10

Pure • Matrix Polynomial Equations

STEP II 1990, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q10

The key idea

Determinants turn a matrix product into a product of ordinary numbers, and a product of real numbers is zero only if one of the factors is. For the cubic, $\mathbf{M}$ commutes with $\mathbf{I}$, so a matrix polynomial in $\mathbf{M}$ factorises exactly like the polynomial in x. Upper triangular matrices then behave very well: the diagonal entries of a product are just the products of the diagonal entries, so the diagonal is quick to deal with and only the top-right corner needs real work.

Solution

Matrices with $\mathbf{AB} = \mathbf{0}$

Since the determinant of a product is the product of the determinants,

$$\det \mathbf{A} \det \mathbf{B} = \det(\mathbf{AB}) = \det \mathbf{0} = 0.$$

The left-hand side is a product of two real numbers, so at least one of them is zero: $\det \mathbf{A} = 0$, or $\det \mathbf{B} = 0$, or both. That is the required result.

If $\det \mathbf{B} \neq 0$ then $\mathbf{B}^{-1}$ exists, and multiplying $\mathbf{AB} = \mathbf{0}$ on the right by $\mathbf{B}^{-1}$ gives

$$\mathbf{A} = \mathbf{ABB}^{-1} = \mathbf{0}\,\mathbf{B}^{-1} = \mathbf{0}.$$

So $\mathbf{A}$ must be the zero matrix.

Both determinants being zero is not enough, though. Take

$$\mathbf{A} = \mathbf{B} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}.$$

Then $\det \mathbf{A} = \det \mathbf{B} = 0$, but $\mathbf{AB} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \neq \mathbf{0}$.

Factorising the cubic

First factorise the polynomial $f(x) = x^3 + 2x^2 - 5x - 6$. Trying small divisors of 6:

$$\begin{aligned} f(-1) &= -1 + 2 + 5 - 6 = 0, \\ f(2) &= 8 + 8 - 10 - 6 = 0, \\ f(-3) &= -27 + 18 + 15 - 6 = 0, \end{aligned}$$

so $f(x) = (x+1)(x-2)(x+3)$, since the leading coefficient is 1.

Now expand $(\mathbf{M} + p\mathbf{I})(\mathbf{M} + q\mathbf{I})(\mathbf{M} + r\mathbf{I})$. Because $\mathbf{I}$ commutes with $\mathbf{M}$ (and $\mathbf{I}^2 = \mathbf{I}$), the expansion is exactly the same as for numbers:

$$\mathbf{M}^3 + (p+q+r)\mathbf{M}^2 + (pq + qr + rp)\mathbf{M} + pqr\,\mathbf{I}.$$

Matching with the polynomial above, we can take

$$p = 1, \qquad q = -2, \qquad r = 3$$

(in any order), and then $\mathbf{M}^3 + 2\mathbf{M}^2 - 5\mathbf{M} - 6\mathbf{I} = (\mathbf{M} + \mathbf{I})(\mathbf{M} - 2\mathbf{I})(\mathbf{M} + 3\mathbf{I})$ for every square matrix $\mathbf{M}$.

> **Ask yourself** Can we now say that one of the three factors must be the zero matrix?

No. The first part shows that a product of matrices can vanish without any factor vanishing; all we learn is that at least one factor is singular. So we have to compute.

The upper triangular solutions

Let $\mathbf{M} = \begin{pmatrix} a & c \\ 0 & b \end{pmatrix}$. Each factor is upper triangular with the same top-right entry c:

$$\mathbf{M} + k\mathbf{I} = \begin{pmatrix} a+k & c \\ 0 & b+k \end{pmatrix}.$$

For three upper triangular matrices with diagonals (d_i, e_i) and top-right entries all equal to c, multiplying out gives

$$\begin{aligned} &\begin{pmatrix} d_1 & c \\ 0 & e_1 \end{pmatrix}\begin{pmatrix} d_2 & c \\ 0 & e_2 \end{pmatrix}\begin{pmatrix} d_3 & c \\ 0 & e_3 \end{pmatrix} \\ &\quad = \begin{pmatrix} d_1d_2d_3 & c\,(d_1d_2 + d_1e_3 + e_2e_3) \\ 0 & e_1e_2e_3 \end{pmatrix}. \end{aligned}$$

With $d_1, d_2, d_3 = a+1, a-2, a+3$ and $e_1, e_2, e_3 = b+1, b-2, b+3$, the equation becomes three conditions:

$$\begin{aligned} (a+1)(a-2)(a+3) &= 0, \\ (b+1)(b-2)(b+3) &= 0, \\ c\left[(a+1)(a-2) + (a+1)(b+3) + (b-2)(b+3)\right] &= 0. \end{aligned}$$

The first two say $a, b \in \{-1, 2, -3\}$. Expanding the bracket in the third,

$$(a^2 - a - 2) + (ab + 3a + b + 3) + (b^2 + b - 6) = a^2 + ab + b^2 + 2(a + b) - 5.$$

Ask yourself Is there a smarter way to see when this bracket is zero than trying all nine pairs?

Yes. If $a \neq b$, the bracket is

$$\frac{a^3 - b^3}{a - b} + 2\frac{a^2 - b^2}{a - b} - 5\frac{a - b}{a - b} = \frac{f(a) - f(b)}{a - b},$$

which is 0 because $f(a) = f(b) = 0$. So when $a \neq b$ the third condition holds whatever c is.

If $a = b$, the bracket is $3a^2 + 4a - 5 = f'(a)$. This is -6, 15 and 10 at $a = -1, 2, -3$ respectively (none is zero, since f has three distinct roots), so the third condition forces $c = 0$.

So the solutions are:

- $\mathbf{M} = \begin{pmatrix} a & c \\ 0 & b \end{pmatrix}$ with a and b *different* numbers from $\{-1, 2, -3\}$ and c *any* real number (six families);
- the three scalar matrices $-\mathbf{I}$, $2\mathbf{I}$ and $-3\mathbf{I}$ (when $a = b$, which forces $c = 0$).

Checks

Take $a = -1$, $b = 2$, $c = 5$. Then $\mathbf{M} + \mathbf{I} = \begin{pmatrix} 0 & 5 \\ 0 & 3 \end{pmatrix}$, $\mathbf{M} - 2\mathbf{I} = \begin{pmatrix} -3 & 5 \\ 0 & 0 \end{pmatrix}$ and $\mathbf{M} + 3\mathbf{I} = \begin{pmatrix} 2 & 5 \\ 0 & 5 \end{pmatrix}$. The product of the first two is $\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$ already, so the whole product vanishes for any c. On the other hand $\mathbf{M} = \begin{pmatrix} 2 & 1 \\ 0 & 2 \end{pmatrix}$ gives $\mathbf{M}^3 + 2\mathbf{M}^2 - 5\mathbf{M} - 6\mathbf{I} = \begin{pmatrix} 0 & 15 \\ 0 & 0 \end{pmatrix} \neq \mathbf{0}$, matching $f'(2) = 15$. A computer search over many rational a, b, c found exactly the solutions above, and the factorisation was checked on random 3×3 integer matrices.

Common slips

- Concluding from $(\mathbf{M} + \mathbf{I})(\mathbf{M} - 2\mathbf{I})(\mathbf{M} + 3\mathbf{I}) = \mathbf{0}$ that $\mathbf{M}$ must be $-\mathbf{I}$, $2\mathbf{I}$ or $-3\mathbf{I}$. The first part is there to warn you against exactly this.
- Forgetting the case $a = b$, or assuming c is free there too.
- Writing the example for the first part with $\mathbf{AB} = \mathbf{0}$ by accident; check that your product really is non-zero.

Beyond STEP

The answer is a first look at the *minimal polynomial*: a matrix satisfies a polynomial with distinct roots exactly when it is diagonalisable with eigenvalues among those roots. Our $a \neq b$ solutions are diagonalisable for every c, while $\begin{pmatrix} a & c \\ 0 & a \end{pmatrix}$ with $c \neq 0$ is not, which is why it fails. The divided difference $\big(f(a) - f(b)\big)/(a - b)$ in the corner, becoming $f'(a)$ when $a = b$, is how functions of matrices are defined in general. Strang's *Introduction to Linear Algebra* covers eigenvalues and diagonalisation clearly.

WORKED SOLUTION

STEP II 1990, Question 11

Mechanics • Conservation of Angular Momentum

STEP II 1990, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q11

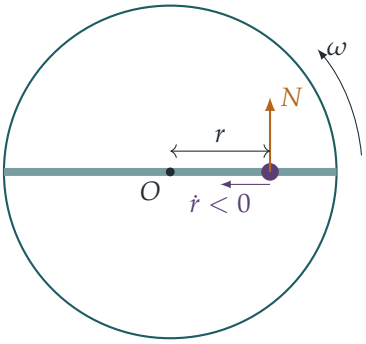
The key idea

Disc and particle together form a system with no external torque about the vertical axis and no external force doing work, so two quantities are conserved: angular momentum about the axis and kinetic energy. Those two equations give ω and $\mathrm{d}r/\mathrm{d}t$ directly. Dividing one rate by the other removes time and leaves a separable equation for the path, r as a function of θ.

Solution

The model

Treat the particle as a point mass in a smooth groove, and assume the axis is smooth (frictionless bearing). Seen from above, the particle at distance r along the groove has velocity $\dot{r}$ along the groove and $r\omega$ perpendicular to it, where $\omega = \dot{\theta}$.



The only horizontal force on the particle is the groove's reaction N, perpendicular to the groove (the groove is smooth, so there is no component along it). The disc feels $-N$.

Angular momentum

Gravity and the vertical reactions are parallel to the axis, and the smooth axis exerts no torque, so the total angular momentum about the axis is constant. The disc contributes $mk^2\omega$ and the particle $m\,r\cdot r\omega = mr^2\omega$. Comparing with $t=0$:

$$mk^2\omega + mr^2\omega = mk^2\Omega + ma^2\Omega \quad\Longrightarrow\quad \omega = \frac{\Omega(k^2+a^2)}{k^2+r^2}.$$

Energy

> **Ask yourself** The reaction N pushes the particle sideways while it moves sideways. Doesn't that change the kinetic energy?

It changes the particle's kinetic energy, but not the total. N does work on the particle at rate $N\cdot r\omega$, while the equal and opposite force on the disc, acting at the same distance r, exerts a torque $-Nr$ and so does work at rate $-Nr\omega$. These cancel, so the total kinetic energy is conserved:

$$\tfrac{1}{2}mk^2\omega^2 + \tfrac{1}{2}m\left(\dot r^2 + r^2\omega^2\right) = \tfrac{1}{2}mk^2\Omega^2 + \tfrac{1}{2}m\left(V^2 + a^2\Omega^2\right).$$

Hence

$$\dot r^2 = V^2 + (k^2+a^2)\Omega^2 - (k^2+r^2)\omega^2.$$

Using the given condition $V^2 = \Omega^2 a^2(k^2+a^2)/k^2$,

$$V^2 + (k^2+a^2)\Omega^2 = \Omega^2(k^2+a^2)\,\frac{a^2+k^2}{k^2} = \frac{\Omega^2(k^2+a^2)^2}{k^2},$$

and from the angular momentum result $(k^2+r^2)\omega^2 = \Omega^2(k^2+a^2)^2/(k^2+r^2)$. Therefore

$$\dot r^2 = \Omega^2(k^2+a^2)^2\left(\frac{1}{k^2} - \frac{1}{k^2+r^2}\right) = \frac{\Omega^2(k^2+a^2)^2\,r^2}{k^2(k^2+r^2)}.$$

While the particle moves towards the axis, $\dot r < 0$, so we take the negative square root (with $r>0$):

$$\frac{\mathrm{d}r}{\mathrm{d}t} = -\frac{\Omega r(k^2+a^2)}{k(k^2+r^2)^{1/2}},$$

as required.

The path

Since $\omega = \mathrm{d}\theta/\mathrm{d}t > 0$, the chain rule gives

$$\frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{\mathrm{d}r/\mathrm{d}t}{\mathrm{d}\theta/\mathrm{d}t} = -\frac{\Omega r(k^2+a^2)}{k(k^2+r^2)^{1/2}}\cdot\frac{k^2+r^2}{\Omega(k^2+a^2)} = -\frac{r(k^2+r^2)^{1/2}}{k},$$

that is, $k\,\mathrm{d}r/\mathrm{d}\theta = -r(k^2+r^2)^{1/2}$. Note that Ω and a have dropped out.

Now put $u = 1/r$, so $\mathrm{d}r/\mathrm{d}\theta = -u^{-2}\,\mathrm{d}u/\mathrm{d}\theta$ and $r(k^2+r^2)^{1/2} = u^{-2}(1+k^2u^2)^{1/2}$ (as $u > 0$). The equation becomes

$$k\frac{\mathrm{d}u}{\mathrm{d}\theta} = \sqrt{1+k^2u^2}.$$

The square root suggests a hyperbolic substitution: let $ku = \sinh\phi$. Then $k\,\mathrm{d}u/\mathrm{d}\theta = \cosh\phi\,\mathrm{d}\phi/\mathrm{d}\theta$ and $\sqrt{1+\sinh^2\phi} = \cosh\phi$, so

$$\frac{\mathrm{d}\phi}{\mathrm{d}\theta} = 1 \quad\Longrightarrow\quad \phi = \theta + \alpha.$$

At $\theta = 0$ we have $r = a$, so $\sinh\alpha = ku = k/a$. Hence $k/r = \sinh(\theta+\alpha)$, i.e.

$$r\sinh(\theta+\alpha) = k, \qquad \sinh\alpha = \frac{k}{a}.$$

The particle never reaches the axis

> **Ask yourself** $r = k/\sinh(\theta+\alpha)$ tends to 0 as $\theta \to \infty$. Could the disc turn through an infinite angle in a finite time?

No. Since $r^2 \geq 0$, $\omega = \Omega(k^2+a^2)/(k^2+r^2) \leq \Omega(k^2+a^2)/k^2$, so by time t the disc has turned through at most $\theta \leq \Omega(k^2+a^2)t/k^2$, which is finite. For any finite θ, $\sinh(\theta+\alpha)$ is finite, so $r = k/\sinh(\theta+\alpha) > 0$. Also $\dot r$ is strictly negative for every $r > 0$, so the particle never stops and turns back: the formulae stay valid for all t, and r decreases towards 0 without ever reaching it.

(Alternatively, straight from the $\mathrm{d}r/\mathrm{d}t$ equation: the time to go from a to r is $\frac{k}{\Omega(k^2+a^2)}\int_r^a \frac{\sqrt{k^2+x^2}}{x}\,\mathrm{d}x$, and near $x = 0$ the integrand behaves like k/x, whose integral diverges logarithmically as $r \to 0$.)

Checks

- At $t = 0$, $r = a$ gives $\omega = \Omega$ and $\dot r = -\Omega a\sqrt{k^2+a^2}/k = -V$, matching the initial data; and $\theta = 0$ gives $r = k/\sinh\alpha = a$.
- For large θ, $r \approx 2k\mathrm{e}^{-(\theta+\alpha)}$: the particle spirals in exponentially, consistent with never arriving.
- Numerically, stepping the equations of motion ($\ddot r = r\omega^2$ with angular momentum fixed) for random k, a, Ω reproduced $\dot r$, the energy and $r\sinh(\theta+\alpha) = k$ to about 10^{-12}. With $k = a = \Omega = 1$, the time to reach $r = 10^{-8}$ is about 9.32, growing like $\frac{1}{2}\ln(1/r)$.

Common slips

- Assuming the particle's energy alone is conserved: the groove's reaction does work on it, and only the total is conserved.
- Forgetting the transverse velocity $r\omega$ in the particle's kinetic energy or angular momentum.
- Concluding the particle never reaches the axis from $r > 0$ for finite θ, without checking that θ stays finite in finite time.

Beyond STEP

The special condition on V puts the particle exactly on the trajectory that creeps towards $r = 0$ for ever: in the rotating frame $r = 0$ is an unstable equilibrium, and this is its *stable manifold*. Any slightly larger V sends the particle through the centre, any smaller one makes it turn back. Ideas like this, along with conservation laws for coupled bodies, appear in Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1990, Question 12

Mechanics • Restitution and Projectiles

STEP II 1990, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

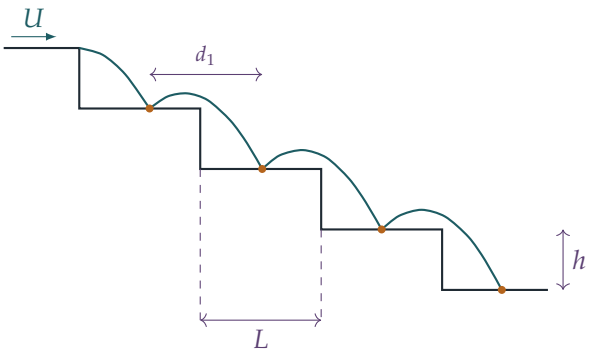
step.maths.org/questions/90-s2-q12

The key idea

The stairs are smooth, so the horizontal velocity never changes: it stays U throughout, and every horizontal distance is just U times a flight time. All the work is in the vertical motion, where each bounce scales the vertical speed by e and each drop of h adds $2gh$ to its square. That gives a simple recurrence for the *square* of the vertical speed at each bounce. For the last part, think about where the bounces must land: one per stair, on a staircase that goes on for ever.

Solution

We model the particle as a point moving under gravity alone between bounces, and the stairs as smooth horizontal surfaces, each a height h below the last, with edges a horizontal distance L apart.



The distance between bounces

Horizontal motion. The stairs are smooth, so each impulse from a stair is purely vertical, and between bounces the only force is gravity. So the horizontal velocity is U

at all times, and a flight lasting time T covers horizontal distance UT.

Vertical motion. Let w_n be the speed (downwards) with which the particle hits the stair at its nth bounce. It leaves the top stair with no vertical velocity and falls h, so

$$w_1^2 = 2gh.$$

At the nth bounce, Newton's law of restitution with a fixed stair sends it upwards at speed ew_n. It then has to end up h lower, on the next stair, so by $v^2 = u^2 + 2as$ (taking downwards as positive, the displacement is $+h$):

$$w_{n+1}^2 = e^2 w_n^2 + 2gh.$$

Starting from $w_1^2 = 2gh$ and applying this repeatedly,

$$\begin{aligned} w_n^2 &= 2gh\left(1 + e^2 + e^4 + \cdots + e^{2(n-1)}\right) \\ &= 2gh\,\frac{1 - e^{2n}}{1 - e^2} = 2gh\,\alpha_n^2, \end{aligned}$$

summing the geometric series (valid as $e^2 \neq 1$). So $w_n = \sqrt{2gh}\,\alpha_n$ for every $n \geq 1$; a quick induction makes this formal, since $e^2 \cdot 2gh\,\alpha_n^2 + 2gh = 2gh\left(e^2\alpha_n^2 + 1\right)$ and $e^2\alpha_n^2 + 1 = \dfrac{e^2 - e^{2n+2} + 1 - e^2}{1 - e^2} = \alpha_{n+1}^2$.

Ask yourself How long is the flight from the nth bounce to the $(n+1)$th?

The vertical velocity changes at the constant rate g (downwards). It starts at ew_n upwards and finishes at w_{n+1} downwards, so it changes by $ew_n + w_{n+1}$ in total, and the flight time is

$$T_n = \frac{ew_n + w_{n+1}}{g} = \frac{\sqrt{2gh}}{g}\left(e\alpha_n + \alpha_{n+1}\right) = \left(\frac{2h}{g}\right)^{\frac{1}{2}}\left(e\alpha_n + \alpha_{n+1}\right).$$

(This avoids solving a quadratic in T_n: we already know the final speed.) Multiplying by the horizontal speed U,

$$d_n = U\left(\frac{2h}{g}\right)^{\frac{1}{2}}\left(e\alpha_n + \alpha_{n+1}\right),$$

as required.

A very long staircase

Since $0 \leq e < 1$, we have $e^{2n} \to 0$, so $\alpha_n \to \dfrac{1}{\sqrt{1 - e^2}}$ as $n \to \infty$, and therefore

$$d_n \to D = U\left(\frac{2h}{g}\right)^{\frac{1}{2}} \frac{1 + e}{\sqrt{1 - e^2}} = U\left(\frac{2h}{g}\right)^{\frac{1}{2}}\left(\frac{1 + e}{1 - e}\right)^{\frac{1}{2}},$$

using $1 - e^2 = (1 - e)(1 + e)$.

Ask yourself Why must this limiting distance be exactly L?

Measure x horizontally from the edge of the top stair. The first stair below occupies $0 \le x \le L$, and in general the nth stair below the top occupies $(n-1)L \le x \le nL$. The particle bounces once on every stair, so its nth bounce, at x_n say, satisfies

$$(n-1)L \le x_n \le nL, \qquad \text{so} \qquad \frac{x_n}{n} \to L \text{ as } n \to \infty.$$

On the other hand, $x_n = x_1 + d_1 + d_2 + \cdots + d_{n-1}$, and $d_k \to D$. If $D > L$, then from some point on every d_k exceeds L by at least a fixed amount, so the particle gains a fixed distance on the stair edges with every bounce and eventually overshoots a stair; if $D < L$ it falls steadily behind and eventually lands twice on the same stair. (Formally, the average x_n/n of terms tending to D tends to D.) Either way, a long enough staircase is not bounced on once per stair. So for very large N we need $D = L$:

$$U\left(\frac{2h}{g}\right)^{\frac{1}{2}}\left(\frac{1+e}{1-e}\right)^{\frac{1}{2}} = L \quad \Longrightarrow \quad U = \left(\frac{L^2 g}{2h}\right)^{\frac{1}{2}}\left(\frac{1-e}{1+e}\right)^{\frac{1}{2}},$$

as required.

Checks

- *Special values.* $\alpha_1 = 1$, so $d_1 = U\sqrt{2h/g}\,(e + \sqrt{1+e^2})$. With $e = 0$ the particle stops dead vertically at each bounce, so each flight is a plain drop of h taking $\sqrt{2h/g}$: indeed $d_n = U\sqrt{2h/g}$, and the final condition becomes $U\sqrt{2h/g} = L$, exactly as it should.
- *Dimensions.* $\sqrt{2h/g}$ is a time, so d_n is a length; $\sqrt{L^2 g/h}$ is a speed.
- *Numerically.* Stepping the motion numerically for random values of U, h and e (with $g = 9.8$) reproduced each d_n to within 10^{-6}. With $e = \frac{1}{2}$, $L = 0.3$ and $h = 0.2$, the formula gives $U \approx 0.857$, and the bounce positions then stayed on successive stairs for 2000 stairs.

Common slips

- Taking every bounce to be a drop of h from rest, or from the height of the previous bounce: the particle rises after each bounce, so its vertical speed builds up from stair to stair.
- Writing $w_{n+1} = ew_n + \sqrt{2gh}$: it is the *squares* of the speeds that add, not the speeds themselves.

Beyond STEP

The last part is a statement about averages: if $d_n \to D$ then the running mean $\frac{1}{n}\sum d_k \to D$ too (Cesàro's theorem), which is a standard early exercise in real analysis. Notice too that the condition on U is *necessary*, not sufficient: the early bounces are shorter than L, and if e is close to 1 the accumulated shortfall can exceed the head start x_1, so no launch speed works at all. For limits of sequences done carefully, see Abbott, *Understanding Analysis*; for the mechanics of impacts, Kleppner and Kolenkow, *An Introduction to Mechanics*.

WORKED SOLUTION

STEP II 1990, Question 13

Mechanics • Equilibrium of a Rod

STEP II 1990, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q13

The key idea

Because the peg is smooth and the string light, the tension is the same all the way along the string, so the rod feels two *equal* pulls, along PO and along QO. Their resultant bisects the angle POQ, and for equilibrium it must balance the weight along the same vertical line. So the vertical through O is the angle bisector of POQ and passes through G, and the angle bisector theorem does the rest. The inequality is then just the statement that the triangle OPQ is a genuine triangle.

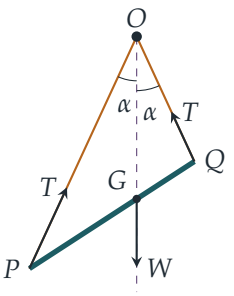
Solution

Finding OP and OQ

Model the rod as a rigid rod with centre of gravity G, where $PG = a + d$ and $GQ = a - d$; we take $0 \leq d < a$, so that G lies on the rod, on the Q side of the midpoint (or at it). The string is light and inextensible and the peg is small and smooth, so the tension T is the same in both parts OP and OQ, and

$$OP + OQ = 2b.$$

The rod is acted on by just three forces: its weight W at G, a tension T at P along PO, and a tension T at Q along QO.



The lines of action of the two tensions meet at O, so (taking moments about O) the weight must also act through O: the vertical through O passes through G. Moreover, the two tensions have equal magnitude, so their resultant points along the sum of two unit vectors, one along PO and one along QO, which is the direction of the bisector of the angle between them. This resultant must be vertical to balance W. Hence the vertical line OG bisects the angle POQ; call each half α.

> **Ask yourself** What does knowing that OG bisects the angle POQ tell us about the lengths?

By the angle bisector theorem in triangle OPQ, the bisector from O divides PQ in the ratio of the adjacent sides:

$$\frac{OP}{OQ} = \frac{PG}{GQ} = \frac{a+d}{a-d}.$$

(To see this directly: triangles OPG and OQG share the height from O, so their areas are in the ratio $PG : GQ$; they also have areas $\frac{1}{2}\,OP \cdot OG \sin\alpha$ and $\frac{1}{2}\,OQ \cdot OG \sin\alpha$.) So $OP = k(a+d)$ and $OQ = k(a-d)$ for some $k > 0$, and $OP + OQ = 2b$ gives $2ka = 2b$, so $k = b/a$:

$$OP = \frac{b(a+d)}{a}, \qquad OQ = \frac{b(a-d)}{a}.$$

The inequality

Let $\theta = \angle POQ = 2\alpha$. Because the rod is not vertical, O is not on the line PQ, so OPQ is a genuine triangle and $0 < \theta < \pi$. The cosine rule with $PQ = 2a$ gives

$$\begin{aligned} 4a^2 &= OP^2 + OQ^2 - 2\,OP \cdot OQ \cos\theta \\ &= \frac{b^2}{a^2}\Big[(a+d)^2 + (a-d)^2 - 2(a^2-d^2)\cos\theta\Big] \\ &= \frac{2b^2}{a^2}\Big[(a^2+d^2) - (a^2-d^2)\cos\theta\Big]. \end{aligned}$$

Now $\cos\theta < 1$ because $\theta \neq 0$, and $a^2 - d^2 > 0$ because $d < a$. So

$$4a^2 > \frac{2b^2}{a^2}\Big[(a^2+d^2) - (a^2-d^2)\Big] = \frac{4b^2d^2}{a^2},$$

that is, $b^2d^2 < a^4$. Taking positive square roots (with $b > 0$, $d \geq 0$), $bd < a^2$, so

$$d < \frac{a^2}{b},$$

as required.

Equivalently, this is the triangle inequality $OP - OQ < PQ$: here $OP - OQ = 2bd/a$, and $2bd/a < 2a$ is the same condition. (The other inequality, $\cos\theta > -1$, gives $OP + OQ > PQ$, i.e. $b > a$, which just says that the string is long enough to go over the peg at all.)

Checks

- If $d = 0$ (uniform rod), $OP = OQ = b$ and the rod hangs horizontally, as symmetry demands; the inequality $0 < a^2/b$ holds automatically.
- As $d \to a^2/b$, $OP - OQ \to 2a = PQ$, so the triangle flattens and the rod becomes vertical: beyond this value the only way to hang is vertically.
- With $a = 1$, $b = 2$, $d = 0.3$: $OP = 2.6$, $OQ = 1.4$, $\cos\theta \approx 0.648$. Placing this triangle so that OG is vertical, the horizontal components of the two tensions cancel and the moments about O balance to machine precision. A scan over all positions of the string confirmed this is the only non-vertical equilibrium, and that none exists once $d \geq a^2/b$.

Common slips

- Assuming the tensions in the two parts of the string differ. Over a smooth peg they are equal, and that is what makes OG the angle bisector.
- Using the angle bisector ratio the wrong way round: the longer string, OP, goes to the end further from G.
- Expecting this equilibrium to be stable. In fact it is unstable: with $d = 0$ the horizontal rod has G a distance $\sqrt{b^2 - a^2}$ below O, while hanging vertically puts it a distance b below, which is lower. The question asks only about equilibrium.

Beyond STEP

The locus of O relative to the rod is an ellipse with foci P and Q, and the equilibrium positions are where the height of G is stationary. That is the principle of virtual work, which leads on to Lagrangian mechanics. It also explains why equal tensions bisect the angle: it is the reflection property of the ellipse. Kleppner and Kolenkow, *An Introduction to Mechanics*, is a good next step for statics and stability.

WORKED SOLUTION

STEP II 1990, Question 14

Mechanics • Elastic Collisions

STEP II 1990, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q14

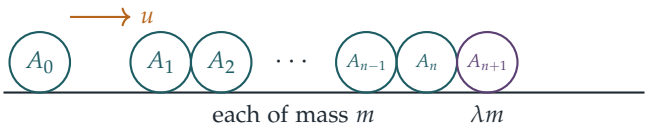
This question has a full guided treatment in *Beyond STEP*, Section 10.1, *Newton's Cradle*: https://beyondstep.keithmcnulty.org/ch10/newtons-cradle.html

The key idea

Without knowing how the impulses pass along the row, we can't follow the collisions one by one. What we can do is test each proposed final motion against three conditions: momentum is conserved, kinetic energy is conserved, and the marbles never pass through one another. The third condition is easy to forget, but it is what decides each part. A marble with a stationary marble directly behind it cannot move backwards, and a marble cannot be moving faster than the marble directly in front of it.

Solution

We model the marbles as particles on a smooth line (the trough), with velocities measured in the direction of A_0's original motion. Before the impact the total momentum is mu and the total kinetic energy is $\frac{1}{2}mu^2$, and both are conserved.



Two facts about any possible final motion follow because the marbles cannot overlap.

- If A_k is at rest and A_{k+1} is directly in front of it, touching it, then A_{k+1} cannot have negative velocity.
- If A_k and A_{k+1} are touching, then A_k's velocity cannot exceed A_{k+1}'s.

Part (i)

Suppose only A_n and A_{n+1} move, with velocities v and w. Dividing through by m,

$$v + \lambda w = u,$$
$$v^2 + \lambda w^2 = u^2.$$

Substituting $v = u - \lambda w$ into the energy equation gives

$$u^2 - 2\lambda uw + \lambda^2 w^2 + \lambda w^2 = u^2 \quad \Longrightarrow \quad \lambda w\big((1+\lambda)w - 2u\big) = 0.$$

The root $w = 0$ gives $v = u$. Then A_n would move forwards into the stationary A_{n+1}, which is impossible. The other root is

$$w = \frac{2u}{1+\lambda}, \qquad v = u - \lambda w = \frac{(1-\lambda)u}{1+\lambda}.$$

If $\lambda < 1$: then $v > 0$, so A_n moves away from the stationary A_{n-1}. Also

$$w - v = \frac{(1+\lambda)u}{1+\lambda} = u > 0,$$

so A_{n+1} moves away from A_n. Both conservation laws hold and nothing overlaps, so this motion is possible, with $A_0, \ldots, A_{n-1}$ all at rest.

If $\lambda > 1$: then $v < 0$, so A_n would have to move backwards. But A_{n-1} is directly behind it, touching it and at rest. So the only motion that satisfies both conservation laws is physically impossible, and no motion of this kind exists when $\lambda > 1$.

Part (ii)

Suppose only A_{n-1}, A_n and A_{n+1} move, with velocities a, b and c. Since $n \geq 3$, the marble A_{n-2} exists, and it is at rest and touching A_{n-1} from behind. So $a \geq 0$. Since the three moving marbles cannot overtake one another, $a \leq b \leq c$. Hence a, b and c are all non-negative.

> **Ask yourself** Both conservation laws give equations in a, b, c. How can we compare them without solving?

We square the momentum equation and subtract the energy equation. Conservation gives $a + b + \lambda c = u$ and $a^2 + b^2 + \lambda c^2 = u^2$, so

$$(a + b + \lambda c)^2 - (a^2 + b^2 + \lambda c^2) = u^2 - u^2 = 0.$$

Expanding the left-hand side,

$$\lambda(\lambda - 1)c^2 + 2ab + 2\lambda c(a + b) = 0.$$

When $\lambda > 1$ and $a, b, c \geq 0$, every term on the left is non-negative, so every term must be zero. In particular $\lambda(\lambda - 1)c^2 = 0$, which gives $c = 0$. Then $0 \leq a \leq b \leq c = 0$, so all three marbles are at rest. That contradicts the momentum equation, since $u > 0$. Hence no such motion exists when $\lambda > 1$.

Part (iii)

The argument in (ii) applies to any set of moving marbles: if every velocity is non-negative and A_{n+1} moves, then the kinetic energy is too small. So when $\lambda > 1$ some marble must move backwards. Every marble except A_0 has a marble directly behind it, so A_0 is the natural candidate for a marble that bounces back.

Ask yourself Which two marbles should we try?

Try A_0 and A_{n+1}, with velocities p and q, and $A_1, \ldots, A_n$ left at rest. These are the same two equations as in (i), with p, q in place of v, w, and the non-trivial solution is

$$p = \frac{(1-\lambda)u}{1+\lambda} = -\frac{(\lambda-1)u}{\lambda+1}, \qquad q = \frac{2u}{1+\lambda}.$$

Momentum and energy are conserved, as in (i). The motion is also physically possible:

- $p < 0$, so A_0 moves backwards away from A_1, and there is nothing behind A_0.
- $q > 0$, so A_{n+1} moves away from the stationary A_n.

So one possible motion is that A_0 rebounds with speed $\dfrac{(\lambda-1)u}{\lambda+1}$, A_{n+1} moves off with speed $\dfrac{2u}{\lambda+1}$, and $A_1, \ldots, A_n$ stay at rest.

This is also what happens if we model the impact as a sequence of separate collisions between neighbouring marbles. Each pair of equal marbles swaps velocities, so the speed u is passed along to A_n. When A_n hits A_{n+1} it rebounds with velocity p, and that backward velocity is then passed back along the row to A_0.

Checks

- When $\lambda = 1$ all the marbles are identical, and both answers become A_{n+1} moving off with speed u while everything else is at rest, which is the familiar "Newton's cradle".
- With $\lambda = 2$ and $u = 3$, part (iii) gives $p = -1$ and $q = 2$. Momentum: $-1 + 2 \times 2 = 3$. Energy: $1 + 2 \times 4 = 9 = 3^2$.
- We stepped through the sequence of pairwise elastic collisions numerically for $n = 3$ and $n = 6$. With $\lambda = 2$ the final velocities were $-\frac{1}{3}u$ and $\frac{2}{3}u$, and with $\lambda = 4$ they were $-0.6u$ and $0.4u$, exactly as predicted. With $\lambda = \frac{1}{2}$ the simulation ends with A_n and A_{n+1} moving at $\frac{1}{3}u$ and $\frac{4}{3}u$, matching (i).

Common slips

- Solving the two conservation equations and stopping there. Each part depends on whether the solution is physically possible, that is, on the signs of the velocities and whether marbles would pass through each other.

- Keeping the root $w = 0$, $v = u$ in (i). That root just describes the motion before the collision.
- Forgetting that $n \geq 3$ is what guarantees there is a stationary A_{n-2} behind A_{n-1} in (ii).

Beyond STEP

For a row of touching marbles, the two conservation laws alone don't fix the outcome. This question is really about which of the many conserved-quantity solutions are physically possible. Choosing between them needs a more detailed model of the contacts, such as a sequence of pairwise impacts or the marbles treated as stiff springs. Modelling the marbles as springs leads to the theory of waves in chains of particles. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers collisions and conservation laws in this spirit.

WORKED SOLUTION

STEP II 1990, Question 15

Probability and Statistics • Order Statistics

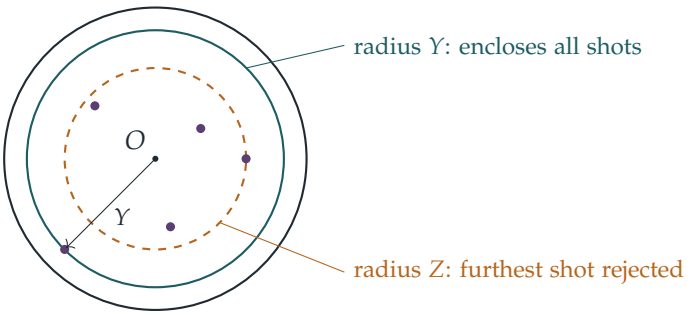
STEP II 1990, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q15

The key idea

The smallest circle centred at O enclosing a set of shots has radius equal to the distance of the *furthest* shot, so Y is a maximum of independent distances. A maximum is at most y exactly when *every* shot is within y, which turns the cumulative distribution function into a product. After rejecting the furthest shot, the radius we want is the *second* largest distance, and its distribution function comes from counting how many shots may lie outside a given circle.



Solution

Let $R_1, \ldots, R_n$ be the distances of the n shots from O. They are independent (the shots are fired independently), and each has distribution function

$$P(R_i \leq t) = t^2, \qquad 0 \leq t \leq 1.$$

The distribution of Y

The smallest circle centred at O containing every shot has radius $Y = \max(R_1, \ldots, R_n)$. For $0 \le y \le 1$,

$$P(Y \le y) = P(R_1 \le y, \ldots, R_n \le y) = \prod_{i=1}^{n} P(R_i \le y) = (y^2)^n = y^{2n},$$

using independence for the middle step. Differentiating, the probability density function of Y is

$$f_Y(y) = \frac{\mathrm{d}}{\mathrm{d}y} y^{2n} = 2ny^{2n-1}, \qquad 0 \le y \le 1,$$

as required. (It is zero outside $[0, 1]$, and it integrates to 1, as it should.)

The area of the circle is πY^2, so

$$E(\pi Y^2) = \pi \int_0^1 y^2 \cdot 2ny^{2n-1}\,\mathrm{d}y = 2n\pi \int_0^1 y^{2n+1}\,\mathrm{d}y = \frac{2n\pi}{2n+2} = \frac{n\pi}{n+1}.$$

Rejecting the furthest shot

We need $n \ge 2$ so that some shots remain. Once the furthest shot is removed, the smallest enclosing circle centred at O has radius Z, the second largest of $R_1, \ldots, R_n$.

> **Ask yourself** How can we describe the event $Z \le z$ in terms of how many shots land outside the circle of radius z?

The second largest distance is at most z exactly when *at most one* shot lies further than z from O. Each shot independently lies within z with probability z^2, so the number of shots outside the circle of radius z is binomial, $\mathrm{B}(n, 1 - z^2)$. Hence, for $0 \le z \le 1$,

$$\begin{aligned} P(Z \le z) &= P(\text{no shot outside}) + P(\text{exactly one shot outside}) \\ &= z^{2n} + n(1 - z^2)z^{2(n-1)} \\ &= nz^{2n-2} - (n-1)z^{2n}. \end{aligned}$$

Differentiating gives the density of Z:

$$f_Z(z) = 2n(n-1)z^{2n-3} - 2n(n-1)z^{2n-1} = 2n(n-1)z^{2n-3}(1 - z^2), \qquad 0 \le z \le 1.$$

The expected area of the new circle is therefore

$$\begin{aligned} E(\pi Z^2) &= 2n(n-1)\pi \int_0^1 \left(z^{2n-1} - z^{2n+1}\right)\mathrm{d}z \\ &= 2n(n-1)\pi \left(\frac{1}{2n} - \frac{1}{2n+2}\right) \\ &= 2n(n-1)\pi \cdot \frac{2}{2n(2n+2)} = \left(\frac{n-1}{n+1}\right)\pi, \end{aligned}$$

which is the required result.

Checks

- With $n = 1$ the first answer is $\pi/2$: a single shot has R^2 uniform on $[0,1]$ (since $P(R^2 \leq u) = u$), so the expected area πR^2 is $\pi/2$.
- With $n = 2$ the second answer is $\pi/3$, the expected area for the *nearer* of two shots; together with the first answer, $2\pi/3$, the two average to $\pi/2$, the single-shot value, as they must.
- A simulation with 2×10^5 trials and $n = 5$ gave expected areas 2.618 and 2.095, against $5\pi/6 \approx 2.618$ and $4\pi/6 \approx 2.094$.

Common slips

- Writing $P(Z \leq z) = z^{2(n-1)}$, as if the remaining shots were a fresh sample of $n-1$; the rejected shot is not random, so the survivors are biased towards O.
- Forgetting the factor n for which shot is the one outside the circle.
- Finding $E(Y)$ and then squaring it: the expected area is $\pi E(Y^2)$, not $\pi\big(E(Y)\big)^2$.

Beyond STEP

These are *order statistics*. Because $U_i = R_i^2$ is uniform on $[0,1]$, the areas are just the order statistics of n uniforms, and the kth smallest of n independent uniforms has a beta distribution with mean $k/(n+1)$; both answers here are the cases $k = n$ and $k = n-1$. The neat picture is that n random points cut $[0,1]$ into $n+1$ gaps of equal expected length. Grimmett and Welsh, *Probability: An Introduction*, covers transformations and order statistics of this kind.

WORKED SOLUTION

STEP II 1990, Question 16

Probability and Statistics • Bayes' Theorem

STEP II 1990, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s2-q16

The key idea

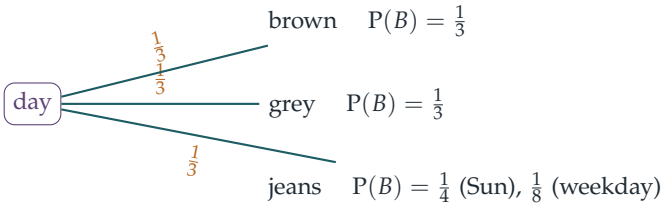
This is Bayes' theorem in disguise. The photograph tells us the black tie was worn, and we want the probability of the day *given* that. So we work forwards first: for each kind of day, find the probability of wearing the black tie, using a tree split by trousers. Then we turn the conditional probability round. The Monday rule only changes one branch of the tree, so the second part needs very little extra work.

Solution

Let S be the event that the photograph was taken on a Sunday, and B the event that I am wearing the black tie. The photograph is equally likely to come from any of the seven days, so before we look at it $\mathrm{P}(S) = \frac{1}{7}$. On every day except (later) Monday, each pair of trousers has probability $\frac{1}{3}$.

The chance of the black tie on each kind of day

- **Brown trousers.** The cravat is ruled out, so the tie is one of black, polka-dot or regimental, chosen at random: $\mathrm{P}(B) = \frac{1}{3}$.
- **Grey trousers.** The polka-dot tie is ruled out, so the choice is from black, regimental or cravat: $\mathrm{P}(B) = \frac{1}{3}$.
- **Jeans on Sunday.** A tie is always worn, picked from all four: $\mathrm{P}(B) = \frac{1}{4}$.
- **Jeans on a weekday.** With probability $\frac{1}{2}$ it's the sweat-shirt and no tie. Otherwise it's one of the four ties: $\mathrm{P}(B) = \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8}$.



Adding along the branches (the law of total probability over the trousers),

$$\mathrm{P}(B \mid \text{Sunday}) = \tfrac{1}{3}\left(\tfrac{1}{3} + \tfrac{1}{3} + \tfrac{1}{4}\right) = \tfrac{1}{3} \cdot \tfrac{11}{12} = \tfrac{11}{36} = \tfrac{22}{72},$$
$$\mathrm{P}(B \mid \text{weekday}) = \tfrac{1}{3}\left(\tfrac{1}{3} + \tfrac{1}{3} + \tfrac{1}{8}\right) = \tfrac{1}{3} \cdot \tfrac{19}{24} = \tfrac{19}{72}.$$

The first answer

By Bayes' theorem, with prior $\frac{1}{7}$ for Sunday and $\frac{6}{7}$ for the six weekdays together,

$$\mathrm{P}(S \mid B) = \frac{\frac{1}{7} \cdot \frac{22}{72}}{\frac{1}{7} \cdot \frac{22}{72} + \frac{6}{7} \cdot \frac{19}{72}} = \frac{22}{22 + 6 \times 19} = \frac{22}{136} = \frac{11}{68},$$

as required. (The common factor $\frac{1}{7 \times 72}$ cancels, which is why it's worth putting everything over 72.)

? **Ask yourself** Why isn't the answer just $\mathrm{P}(B \mid \text{Sunday})$, or $\frac{1}{7}$?

$\mathrm{P}(B \mid \text{Sunday}) = \frac{11}{36}$ answers the wrong question: it's the chance of the tie given the day, not the chance of the day given the tie. And $\frac{1}{7}$ ignores the evidence. The black tie is slightly more likely on a Sunday ($\frac{22}{72}$ against $\frac{19}{72}$), because there's no tieless option then, so the photograph nudges the probability of Sunday up from $\frac{1}{7} = \frac{11}{77}$ to $\frac{11}{68}$.

With the Monday rule

Now on Monday the jeans are certain, and Monday is still a weekday, so the sweat-shirt option still applies:

$$\mathrm{P}(B \mid \text{Monday}) = 1 \cdot \tfrac{1}{2} \cdot \tfrac{1}{4} = \tfrac{1}{8} = \tfrac{9}{72}.$$

Sunday and the other five weekdays are unchanged. Each day still has prior probability $\frac{1}{7}$, which cancels, so

$$\mathrm{P}(S \mid B) = \frac{\frac{22}{72}}{\frac{22}{72} + \frac{9}{72} + 5 \cdot \frac{19}{72}} = \frac{22}{22 + 9 + 95} = \frac{22}{126} = \frac{11}{63}.$$

This is a little larger than $\frac{11}{68}$. That makes sense: on Monday the black tie becomes less likely ($\frac{1}{8}$ rather than $\frac{19}{72}$), so a black-tie photograph points away from Monday and slightly more towards Sunday.

Checks

- The tie probabilities for each day, together with "no tie", add to 1. For example, on a weekday the chance of no tie is $\frac{1}{3} \cdot \frac{1}{2} = \frac{1}{6}$.
- A simulation of 4×10^5 random days gives the proportion of black-tie days that are Sundays as about 0.161 (exact $\frac{11}{68} \approx 0.1618$), and about 0.174 with the Monday rule (exact $\frac{11}{63} \approx 0.1746$).

Common slips

- Forgetting that on Monday the jeans still come with the half-chance of the sweat-shirt: Monday is a weekday. Using $\frac{1}{4}$ instead of $\frac{1}{8}$ for Monday gives $\frac{22}{135}$.
- Choosing among all four ties with the brown or grey trousers. Each of those leaves only three ties.

Beyond STEP

Turning "probability of the evidence given the hypothesis" into "probability of the hypothesis given the evidence" is the heart of Bayesian inference. A neat way to say it: posterior odds equal prior odds times the likelihood ratio. Here the prior odds of Sunday against a weekday are $1 : 6$ and the likelihood ratio is $22 : 19$, which gives $22 : 114$, i.e. $11 : 57$. University courses build this up into Bayesian statistics, where whole distributions are updated in the same way. Blitzstein and Hwang, *Introduction to Probability*, has an excellent chapter on conditional probability with many examples like this one.

WORKED SOLUTION

STEP III 1990, Question 1

Pure • Multiple Angles and Polynomial Roots

STEP III 1990, Question 1

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q1

The key idea

Nine is 3×3, so we can get $\tan 9\theta$ by using the triple-angle formula for tangent twice. This is much less work than expanding $(\cos\theta + \mathrm{i}\sin\theta)^9$. The factorised form also tells us which factor vanishes at $\theta = \pi/9,\ 2\pi/9,\ 4\pi/9$: these are exactly the angles where $\tan^2 3\theta = 3$. That gives a cubic in $u = \tan^2\theta$, and Vieta's formulas (the links between a polynomial's roots and its coefficients) do the rest.

Solution

The formula for $\tan 9\theta$

Write $t = \tan\theta$ and $T = \tan 3\theta$. The triple-angle formula, which you can get from $\tan(2\theta + \theta)$ or from de Moivre with $n = 3$, gives

$$T = \frac{3t - t^3}{1 - 3t^2} = \frac{t(t^2 - 3)}{3t^2 - 1}, \qquad \tan 9\theta = \frac{3T - T^3}{1 - 3T^2} = \frac{T(3 - T^2)}{1 - 3T^2}.$$

We now write $3 - T^2$ and $1 - 3T^2$ over the common denominator $(1 - 3t^2)^2$:

$$\begin{aligned}(1 - 3t^2)^2(3 - T^2) &= 3(1 - 3t^2)^2 - t^2(3 - t^2)^2 \\ &= (3 - 18t^2 + 27t^4) - (9t^2 - 6t^4 + t^6) \\ &= -(t^6 - 33t^4 + 27t^2 - 3),\end{aligned}$$

$$\begin{aligned}(1 - 3t^2)^2(1 - 3T^2) &= (1 - 3t^2)^2 - 3t^2(3 - t^2)^2 \\ &= (1 - 6t^2 + 9t^4) - (27t^2 - 18t^4 + 3t^6) \\ &= -(3t^6 - 27t^4 + 33t^2 - 1).\end{aligned}$$

The factors $-(1 - 3t^2)^{-2}$ cancel in the ratio, so

$$\tan 9\theta = \frac{t(t^2 - 3)}{3t^2 - 1} \cdot \frac{t^6 - 33t^4 + 27t^2 - 3}{3t^6 - 27t^4 + 33t^2 - 1},$$

which is the required result.

Using de Moivre instead, $\tan 9\theta = \operatorname{Im}(1+it)^9 / \operatorname{Re}(1+it)^9$. One then has to check that

$$9t - 84t^3 + 126t^5 - 36t^7 + t^9 \quad \text{and} \quad 1 - 36t^2 + 126t^4 - 84t^6 + 9t^8$$

factorise as the numerator and the denominator above. They do, but the route through $\tan 3\theta$ hands us the factorisation for free.

The cubic

Let $P(t) = t^6 - 33t^4 + 27t^2 - 3$. The first calculation above says

$$3 - \tan^2 3\theta = -\frac{P(t)}{(1-3t^2)^2} \qquad \text{whenever } 3t^2 \neq 1.$$

Now take $\theta = \pi/9,\ 2\pi/9,\ 4\pi/9$. Then $3\theta = \pi/3,\ 2\pi/3,\ 4\pi/3$, and in each case $\tan 3\theta = \pm\sqrt{3}$, so $\tan^2 3\theta = 3$. In each case 3θ is not an odd multiple of $\pi/6$, so $3t^2 \neq 1$ (otherwise $\tan 3\theta$ would be undefined). Hence $P(\tan\theta) = 0$ for all three angles.

These are exactly the angles in $(0, \pi/2)$ at which $\tan 9\theta = 0$ through the factor P rather than through t or $t^2 - 3$. Indeed, $t \neq 0$, and $t^2 \neq 3$ because $\theta \neq \pi/3$.

> **Ask yourself** How do we know that these three numbers are *all* the roots of the cubic, with none repeated?

Put $u = t^2$. Each of $\tan^2(\pi/9)$, $\tan^2(2\pi/9)$, $\tan^2(4\pi/9)$ is a root of

$$u^3 - 33u^2 + 27u - 3 = 0.$$

The three values are distinct because tan is positive and strictly increasing on $(0, \pi/2)$, and $\pi/9 < 2\pi/9 < 4\pi/9 < \pi/2$. A cubic has at most three roots, so these are precisely its roots. This is the required cubic with integer coefficients.

The product

By Vieta's formulas, the product of the roots of $u^3 - 33u^2 + 27u - 3$ is 3:

$$\tan^2\left(\tfrac{\pi}{9}\right)\tan^2\left(\tfrac{2\pi}{9}\right)\tan^2\left(\tfrac{4\pi}{9}\right) = 3.$$

All three angles lie in $(0, \pi/2)$, so each tangent is positive, and we take the positive square root:

$$\tan\left(\tfrac{\pi}{9}\right)\tan\left(\tfrac{2\pi}{9}\right)\tan\left(\tfrac{4\pi}{9}\right) = \sqrt{3}.$$

The sum of sixth powers

Let the roots be u_1, u_2, u_3. From the cubic,

$$e_1 = u_1 + u_2 + u_3 = 33, \qquad e_2 = u_1u_2 + u_2u_3 + u_3u_1 = 27, \qquad e_3 = u_1u_2u_3 = 3.$$

We want $u_1^3 + u_2^3 + u_3^3$, which is the sum of the sixth powers of the tangents.

Ask yourself How do we get a sum of cubes from e_1, e_2, e_3 without finding the roots?

Each root satisfies $u_i^3 = 33u_i^2 - 27u_i + 3$. Adding these three equations gives

$$\sum u_i^3 = 33\sum u_i^2 - 27\sum u_i + 9.$$

Here $\sum u_i = 33$, and

$$\sum u_i^2 = e_1^2 - 2e_2 = 1089 - 54 = 1035.$$

Therefore

$$\begin{aligned}\sum u_i^3 &= 33 \times 1035 - 27 \times 33 + 9\\ &= 34155 - 891 + 9\\ &= 33273.\end{aligned}$$

That is,

$$\tan^6\left(\tfrac{\pi}{9}\right) + \tan^6\left(\tfrac{2\pi}{9}\right) + \tan^6\left(\tfrac{4\pi}{9}\right) = 33273,$$

as required. Equivalently, $\sum u_i^3 = e_1^3 - 3e_1e_2 + 3e_3 = 35937 - 2673 + 9$.

Checks

- Numerically, $\tan^2(\pi/9) \approx 0.1325$, $\tan^2(2\pi/9) \approx 0.7041$ and $\tan^2(4\pi/9) \approx 32.163$. Their sum is 33.000, and their product is 3.000.
- The product of the three tangents is $1.7320508\ldots = \sqrt{3}$.
- The sum of the cubes of the three values above is 33273.0. Almost all of it comes from $\tan^6(4\pi/9) \approx 33272.4$.
- The formula for $\tan 9\theta$ agrees with a calculator to about twelve significant figures at thousands of random values of θ.
- Small case: put $t = 1$, so $\theta = \pi/4$. The formula gives $\frac{1\cdot(-2)(-8)}{2\cdot 8} = 1 = \tan(9\pi/4)$.

Common slips

- Forgetting to check that $3t^2 \neq 1$ (or that the denominator is non-zero) before concluding that the sextic factor vanishes.
- Taking $-\sqrt{3}$ for the product. Every angle is acute, so every tangent is positive.
- Writing $\sum u_i^3 = e_1^3 - 3e_1e_2$ without the $+3e_3$ term, which gives 33264.

Beyond STEP

The cubic $u^3 - 33u^2 + 27u - 3$ has no rational roots, so it is irreducible over $\mathbb{Q}$. It also has three real roots. Galois theory shows that the roots of such a cubic cannot be written

using real radicals alone (the *casus irreducibilis*). That is why trigonometric expressions like $\tan^2(\pi/9)$ are the natural way to write them down. Sums of powers of roots are governed in general by Newton's identities. For the geometric side of de Moivre's theorem, read Needham, *Visual Complex Analysis*. For the algebra, read Allenby, *Rings, Fields and Groups*.

WORKED SOLUTION

STEP III 1990, Question 2

Pure • Vectors in Three Dimensions

STEP III 1990, Question 2

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q2

The key idea

All four points lie on the single curve $t \mapsto (t^3, t^2, t)$, the *twisted cubic*. Two lines that meet lie in a common plane, so part (i) reduces to showing that O, A, B, C are never coplanar, which is a determinant that factorises like a Vandermonde determinant. In part (ii), the condition $ab = 1$ makes the scalar product and the lengths collapse into a function of the single quantity $a^2 + 1/a^2$.

Solution

First, what does "distinct" tell us? The third coordinate of the point with parameter t is t itself, so different points have different parameters. Hence a, b, c are distinct, and since O is the point with $t = 0$, none of a, b, c is zero.

Part (i)

Suppose the lines OA and BC did meet. Two lines through a common point lie in a common plane (if they happen to coincide, any plane containing them will do). So O, A, B, C would all lie in one plane through the origin, and the position vectors $\mathbf{a} = \overrightarrow{OA}$, $\mathbf{b} = \overrightarrow{OB}$, $\mathbf{c} = \overrightarrow{OC}$ would be linearly dependent. That happens exactly when

$$D = \begin{vmatrix} a^3 & a^2 & a \\ b^3 & b^2 & b \\ c^3 & c^2 & c \end{vmatrix} = 0.$$

Taking the factor a out of the first row, b out of the second and c out of the third,

$$D = abc \begin{vmatrix} a^2 & a & 1 \\ b^2 & b & 1 \\ c^2 & c & 1 \end{vmatrix}.$$

Subtract the third row from each of the first two, then expand down the last column:

$$D = abc\begin{vmatrix}(a-c)(a+c) & a-c & 0\\ (b-c)(b+c) & b-c & 0\\ c^2 & c & 1\end{vmatrix} = abc\,(a-c)(b-c)\begin{vmatrix}a+c & 1\\ b+c & 1\end{vmatrix}$$
$$= abc\,(a-b)(a-c)(b-c).$$

Every factor is non-zero: $a, b, c \neq 0$ and a, b, c are distinct. So $D \neq 0$, the four points are not coplanar, and therefore the lines OA and BC cannot intersect. (They are not parallel either, so they are skew.)

Ask yourself Why is it enough to show that the four points are not coplanar?

Because intersecting lines always span a plane. If OA and BC met at a point P, the plane through P containing both lines would contain O, A, B and C. We have shown that no such plane exists.

Part (ii)

Put $b = 1/a$. Then $a \neq 0$, and $a \neq \pm 1$, since otherwise $b = a$ and A, B would coincide. The scalar product is

$$\mathbf{a}\cdot\mathbf{b} = (ab)^3 + (ab)^2 + ab = 3,$$

and the squared lengths are

$$|\mathbf{a}|^2 = a^2(a^4+a^2+1), \qquad |\mathbf{b}|^2 = \frac{1}{a^6}+\frac{1}{a^4}+\frac{1}{a^2} = \frac{a^4+a^2+1}{a^6}.$$

Multiplying, $|\mathbf{a}|^2|\mathbf{b}|^2 = (a^4+a^2+1)^2/a^4$, so $|\mathbf{a}||\mathbf{b}| = (a^4+a^2+1)/a^2$, which is positive. Writing $\theta = \angle AOB$,

$$\cos\theta = \frac{3a^2}{a^4+a^2+1} = \frac{3}{u+1}, \qquad \text{where } u = a^2 + \frac{1}{a^2}.$$

Which values can occur. By AM–GM (or since $a^2 + a^{-2} - 2 = (a - a^{-1})^2$), $u \geq 2$, with equality only when $a^2 = 1$, which is excluded. So $u > 2$, and therefore

$$0 < \cos\theta = \frac{3}{u+1} < \frac{3}{3} = 1,$$

which gives $0 < \theta < \frac{1}{2}\pi$. No other angle is possible.

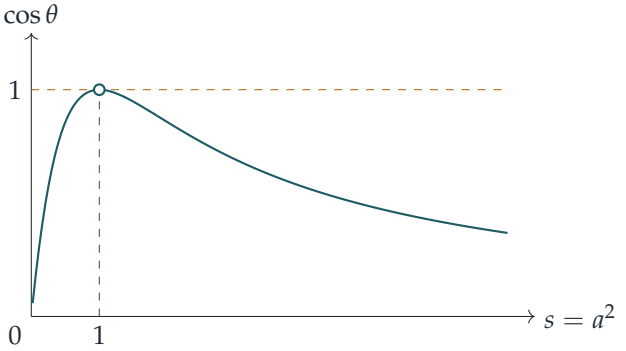
Every such value occurs. Take any θ with $0 < \theta < \frac{1}{2}\pi$, so $0 < \cos\theta < 1$. We need $u = 3/\cos\theta - 1$, which is greater than 2. The equation $a^2 + 1/a^2 = u$ is the quadratic $(a^2)^2 - u\,a^2 + 1 = 0$ in a^2, with root

$$a^2 = \frac{u + \sqrt{u^2-4}}{2} > 1,$$

which is real because $u > 2$. Choose a to be the positive square root and $b = 1/a$. Then $a > 1 > b > 0$, so $ab = 1$, $A \neq B$, neither is O, and $\cos\angle AOB = 3/(u+1) = \cos\theta$. Hence $\angle AOB = \theta$, since both angles lie in $[0, \pi]$, where cosine is one-to-one.

So $\angle AOB$ takes every value in $0 < \angle AOB < \frac{1}{2}\pi$ and no others, as required.

The graph shows $\cos\theta$ against $s = a^2$. It rises to 1 at the excluded value $s = 1$ and tends to 0 at both ends, so its values fill $(0, 1)$ exactly.



Checks

- Try $a = 2$, $b = \frac{1}{2}$: $A = (8, 4, 2)$ and $B = (\frac{1}{8}, \frac{1}{4}, \frac{1}{2})$ give $\mathbf{a} \cdot \mathbf{b} = 1 + 1 + 1 = 3$, and the formula gives $\cos\theta = 12/21 = 4/7$, so $\theta \approx 55.2°$, which lies inside the interval.
- As $a \to 1$, the points A and B merge and $\theta \to 0$. As $a \to \infty$, A heads off almost along the x-axis and B approaches O almost along the z-axis, so $\theta \to \frac{1}{2}\pi$. These are the two ends of the interval.
- The factorisation of D was checked exactly for thousands of random rational a, b, c, and the angle formula and its range were checked numerically.

Common slips

- Forgetting to say why a, b, c are distinct and non-zero. Those are exactly the facts that make $D \neq 0$.
- Concluding from $\cos\theta < 1$ alone, without excluding $a = \pm 1$. The strict inequality $u > 2$ is what rules out $\theta = 0$.
- Proving only "no other values". The question also asks you to show that every value in the interval is attained, which needs an explicit a.

Beyond STEP

The curve (t^3, t^2, t) is the twisted cubic, a standard example in algebraic geometry. Part (i) is the fact that no four of its points are coplanar, and the determinant you computed is a Vandermonde determinant, which reappears in polynomial interpolation and in linear independence arguments. Strang's *Introduction to Linear Algebra* covers determinants and linear independence clearly.

WORKED SOLUTION

STEP III 1990, Question 3

Pure • Group Axioms and Conjugation

STEP III 1990, Question 3

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q3

The key idea

All three parts come from the group axioms alone, and in each one the useful move is to rewrite one element in terms of another. An element of order 2 is its own inverse, which turns (i) into a one-line calculation. In (ii) and (iii) the tool is *conjugation*: $x \mapsto c^{-1}xc$ respects products, so it carries powers to powers.

Throughout we write xy for $x * y$ and e for the identity.

Solution

Part (i)

An element x has order 2 exactly when $x \neq e$ and $x^2 = e$; the second condition says $x^{-1} = x$. So

$$a^{-1} = a, \qquad b^{-1} = b, \qquad (ab)^{-1} = ab.$$

For any group elements, $(ab)^{-1} = b^{-1}a^{-1}$, since $(ab)(b^{-1}a^{-1}) = a(bb^{-1})a^{-1} = e$. Hence

$$ab = (ab)^{-1} = b^{-1}a^{-1} = ba,$$

so a and b commute.

Part (ii)

First note that dc is a conjugate of cd:

$$c^{-1}(cd)\,c = (c^{-1}c)\,dc = dc.$$

? **Ask yourself** Why does writing $dc = c^{-1}(cd)c$ help with orders?

Because conjugation passes through powers. For any x and any positive integer n,

$$(c^{-1}xc)^n = (c^{-1}xc)(c^{-1}xc)\cdots(c^{-1}xc) = c^{-1}x^nc,$$

since each adjacent pair $c\,c^{-1}$ in the middle cancels. With $x = cd$ this gives $(dc)^n = c^{-1}(cd)^n c$. Now

$$(dc)^n = e \iff c^{-1}(cd)^n c = e \iff (cd)^n = cec^{-1} = e,$$

where the second step multiplies on the left by c and on the right by c^{-1} (and the converse multiplies by c^{-1} and c). So the positive integers n with $(cd)^n = e$ are exactly those with $(dc)^n = e$. The order is the least such n, so cd and dc have the same order. If there is no such n, both have infinite order, so the statement holds in that case too.

Part (iii)

We are given $c^{-1}bc = b^r$. For a positive integer s, the telescoping argument from (ii) gives

$$c^{-1}b^s c = (c^{-1}bc)^s = (b^r)^s = b^{sr}.$$

The same holds for $s = 0$ (both sides are e). For negative s, write $s = -t$ with $t > 0$; then $c^{-1}b^s c = \left(c^{-1}b^t c\right)^{-1}$, because $(c^{-1}b^t c)(c^{-1}b^{-t}c) = e$. So

$$c^{-1}b^s c = \left(b^{tr}\right)^{-1} = b^{-tr} = b^{sr}.$$

Thus $c^{-1}b^s c = b^{sr}$ for every integer s.

For the second result we use induction on $n \geq 1$, for every integer s at once. The case $n = 1$ is what we have just proved. Suppose $c^{-n}b^s c^n = b^{sr^n}$ for all integers s. Then

$$\begin{aligned} c^{-(n+1)}\, b^s\, c^{n+1} &= c^{-1}\left(c^{-n}b^s c^n\right)c \\ &= c^{-1}\, b^{sr^n}\, c \\ &= b^{(sr^n)r} = b^{sr^{n+1}}, \end{aligned}$$

where the last line uses the first result with the integer sr^n in place of s. This completes the induction, so $c^{-n}b^s c^n = b^{sr^n}$ for all positive integers n.

Ask yourself Why did we bother proving the first result for negative s?

If r is negative (for example $c^{-1}bc = b^{-1}$ in a dihedral group), then sr^n can be negative even when s is positive, and the inductive step applies the first result with sr^n in place of s. Proving it for all integers s makes the induction work whatever the sign of r.

Checks

Test on small groups. In the symmetries of a square, the reflections in the two diagonals and their product (a half-turn) all have order 2, and the two reflections do commute, as (i) predicts. In S_3, take $c = (1\,2)$ and $d = (1\,3)$: then cd and dc are the two 3-cycles, both of order 3. For (iii), in the dihedral group a reflection c and rotation b satisfy $c^{-1}bc = b^{-1}$, so $r = -1$ and the formula predicts $c^{-n}b^s c^n = b^{(-1)^n s}$, which is right: conjugating by a reflection twice does nothing. We also checked all three statements by computer, exhaustively on all suitable elements of S_4 (and of S_5 for (i)), and in the group of maps $x \mapsto ux + v$ modulo 13.

Common slips

- Writing $(ab)^{-1} = a^{-1}b^{-1}$. The order reverses; that reversal is exactly what makes (i) work.
- In (ii), proving only that $(cd)^n = e \Rightarrow (dc)^n = e$. You need both directions (or symmetry: swap the roles of c and d), and the infinite-order case should be mentioned.
- In (iii), giving a "proof" by writing three dots for the second result instead of an induction.

Beyond STEP

The map $x \mapsto c^{-1}xc$ is an *automorphism* of the group: a bijection from G to itself that respects the operation, so it preserves orders and sends powers to powers. Part (iii) is the key relation in *semidirect products*: a group generated by b and c with $c^{-1}bc = b^r$ includes the dihedral groups ($r = -1$) and the groups of maps $x \mapsto ux + v$ modulo a prime. Allenby's *Rings, Fields and Groups* develops conjugation and these examples carefully.

WORKED SOLUTION

STEP III 1990, Question 4

Pure • Trigonometric Series

STEP III 1990, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q4

The key idea

Both series are sums of cosines whose angles go up in equal steps, so there are two natural tools. For the plain sum, multiply by $2\sin\beta$ and use a product-to-sum formula: every term splits into a difference of two sines and the sum telescopes. For the binomial sum, and for the final identity, think of $\cos\phi$ as the real part of $e^{i\phi}$. Then the binomial theorem or the geometric-series formula does the work, and the answer comes out by taking real or imaginary parts at the end.

Solution

The first series

Call the sum S. By the product-to-sum formula $2\sin A\cos B = \sin(B+A) - \sin(B-A)$,

$$2\sin\beta\cos(\alpha+2r\beta) = \sin\big(\alpha+(2r+1)\beta\big) - \sin\big(\alpha+(2r-1)\beta\big).$$

Adding these for $r = 0, 1, \ldots, n$, the sum telescopes: the first sine for r cancels the second sine for $r+1$. So only two terms are left:

$$\begin{aligned} 2\sin\beta\, S &= \sin\big(\alpha+(2n+1)\beta\big) - \sin(\alpha-\beta) \\ &= 2\cos(\alpha+n\beta)\sin\big((n+1)\beta\big), \end{aligned}$$

using $\sin P - \sin Q = 2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$ with $\frac{P+Q}{2} = \alpha+n\beta$ and $\frac{P-Q}{2} = (n+1)\beta$. Since $\sin\beta \neq 0$ we may divide, and

$$\boxed{\sum_{r=0}^{n}\cos(\alpha+2r\beta) = \frac{\sin(n+1)\beta\,\cos(\alpha+n\beta)}{\sin\beta}}$$

Notice that $\alpha+n\beta$ is the average of the first and last angles, which is a good sanity check on the form.

The binomial series

Call this sum T. Since $\cos\phi = \operatorname{Re} e^{i\phi}$ and the binomial coefficients are real,

$$T = \operatorname{Re} \sum_{r=0}^{n} \binom{n}{r} e^{i\alpha} \left(e^{2i\beta}\right)^r = \operatorname{Re}\left[e^{i\alpha}\left(1 + e^{2i\beta}\right)^n\right]$$

by the binomial theorem. Now take out the "half-angle" factor:

$$1 + e^{2i\beta} = e^{i\beta}\left(e^{-i\beta} + e^{i\beta}\right) = 2\cos\beta\, e^{i\beta}.$$

Hence $e^{i\alpha}(1 + e^{2i\beta})^n = 2^n \cos^n\beta\, e^{i(\alpha+n\beta)}$, and since $2^n\cos^n\beta$ is real,

$$\boxed{\sum_{r=0}^{n} \binom{n}{r} \cos(\alpha + 2r\beta) = 2^n \cos^n\beta\, \cos(\alpha + n\beta)}$$

(This one doesn't actually need $\sin\beta \neq 0$.)

The final identity

The secants mean we need $\cos\theta \neq 0$, which is implicit in the question. The rth term is $\cos r\theta\, \sec^r\theta = \operatorname{Re}\left(e^{i\theta}\sec\theta\right)^r$, because $\sec^r\theta$ is real. So put

$$w = e^{i\theta}\sec\theta = \frac{\cos\theta + i\sin\theta}{\cos\theta} = 1 + i\tan\theta.$$

The left-hand side is $\operatorname{Re}(1 + w + \cdots + w^n)$.

> **Ask yourself** Why is it safe to use the geometric-series formula here, and why does w help?

The formula needs $w \neq 1$, and $w - 1 = i\tan\theta \neq 0$ precisely because $\sin\theta \neq 0$. That's where the condition is used. The point of w is that $w - 1$ is purely imaginary, so dividing by it is easy, while w^{n+1} is easy to compute from the exponential form $w = \sec\theta\, e^{i\theta}$. Writing $t = \tan\theta$,

$$1 + w + \cdots + w^n = \frac{w^{n+1} - 1}{w - 1} = \frac{\sec^{n+1}\theta\, e^{i(n+1)\theta} - 1}{it}.$$

For any complex z and real $t \neq 0$ we have $\operatorname{Re}\dfrac{z}{it} = \operatorname{Re}\dfrac{-iz}{t} = \dfrac{\operatorname{Im} z}{t}$. The -1 in the numerator has zero imaginary part, so

$$\begin{aligned}\sum_{r=0}^{n} \cos r\theta\, \sec^r\theta &= \frac{\sec^{n+1}\theta\, \sin(n+1)\theta}{\tan\theta} = \sec^{n+1}\theta\, \sin(n+1)\theta \cdot \frac{\cos\theta}{\sin\theta} \\ &= \frac{\sin(n+1)\theta\, \sec^n\theta}{\sin\theta},\end{aligned}$$

which is the required result.

Link with the first part. Multiplying through by $\cos^n\theta$, the identity says

$$\sum_{r=0}^{n} \cos r\theta \cos^{n-r}\theta = \frac{\sin(n+1)\theta}{\sin\theta}.$$

You can also prove it in that form by induction on n: if C_n is the left-hand side then $C_n = \cos\theta\, C_{n-1} + \cos n\theta$, and $\sin n\theta\cos\theta + \cos n\theta\sin\theta = \sin(n+1)\theta$ finishes the step.

Checks

- $n = 0$: all three formulae reduce to their single first term ($\cos\alpha$, $\cos\alpha$ and 1).
- In the first sum, $\beta \to 0$ gives $\frac{\sin(n+1)\beta}{\sin\beta} \to n+1$, so $S \to (n+1)\cos\alpha$, which is right for $n+1$ equal terms. In the second, $\beta = 0$ gives $2^n\cos\alpha$, since $\sum \binom{n}{r} = 2^n$.
- $\theta = \pi/4$, $n = 2$: the left-hand side is $1 + \frac{1}{\sqrt{2}}\cdot\sqrt{2} + 0 = 2$, and the right-hand side is $\sin\frac{3\pi}{4}\cdot 2/\sin\frac{\pi}{4} = 2$.
- Numerically, all three formulae agree with direct summation to within 10^{-9} (relative) over 20 000 random choices of α, β, θ and $n \le 15$.

Common slips

- Forgetting that the first series has $n+1$ terms ($r = 0$ to n), which gives $\sin n\beta$ instead of $\sin(n+1)\beta$.
- Writing $1 + e^{2i\beta}$ as $2\cos\beta$ and losing the factor $e^{i\beta}$, which shifts the angle in the answer.
- In the last part, taking the real part of the numerator and the denominator separately. You must first make the denominator real.

Beyond STEP

Writing cos as the real part of an exponential turns trigonometric sums into geometric or binomial sums. This is the starting point of Fourier analysis: the first sum is essentially the Dirichlet kernel $\sin\big((n+\frac{1}{2})x\big)/\sin\frac{x}{2}$, which controls how Fourier series converge. Needham's *Visual Complex Analysis* gives a beautifully geometric view of why multiplying by $e^{i\theta}$ means rotation, and why sums like these close up into neat formulae.

WORKED SOLUTION

STEP III 1990, Question 5

Pure • Leibniz's Theorem

STEP III 1990, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q5

The key idea

Pascal's rule is just fractions with factorials, once we put both terms over a common denominator. Leibniz's formula for the nth derivative of a product is then a proof by induction on n: differentiating the nth-order formula once more produces each mixed term twice, and Pascal's rule is exactly what merges each pair. For the last part, differentiating $y = \sin^{-1} x$ directly n times is hopeless. Instead we find a simple second-order equation that y satisfies, and apply Leibniz to that.

(The printed question says $y = \sin^{-1} y$. This is clearly a misprint for $y = \sin^{-1} x$, and that is what we use.)

Solution

Pascal's rule

Recall $\binom{n}{r} = \dfrac{n!}{r!\,(n-r)!}$, with $0! = 1$. For $1 \le r \le n$ both $r-1$ and $n-r$ are non-negative, so every factorial below makes sense. The common denominator is $r!\,(n-r+1)!$, since $r! = r\,(r-1)!$ and $(n-r+1)! = (n-r+1)\,(n-r)!$:

$$\begin{aligned}\binom{n}{r} + \binom{n}{r-1} &= \frac{n!}{r!\,(n-r)!} + \frac{n!}{(r-1)!\,(n-r+1)!} \\ &= \frac{n!\,(n-r+1)}{r!\,(n-r+1)!} + \frac{n!\,r}{r!\,(n-r+1)!} \\ &= \frac{n!\,(n+1)}{r!\,(n+1-r)!} = \frac{(n+1)!}{r!\,(n+1-r)!} = \binom{n+1}{r}.\end{aligned}$$

Leibniz's formula

We assume u and v can be differentiated as often as needed. In summation notation the claim is

$$(uv)^{(n)} = \sum_{r=0}^{n} \binom{n}{r} u^{(n-r)} v^{(r)},$$

where $z^{(0)}$ means z itself. We prove this by induction on $n \geq 1$.

Base case $n = 1$**.** The right-hand side is $u^{(1)}v + uv^{(1)}$, which is $(uv)'$ by the product rule.

Inductive step. Suppose the formula holds for some $n \geq 1$. Differentiate it, using the product rule on each term:

$$(uv)^{(n+1)} = \sum_{r=0}^{n} \binom{n}{r} u^{(n+1-r)} v^{(r)} + \sum_{r=0}^{n} \binom{n}{r} u^{(n-r)} v^{(r+1)}.$$

In the second sum, put $s = r + 1$, so it becomes $\sum_{s=1}^{n+1} \binom{n}{s-1} u^{(n+1-s)} v^{(s)}$. Now both sums are made of terms $u^{(n+1-r)}v^{(r)}$. Split off the $r = 0$ term of the first sum and the $s = n + 1$ term of the second:

$$\begin{aligned}(uv)^{(n+1)} &= u^{(n+1)}v + \sum_{r=1}^{n} \left[\binom{n}{r} + \binom{n}{r-1} \right] u^{(n+1-r)} v^{(r)} + uv^{(n+1)} \\ &= u^{(n+1)}v + \sum_{r=1}^{n} \binom{n+1}{r} u^{(n+1-r)} v^{(r)} + uv^{(n+1)},\end{aligned}$$

by Pascal's rule, which applies because $1 \leq r \leq n$. Since $\binom{n+1}{0} = \binom{n+1}{n+1} = 1$, this is

$$(uv)^{(n+1)} = \sum_{r=0}^{n+1} \binom{n+1}{r} u^{(n+1-r)} v^{(r)},$$

the formula for $n + 1$. By induction it holds for every $n \geq 1$.

The derivatives of $\sin^{-1} x$

Let $y = \sin^{-1} x$ for $-1 < x < 1$. Then

$$y^{(1)} = \frac{1}{\sqrt{1 - x^2}}, \qquad \text{so} \qquad (1 - x^2)\left(y^{(1)}\right)^2 = 1.$$

Differentiating this with respect to x,

$$(1 - x^2) \cdot 2y^{(1)} y^{(2)} - 2x\left(y^{(1)}\right)^2 = 0.$$

Since $y^{(1)} > 0$ on $(-1, 1)$, we may divide by $2y^{(1)}$:

$$(1 - x^2)\, y^{(2)} - x\, y^{(1)} = 0. \qquad (*)$$

This is the required result for $n = 0$.

Ask yourself Why square $y^{(1)}$ before differentiating, rather than just differentiating $(1-x^2)^{-1/2}$?

Differentiating $(1-x^2)^{-1/2}$ directly gives $y^{(2)} = x(1-x^2)^{-3/2}$, and multiplying by $1-x^2$ gives $(*)$ just as well. Either way works. The point is to reach an equation whose coefficients are *polynomials* in x and which is *linear* in the derivatives of y, because that is what Leibniz handles neatly: differentiating a polynomial coefficient soon gives zero.

Now differentiate $(*)$ n times, for $n \geq 1$, applying Leibniz's formula to each product. For the first, take $u = y^{(2)}$ and $v = 1-x^2$. Then $v^{(1)} = -2x$, $v^{(2)} = -2$ and all higher derivatives vanish, so only three terms survive:

$$\left((1-x^2)\,y^{(2)}\right)^{(n)} = (1-x^2)\,y^{(n+2)} + \binom{n}{1}(-2x)\,y^{(n+1)} + \binom{n}{2}(-2)\,y^{(n)}.$$

(When $n = 1$ the last term is absent, which agrees with $\binom{1}{2} = 0$.) For the second, take $u = y^{(1)}$ and $v = x$, so only two terms survive:

$$\left(x\,y^{(1)}\right)^{(n)} = x\,y^{(n+1)} + n\,y^{(n)}.$$

Subtracting, the nth derivative of $(*)$ is

$$(1-x^2)\,y^{(n+2)} - 2nx\,y^{(n+1)} - n(n-1)\,y^{(n)} - x\,y^{(n+1)} - n\,y^{(n)} = 0.$$

Collecting terms, $-2nx - x = -(2n+1)x$ and $-n(n-1) - n = -n^2$, so

$$(1-x^2)\,y^{(n+2)} - (2n+1)x\,y^{(n+1)} - n^2 y^{(n)} = 0,$$

as required. It holds for all $n \geq 0$ and all x in $(-1,1)$.

Checks

- Take $n = 1$ in Leibniz: we get the product rule. With $n = 2$: $(uv)'' = u''v + 2u'v' + uv''$, which you can confirm by differentiating $u'v + uv'$.
- At $x = 0$ the final result says $y^{(n+2)}(0) = n^2 y^{(n)}(0)$. With $y^{(1)}(0) = 1$ this gives $y^{(3)}(0) = 1$ and $y^{(5)}(0) = 9$, so the Maclaurin series begins $x + \frac{x^3}{6} + \frac{9x^5}{120} = x + \frac{x^3}{6} + \frac{3x^5}{40}$, which is the known series for $\sin^{-1} x$.
- We checked numerically: Pascal's rule for all $n < 60$; Leibniz exactly on 300 random pairs of polynomials; and the final identity for n up to 27 at six points in $(-1,1)$, using Taylor coefficients of $(1-x^2)^{-1/2}$. The relative residual was below 10^{-15}.

Common slips

- Getting the binomial coefficient wrong after reindexing the second sum in the inductive step. Write out the $r = 0$ and $r = n+1$ end terms separately before applying Pascal's rule.

- Trying to differentiate $\sin^{-1} x$ n times directly, instead of first finding the equation $(*)$.
- Forgetting the factor $\binom{n}{2}$, or the minus signs, when differentiating $1 - x^2$.

Beyond STEP

The trick of finding a linear differential equation with polynomial coefficients and then differentiating it n times is a standard way to get recurrences for Taylor coefficients. It leads directly to the method of power-series solutions of ODEs, which is how Legendre and Bessel functions arise. Leibniz's formula is also the calculus version of the binomial theorem: if $u = \mathrm{e}^{ax}$ and $v = \mathrm{e}^{bx}$, it reduces to $(a+b)^n = \sum \binom{n}{r} a^{n-r} b^r$. For more on series solutions, see Boyce and DiPrima, *Elementary Differential Equations*.

WORKED SOLUTION

STEP III 1990, Question 6

Pure • Linear Transformations and Ellipses

STEP III 1990, Question 6

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q6

This question has a full guided treatment in *Beyond STEP*, Section 7.2, *Stretching Without Turning*: https://beyondstep.keithmcnulty.org/ch07/eigenvectors.html

The key idea

The two special vectors are eigenvectors of the matrix, and they tell us the whole geometry: T stretches by one factor along one line and by another factor along a perpendicular line, so the unit circle becomes an ellipse with its axes along those lines. For the algebra, the cleanest route to the equation of E is to invert T, write x and y in terms of X and Y, and substitute into $x^2 + y^2 = 1$. Areas scale by the determinant, and because T maps lines to lines, the tangent to E is simply the image of the tangent to C.

Solution

Write $M = \frac{2}{5}\begin{pmatrix} 9 & -2 \\ -2 & 6 \end{pmatrix}$, so that T is multiplication by M.

The two special directions

We compute

$$M\begin{pmatrix} 1 \\ 2 \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 9-4 \\ -2+12 \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 5 \\ 10 \end{pmatrix} = 2\begin{pmatrix} 1 \\ 2 \end{pmatrix},$$
$$M\begin{pmatrix} 2 \\ -1 \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 18+2 \\ -4-6 \end{pmatrix} = \frac{2}{5}\begin{pmatrix} 20 \\ -10 \end{pmatrix} = 4\begin{pmatrix} 2 \\ -1 \end{pmatrix}.$$

So $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ keeps its direction and is multiplied by 2, and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ keeps its direction and is multiplied by 4, as required. Notice that these two directions are perpendicular, since $1 \cdot 2 + 2 \cdot (-1) = 0$.

The equation of E

The determinant of M is

$$\det M = \left(\tfrac{2}{5}\right)^2 (9\cdot 6 - (-2)(-2)) = \tfrac{4}{25}\cdot 50 = 8,$$

which is non-zero, so T is invertible and

$$M^{-1} = \frac{1}{8}\cdot\frac{2}{5}\begin{pmatrix} 6 & 2 \\ 2 & 9 \end{pmatrix} = \frac{1}{20}\begin{pmatrix} 6 & 2 \\ 2 & 9 \end{pmatrix}.$$

Hence a point (X, Y) is the image of (x, y) exactly when

$$x = \frac{3X + Y}{10}, \qquad y = \frac{2X + 9Y}{20}.$$

A point (X, Y) lies on E if and only if the point it came from lies on C, that is, if and only if $x^2 + y^2 = 1$. Substituting,

$$\begin{aligned} \frac{(3X+Y)^2}{100} + \frac{(2X+9Y)^2}{400} = 1 &\iff 4(9X^2 + 6XY + Y^2) + (4X^2 + 36XY + 81Y^2) = 400 \\ &\iff 40X^2 + 60XY + 85Y^2 = 400 \\ &\iff 8X^2 + 12XY + 17Y^2 = 80, \end{aligned}$$

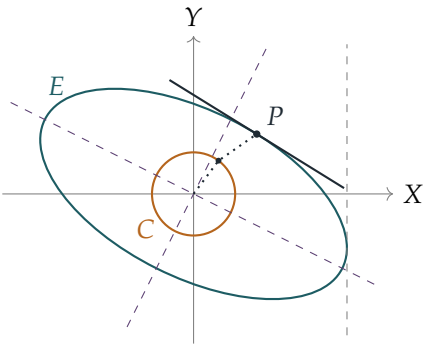
which is the required equation of E.

The area

A linear map multiplies every area by $|\det M|$. The disc bounded by C has area π, so the region bounded by E has area

$$8\pi.$$

This agrees with the geometry: E is an ellipse with semi-axes 2 (along $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$) and 4 (along $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$), and $\pi \cdot 2 \cdot 4 = 8\pi$.



The figure shows C, the ellipse E with its axes (dashed), the vertical line $X = 2\sqrt{17/5}$ touching E, and the tangent at P, the image of $\frac{1}{5}(3, 4)$.

The greatest value of X

Regard the equation of E as a quadratic in Y:

$$17Y^2 + 12XY + (8X^2 - 80) = 0.$$

A given value of X occurs on E if and only if this has a real root Y, that is, if and only if its discriminant is non-negative:

$$\begin{aligned} 144X^2 - 4 \cdot 17(8X^2 - 80) \geq 0 &\iff 144X^2 - 544X^2 + 5440 \geq 0 \\ &\iff 400X^2 \leq 5440 \\ &\iff X^2 \leq \tfrac{68}{5}. \end{aligned}$$

So every point of E has $X \leq \sqrt{68/5} = 2\sqrt{17/5}$, and equality is attained (the discriminant is then zero, giving the single point with $Y = -\frac{6X}{17}$). Hence the greatest value of X on E is $2\sqrt{17/5}$, as required.

> **Ask yourself** Is there a way to see this straight from the circle, without the discriminant?

Yes. On E we have $X = \frac{2}{5}(9x - 2y)$ with $x^2 + y^2 = 1$. Writing $(x, y) = (\cos\theta, \sin\theta)$, the expression $9\cos\theta - 2\sin\theta$ has greatest value $\sqrt{81 + 4} = \sqrt{85}$ (the $R\cos(\theta + \alpha)$ form). So the greatest X is $\frac{2}{5}\sqrt{85} = 2\sqrt{85/25} = 2\sqrt{17/5}$, agreeing with the above.

The tangent

The point $\frac{1}{5}(3, 4)$ on C maps to

$$P = \frac{2}{5}\begin{pmatrix} 9 \cdot \frac{3}{5} - 2 \cdot \frac{4}{5} \\ -2 \cdot \frac{3}{5} + 6 \cdot \frac{4}{5} \end{pmatrix} = \frac{2}{5}\begin{pmatrix} \frac{19}{5} \\ \frac{18}{5} \end{pmatrix} = \begin{pmatrix} \frac{38}{25} \\ \frac{36}{25} \end{pmatrix}.$$

The tangent to C at $\frac{1}{5}(3, 4)$ is perpendicular to the radius there, so it is

$$3x + 4y = 5.$$

Since T is linear and invertible, it maps straight lines to straight lines, and a line meeting C at exactly one point maps to a line meeting E at exactly one point. So the tangent to E at P is the image of this line. Substituting $x = \frac{3X+Y}{10}$ and $y = \frac{2X+9Y}{20}$:

$$\begin{aligned} \frac{3(3X + Y)}{10} + \frac{4(2X + 9Y)}{20} = 5 &\iff 6(3X + Y) + 4(2X + 9Y) = 100 \\ &\iff 26X + 42Y = 100, \end{aligned}$$

so the tangent to E at P is

$$13X + 21Y = 50.$$

As a check, $13 \cdot \frac{38}{25} + 21 \cdot \frac{36}{25} = \frac{494+756}{25} = 50$, so P is on the line. Alternatively, differentiating $8X^2 + 12XY + 17Y^2 = 80$ implicitly, the normal at P is along $(16X + 12Y,\ 12X + 34Y) = \frac{1}{25}(1040, 1680)$, which is parallel to $(13, 21)$, giving the same line.

Checks

- The image of $(1,0)$ is $\frac{2}{5}(9,-2)$, and $\frac{4}{25}\left(8\cdot 81-12\cdot 18+17\cdot 4\right)=\frac{4}{25}\cdot 500=80$, so it lies on E.
- The ends of the axes, $\pm 2\cdot\frac{1}{\sqrt{5}}(1,2)$ and $\pm 4\cdot\frac{1}{\sqrt{5}}(2,-1)$, satisfy the equation of E; for instance $\frac{4}{5}(8+24+68)=80$.
- Numerically, sampling 200 000 points of C gives greatest $X\approx 3.687818$, matching $2\sqrt{17/5}\approx 3.687818$, and the shoelace area of the image polygon is $25.13274\approx 8\pi$.

Common slips

- Applying M rather than M^{-1} when substituting into $x^2+y^2=1$: the equation of the image curve needs the old coordinates in terms of the new ones.
- Forgetting the factor $\frac{2}{5}$ when computing the determinant: it contributes $\left(\frac{2}{5}\right)^2$, not $\frac{2}{5}$.
- Finding the image point P but then using the gradient of C at $\frac{1}{5}(3,4)$ for the tangent to E: linear maps preserve tangency but not slopes.

Beyond STEP

The matrix here is symmetric, and that is why its eigenvectors are perpendicular: this is the two-dimensional case of the spectral theorem, which says every real symmetric matrix can be diagonalised by a rotation. It explains why any quadratic form $ax^2+2hxy+by^2$ describes a conic with perpendicular principal axes, and the eigenvalues of the form's matrix tell you its shape at once. Strang's *Introduction to Linear Algebra* develops eigenvalues, symmetric matrices and quadratic forms with plenty of geometric pictures.

WORKED SOLUTION

STEP III 1990, Question 7

Pure • Implicit Curve Sketching

STEP III 1990, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q7

The key idea

Implicit differentiation, done twice, gives $\mathrm{d}y/\mathrm{d}x$ and $\mathrm{d}^2y/\mathrm{d}x^2$ at each point, and these are exactly the coefficients in the Taylor series. For the inequality, the trick is to notice that the equation is only a *quadratic* in y: a point (x, y) can exist only if that quadratic has real roots, so its discriminant is non-negative. Solving the quadratic explicitly then shows the whole curve, two branches that meet where the discriminant vanishes.

Solution

Throughout, write $F(x, y) = xy^2 + x^3 + a^2y - a^3$. Substituting shows that $F = 0$ at P, Q and R; for instance at R we get $a \cdot a^2 + a^3 - a^3 - a^3 = 0$.

The Taylor series at P, Q and R

Differentiating $F(x, y) = 0$ with respect to x, treating y as a function of x,

$$y^2 + 2xyy' + 3x^2 + a^2y' = 0, \qquad \text{so} \qquad y' = -\frac{y^2 + 3x^2}{2xy + a^2}.$$

Differentiating the first equation again (the term $2xyy'$ gives $2yy' + 2xy'^2 + 2xyy''$):

$$4yy' + 2xy'^2 + 6x + (2xy + a^2)\, y'' = 0.$$

At each point the denominator $2xy + a^2$ is non-zero, so these equations determine y' and y'', and then $y = y_0 + y'h + \frac{1}{2}y''h^2 + \cdots$.

At $P(0, a)$: $y' = -a^2/a^2 = -1$. Then $4a(-1) + a^2y'' = 0$, so $y'' = 4/a$ and

$$y = a - h + \frac{2}{a}h^2 + \cdots.$$

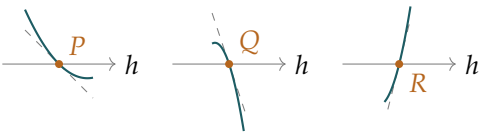
At $Q(a, 0)$: $y' = -3a^2/a^2 = -3$. Then $0 + 2a(9) + 6a + a^2y'' = 0$, so $y'' = -24/a$ and

$$y = -3h - \frac{12}{a}h^2 + \cdots.$$

At $R(a,-a)$: here $2xy + a^2 = -a^2$, so $y' = -4a^2/(-a^2) = 4$. Then $4(-a)(4) + 2a(16) + 6a - a^2 y'' = 0$, that is $22a = a^2 y''$, so $y'' = 22/a$ and

$$y = -a + 4h + \frac{11}{a}h^2 + \cdots .$$

So near P the curve is falling with gradient -1 and is concave up (bending above its tangent); near Q it falls steeply with gradient -3 and is concave down; near R it rises with gradient 4 and is concave up.



(Local shapes with $a = 1$; the dashed lines are the tangents.)

The inequality

Write the equation of C as a quadratic in y:

$$x\,y^2 + a^2\,y + (x^3 - a^3) = 0.$$

If $x \neq 0$ and (x,y) lies on C, this quadratic has the real root y, so its discriminant is non-negative:

$$a^4 - 4x(x^3 - a^3) \geq 0 \quad \Longleftrightarrow \quad 4x^4 - 4a^3x - a^4 \leq 0.$$

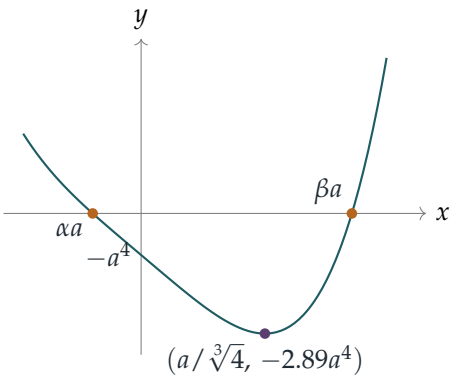
If $x = 0$ the left-hand side is $-a^4 < 0$, so the inequality holds there too. This proves the result.

The graph of $y = 4x^4 - 4a^3x - a^4$

The printed question has $4x^4 - 4a^3 - a^4$, which is surely a misprint for the quartic $g(x) = 4x^4 - 4a^3x - a^4$ we have just met. We have $g(0) = -a^4$ and $g'(x) = 16x^3 - 4a^3$, which vanishes only at $x_0 = a/\sqrt[3]{4} \approx 0.63a$, where g has its minimum. Using $x_0^3 = a^3/4$,

$$g(x_0) = a^3x_0 - 4a^3x_0 - a^4 = -a^4\left(1 + 3\cdot 4^{-1/3}\right) \approx -2.89a^4.$$

Since $g \to +\infty$ as $x \to \pm\infty$ and g is decreasing then increasing, it has exactly two zeros, $x = \alpha a$ and $x = \beta a$, where $\alpha \approx -0.246$ and $\beta \approx 1.072$ are the roots of $4t^4 - 4t - 1 = 0$ (there's no neat closed form, and we don't need one). Also $g''(x) = 48x^2 \geq 0$, so the graph is convex.



(Drawn with $a = 1$.) So every point of C has $\alpha a \le x \le \beta a$: the curve lives in a vertical strip.

The curve C

For $x \neq 0$, solving the quadratic gives the two branches

$$y_\pm = \frac{-a^2 \pm \sqrt{D}}{2x}, \qquad D = a^4 + 4a^3x - 4x^4,$$

which are real exactly on the strip and meet where $D = 0$, at $x = \alpha a$ and $x = \beta a$. There $y = -a^2/(2x)$, and the tangent is vertical, since that is where the denominator $2xy + a^2$ of y' vanishes. This gives the points $(\beta a, -0.47a)$ and $(\alpha a, 2.03a)$.

? **Ask yourself** What happens to the two branches as $x \to 0$?

Multiplying top and bottom by $\sqrt{D} + a^2$, and using $D - a^4 = 4x(a^3 - x^3)$, we get

$$y_+ = \frac{2(a^3 - x^3)}{a^2 + \sqrt{D}},$$

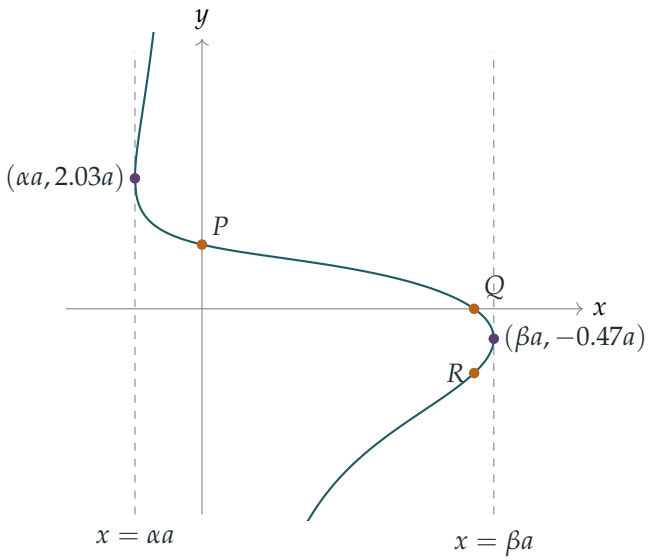
which is perfectly well behaved at $x = 0$ and equals a there. So y_+ is a single smooth branch through P. The other branch satisfies $y_- \approx -a^2/x$ for small x, so $y_- \to -\infty$ as $x \to 0^+$ and $y_- \to +\infty$ as $x \to 0^-$. This is the asymptote along the y-axis.

Some more facts help to pin down the sketch.

- There are no horizontal tangents: $y' = 0$ would need $y^2 + 3x^2 = 0$, that is $(0, 0)$, which is not on C.
- The curve meets $y = 0$ only where $x^3 = a^3$, at Q, and meets $x = 0$ only at P.
- At $x = a$, $D = a^4$, so $y_+ = 0$ (the point Q) and $y_- = -a$ (the point R).

Putting it together, C is one connected curve. It comes down from $+\infty$ just to the left of the y-axis, along y_-, to the vertical tangent at $(\alpha a, 2.03a)$. It turns back along y_+ through P (gradient -1), falls through Q (gradient -3) to the vertical tangent at

$(\beta a, -0.47a)$, then turns back along y_- through R (gradient 4) and goes down to $-\infty$ just to the right of the y-axis. It is concave up at P and concave down at Q, so there is a point of inflection between them.



(Drawn with $a = 1$; the dashed lines bound the strip found above.)

Checks

- With $a = 1.7$ we solved the quadratic for y at $x = x_0 + h$ near each point. The error in each two-term Taylor series fell by a factor of about 8 each time h was halved, which is the h^3 behaviour we'd expect.
- Taking 10^5 random values of x, the quadratic had real roots exactly when $4x^4 - 4a^3x - a^4 \le 0$.
- Bisection gives $\alpha \approx -0.2463$ and $\beta \approx 1.0724$. At $x = \pm 0.001$ (with $a = 1.7$), one root is close to a and the other close to $-a^2/x \approx \mp 2890$.

Common slips

- Forgetting the factor $\frac{1}{2}$: the h^2 coefficient is $\frac{1}{2}y''$, not y''.
- Dividing by x to use the discriminant without checking $x = 0$ separately.
- Missing that y_+ passes smoothly through P. Written as $(-a^2 + \sqrt{D})/(2x)$ it looks singular at $x = 0$, but it isn't.

Beyond STEP

Here we found the curve's behaviour near a point from a Taylor expansion of an implicitly defined function. The theorem that guarantees y is a smooth function of x wherever $\partial F/\partial y \neq 0$ is the *implicit function theorem*, a key result of multivariable analysis. The vertical tangents at the ends of the strip are exactly the points where $\partial F/\partial y = 2xy + a^2$ vanishes. This curve is a cubic in the plane, and the study of such curves leads on to algebraic geometry and elliptic curves. For more problems in this style, try Siklos, *Advanced Problems in Mathematics*.

WORKED SOLUTION

STEP III 1990, Question 8

Pure • Exact Differential Equations

STEP III 1990, Question 8

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q8

The key idea

Expand $\dfrac{\mathrm{d}}{\mathrm{d}x}(py' + qy)$ and compare it with $Py'' + Qy' + Ry$: this pins down p and q completely, and the condition $P'' - Q' + R = 0$ is exactly what makes the last coefficient fit. For the differential equation, the condition holds, so one integration takes us straight from a second-order equation to a first-order linear one, which an integrating factor then finishes.

Solution

The exactness criterion

For any functions p, q and y, the product rule gives

$$\frac{\mathrm{d}}{\mathrm{d}x}(py' + qy) = py'' + (p' + q)y' + q'y. \qquad (*)$$

Only if. Suppose p and q (which don't depend on y) make $Py'' + Qy' + Ry$ equal to $(*)$ for every y. Taking $y = 1$ gives $R = q'$. Taking $y = x$ gives $Q + Rx = (p' + q) + q'x$, so $Q = p' + q$. Taking $y = \frac{1}{2}x^2$ gives $P + Qx + \frac{1}{2}Rx^2 = p + (p' + q)x + \frac{1}{2}q'x^2$, so $P = p$. Therefore

$$P'' - Q' + R = p'' - (p'' + q') + q' = 0.$$

If. Suppose $P'' - Q' + R = 0$. Choose

$$p = P, \qquad q = Q - P'.$$

Then $p' + q = P' + Q - P' = Q$ and $q' = Q' - P'' = R$, using the condition. By $(*)$, $\dfrac{\mathrm{d}}{\mathrm{d}x}(py' + qy) = Py'' + Qy' + Ry$ for every y, as required.

The working also shows that p and q are forced to be P and $Q - P'$, which is how we'll find them below.

The differential equation

Here $P = x - x^4$, $Q = 1 - 7x^3$ and $R = -9x^2$, so

$$P'' - Q' + R = -12x^2 - (-21x^2) - 9x^2 = 0.$$

The criterion applies, with $p = P = x - x^4$ and $q = Q - P' = (1 - 7x^3) - (1 - 4x^3) = -3x^3$. The equation becomes

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[(x - x^4)y' - 3x^3 y\right] = (x^3 + 3x)\mathrm{e}^x.$$

Integrating by parts repeatedly, $\int x^3\mathrm{e}^x\,\mathrm{d}x = (x^3 - 3x^2 + 6x - 6)\mathrm{e}^x$ and $\int 3x\mathrm{e}^x\,\mathrm{d}x = (3x - 3)\mathrm{e}^x$, so

$$(x - x^4)y' - 3x^3 y = (x^3 - 3x^2 + 9x - 9)\mathrm{e}^x + C.$$

At $x = 2$ we have $y = 2\mathrm{e}^2$ and $y' = 0$, so the left side is $-3 \cdot 8 \cdot 2\mathrm{e}^2 = -48\mathrm{e}^2$ and the right side is $5\mathrm{e}^2 + C$. Hence $C = -53\mathrm{e}^2$.

Ask yourself The left side $(x - x^4)y' - 3x^3 y$ isn't the derivative of $(x - x^4)y$. How do we integrate a second time?

Factor out x: the left side is $x\left[(1 - x^3)y' - 3x^2 y\right] = x\,\dfrac{\mathrm{d}}{\mathrm{d}x}\left[(1 - x^3)y\right]$. (This is the usual integrating factor: $y' - \frac{3x^2}{1-x^3}y$ has factor $\exp\int \frac{-3x^2}{1-x^3}\,\mathrm{d}x = 1 - x^3$.) Dividing by x, which is allowed since the solution lives near $x = 2$,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left[(1 - x^3)y\right] = (x^2 - 3x + 9)\mathrm{e}^x - \frac{9\mathrm{e}^x}{x} - \frac{53\mathrm{e}^2}{x}.$$

Integrating by parts again, $\int (x^2 - 3x + 9)\mathrm{e}^x\,\mathrm{d}x = (x^2 - 5x + 14)\mathrm{e}^x$. The term e^x/x has no elementary antiderivative, so we keep it as an integral from the starting point $x = 2$:

$$(1 - x^3)y = (x^2 - 5x + 14)\mathrm{e}^x - 53\mathrm{e}^2 \ln\frac{x}{2} - 9\int_2^x \frac{\mathrm{e}^t}{t}\,\mathrm{d}t + D.$$

At $x = 2$ the left side is $-7 \cdot 2\mathrm{e}^2 = -14\mathrm{e}^2$ and the right side is $8\mathrm{e}^2 + D$, so $D = -22\mathrm{e}^2$. Therefore

$$\boxed{y = \frac{1}{1 - x^3}\left[(x^2 - 5x + 14)\mathrm{e}^x - 22\mathrm{e}^2 - 53\mathrm{e}^2 \ln\frac{x}{2} - 9\int_2^x \frac{\mathrm{e}^t}{t}\,\mathrm{d}t\right]}$$

for $x > 1$. The coefficient $x - x^4$ of y'' vanishes at $x = 1$, so the solution through $x = 2$ is defined on $x > 1$ and needn't extend past $x = 1$.

Checks

- At $x = 2$ the bracket is $8\mathrm{e}^2 - 22\mathrm{e}^2 = -14\mathrm{e}^2$, and dividing by $1 - 8 = -7$ gives $y = 2\mathrm{e}^2$, as required.

- Stepping the original equation numerically from $x = 2$ (fourth-order Runge–Kutta) gives $y(3) \approx 7.90306$, which matches the formula with the integral done by Simpson's rule. The two also agree at $x = 1.2$, 1.5 and 4.
- Expanding $\frac{\mathrm{d}}{\mathrm{d}x}\left[(x - x^4)y' - 3x^3y\right]$ gives back $(x - x^4)y'' + (1 - 7x^3)y' - 9x^2y$.

Common slips

- Forgetting the constant C after the first integration. It isn't zero here, and it produces the logarithm.
- Trying to force $\int \mathrm{e}^x/x\,\mathrm{d}x$ into closed form. It can't be done, so leave it as a definite integral with lower limit 2, which also makes the initial condition easy to apply.

Beyond STEP

An expression $Py'' + Qy' + Ry$ that is a perfect derivative is called *exact*. The expression $P'' - Q' + R$ is the left side of the *adjoint* equation $(Pz)'' - (Qz)' + Rz = 0$ evaluated at $z = 1$. In general, any solution z of the adjoint equation is an integrating factor that makes $z(Py'' + Qy' + Ry)$ exact. Adjoints lead on to self-adjoint (Sturm–Liouville) problems and Green's functions. Boyce and DiPrima, *Elementary Differential Equations*, covers exact and adjoint equations in its exercises on second-order linear equations.

WORKED SOLUTION

STEP III 1990, Question 9

Pure • Hyperbolic Functions

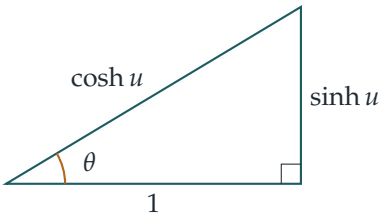
STEP III 1990, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q9

The key idea

The relation $\tan\theta = \sinh u$ is the same as $\sec^2\theta = 1 + \sinh^2 u = \cosh^2 u$, so the trigonometric and hyperbolic functions of the two variables match up exactly: $\cos\theta = \operatorname{sech} u$, $\sin\theta = \tanh u$, $\sec\theta = \cosh u$. A right-angled triangle with sides 1, $\sinh u$ and $\cosh u$ shows all of these at a glance. Once we have that dictionary, each part is a short differentiation followed by a translation back.



Solution

Throughout, since $0 \le \theta < \frac{1}{2}\pi$ we have $\tan\theta \ge 0$, so $\sinh u \ge 0$ and $u \ge 0$. Also $\cos\theta > 0$ and $\cosh u > 0$.

Part (i)

Using $1 + \tan^2\theta = \sec^2\theta$ and $1 + \sinh^2 u = \cosh^2 u$,

$$\sec^2\theta = 1 + \tan^2\theta = 1 + \sinh^2 u = \cosh^2 u.$$

Both $\sec\theta$ and $\cosh u$ are positive, so $\sec\theta = \cosh u$, that is, $\cos\theta = \operatorname{sech} u = v$, as required.

It is worth recording the companion fact: $\sin\theta = \tan\theta\cos\theta = \sinh u \operatorname{sech} u = \tanh u$.

Part (ii)

Differentiating $\tan\theta = \sinh u$ with respect to u (implicitly, by the chain rule),

$$\sec^2\theta\,\frac{\mathrm{d}\theta}{\mathrm{d}u} = \cosh u \quad\Longrightarrow\quad \frac{\mathrm{d}\theta}{\mathrm{d}u} = \cosh u\cos^2\theta = \cosh u\,\operatorname{sech}^2 u = \operatorname{sech} u = v,$$

using part (i) for $\cos\theta$.

Part (iii)

Since $\frac{\mathrm{d}}{\mathrm{d}u}\cosh u = \sinh u$,

$$\frac{\mathrm{d}v}{\mathrm{d}u} = \frac{\mathrm{d}}{\mathrm{d}u}(\cosh u)^{-1} = -\frac{\sinh u}{\cosh^2 u} = -\operatorname{sech} u\tanh u.$$

With the dictionary, $\operatorname{sech} u\tanh u = \cos\theta\sin\theta = \frac{1}{2}\sin 2\theta$, so

$$\sin 2\theta = -2\,\frac{\mathrm{d}v}{\mathrm{d}u}.$$

For the second derivative, use the product rule together with $\frac{\mathrm{d}}{\mathrm{d}u}\tanh u = \operatorname{sech}^2 u$:

$$\begin{aligned}\frac{\mathrm{d}^2v}{\mathrm{d}u^2} &= -\Big[(-\operatorname{sech} u\tanh u)\tanh u + \operatorname{sech} u\cdot\operatorname{sech}^2 u\Big]\\ &= \operatorname{sech} u\big(\tanh^2 u - \operatorname{sech}^2 u\big).\end{aligned}$$

Multiplying by $-\cosh u$ cancels the $\operatorname{sech} u$:

$$-\cosh u\,\frac{\mathrm{d}^2v}{\mathrm{d}u^2} = \operatorname{sech}^2 u - \tanh^2 u = \cos^2\theta - \sin^2\theta = \cos 2\theta,$$

which is the second result.

Part (iv)

Now treat u and v as functions of θ. From part (i), $v = \cos\theta$ and $\cosh u = \sec\theta$, and part (ii) gives $\frac{\mathrm{d}u}{\mathrm{d}\theta} = 1/\frac{\mathrm{d}\theta}{\mathrm{d}u} = 1/v$ (valid because $v > 0$). So

$$\frac{\mathrm{d}u}{\mathrm{d}\theta} = \sec\theta, \qquad \frac{\mathrm{d}^2u}{\mathrm{d}\theta^2} = \sec\theta\tan\theta,$$
$$\frac{\mathrm{d}v}{\mathrm{d}\theta} = -\sin\theta, \qquad \frac{\mathrm{d}^2v}{\mathrm{d}\theta^2} = -\cos\theta.$$

Ask yourself Before grinding through, does the identity in (iv) look exactly right as printed?

It does not. As printed, the first term has $\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2}$, and then the left-hand side is

$$\begin{aligned}\sec\theta\cdot\sec\theta\tan\theta - \sin\theta\sec\theta\tan\theta + \sec^2\theta &= \sec^2\theta\tan\theta + (\sec^2\theta - \tan^2\theta)\\ &= \sec^2\theta\tan\theta + 1 \;\geq 1,\end{aligned}$$

which is never zero. The intended identity has $\frac{\mathrm{d}^2v}{\mathrm{d}\theta^2}$ in the first term (a misprint of u for v):

$$\frac{\mathrm{d}u}{\mathrm{d}\theta}\frac{\mathrm{d}^2v}{\mathrm{d}\theta^2}+\frac{\mathrm{d}v}{\mathrm{d}\theta}\frac{\mathrm{d}^2u}{\mathrm{d}\theta^2}+\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2=0.$$

Substituting the four derivatives,

$$\sec\theta(-\cos\theta)+(-\sin\theta)\sec\theta\tan\theta+\sec^2\theta=-1-\tan^2\theta+\sec^2\theta=0,$$

since $\sec^2\theta=1+\tan^2\theta$. This proves the corrected result.

A neater way to see it: the first two terms are $\frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\frac{\mathrm{d}v}{\mathrm{d}\theta}\right)$ by the product rule, and $\frac{\mathrm{d}u}{\mathrm{d}\theta}\frac{\mathrm{d}v}{\mathrm{d}\theta}=-\sec\theta\sin\theta=-\tan\theta$, whose derivative is $-\sec^2\theta=-\left(\frac{\mathrm{d}u}{\mathrm{d}\theta}\right)^2$.

Checks

Take $u=\ln 2$, so $\sinh u=\frac{3}{4}$, $\cosh u=\frac{5}{4}$ and $\tan\theta=\frac{3}{4}$: the 3–4–5 triangle gives $\cos\theta=\frac{4}{5}=\operatorname{sech}u$ and $\sin 2\theta=\frac{24}{25}$, while $-2\,\mathrm{d}v/\mathrm{d}u=2\cdot\frac{4}{5}\cdot\frac{3}{5}=\frac{24}{25}$. Numerically, finite-difference derivatives at 2000 random points agree with (i)–(iii) and the corrected (iv) to about 10^{-7}, and confirm that the printed left-hand side of (iv) equals $1+\sec^2\theta\tan\theta$.

Common slips

- Taking $\cos\theta=\pm\operatorname{sech}u$ without saying why the positive sign is right: it is the range $0\le\theta<\frac{1}{2}\pi$ that settles it.
- Forgetting that $\frac{\mathrm{d}u}{\mathrm{d}\theta}$ is just the reciprocal of $\frac{\mathrm{d}\theta}{\mathrm{d}u}$, and differentiating $\operatorname{arsinh}(\tan\theta)$ from scratch.

Beyond STEP

The function $\theta(u)=\arctan(\sinh u)$ is the *Gudermannian* function, which links circular and hyperbolic functions without complex numbers; its inverse $u=\ln(\sec\theta+\tan\theta)=\int_0^\theta\sec t\,\mathrm{d}t$ is exactly the vertical coordinate on a Mercator map. The same pairing appears when you write $\cosh(\mathrm{i}x)=\cos x$, and Needham's *Visual Complex Analysis* is a beautiful place to see these connections geometrically.

WORKED SOLUTION

STEP III 1990, Question 10

Pure • Roots and Asymptotic Approximation

STEP III 1990, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q10

The key idea

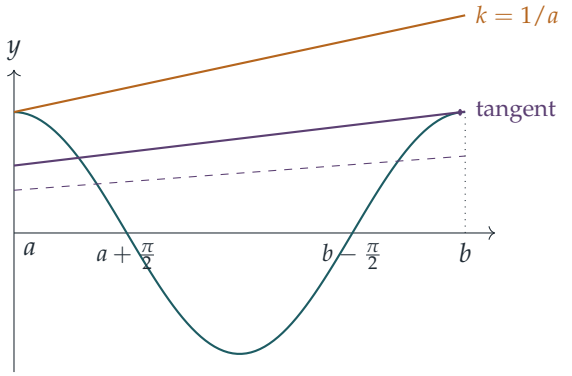
Over the given interval, $\sin x$ runs through one full period, starting and ending at a peak, while the line $y = kx$ is positive and climbs slowly. Roots can only occur where $\sin x > 0$, which happens on two separate stretches: one where $\sin x$ is falling (at most one crossing) and one where it is rising and concave (at most two crossings). The interesting borderline is when the line just touches the second arch, and writing the point of contact as a small distance δ to the left of the end of the interval turns the tangency condition into the given equation.

Solution

Throughout, write

$$a = (4n+1)\frac{\pi}{2}, \qquad b = (4n+5)\frac{\pi}{2} = a + 2\pi,$$

so the interval is $a < x < b$. Since $\sin a = \sin b = 1$, the graph of $\sin x$ starts at a peak, falls to 0 at $x = a + \frac{\pi}{2}$, is negative until $x = a + \frac{3\pi}{2} = b - \frac{\pi}{2}$, and then rises back to the peak at b.



The figure (drawn to scale for $n = 1$, with the origin far off to the left) shows $\sin x$ in teal, the line through $(a, 1)$ in amber, the tangent line in solid plum and a shallower line, which meets the curve three times, dashed.

The possible numbers of roots

Since $k > 0$ and $x > 0$, we have $kx > 0$, so every root lies where $\sin x > 0$: either in $I_1 = (a,\, a + \frac{\pi}{2})$ or in $I_2 = (b - \frac{\pi}{2},\, b)$.

On I_1. Here $\sin x$ is strictly decreasing and kx is strictly increasing, so $kx - \sin x$ is strictly increasing and there is at most one root. At the right-hand end $kx - \sin x > 0$; at $x = a$ it equals $ka - 1$. So there is exactly one root in I_1 if $ka < 1$, and none if $ka \geq 1$ (then $kx - \sin x > 0$ throughout).

On I_2. Let $f(x) = \sin x - kx$. Here $f''(x) = -\sin x < 0$, so f is concave and has at most two zeros on I_2. Also $f < 0$ at the left-hand end, and $f(b) = 1 - kb$.

- If $kb < 1$, then f goes from negative to positive, and a concave function can change sign only once on the way up, so there is exactly one root. The same holds when $kb = 1$: then $f(b) = 0$ and $f'(b) = \cos b - k = -k < 0$, so $f > 0$ just to the left of b, giving one root inside I_2 (the zero at b itself is excluded).
- If $kb > 1$, then $f < 0$ at both ends, so there are 0, 1 or 2 roots according as the maximum of f on I_2 is negative, zero or positive. The value 0 is the tangent case: the line touches the arch at some point $t \in I_2$, with slope k_t say. As k decreases from large values the line falls, so there are no roots for $k > k_t$, one for $k = k_t$ and two for $\frac{1}{b} < k < k_t$.

At the tangent point $k_t t = \sin t < 1$ and $t > a$, so $k_t < 1/a$; and $k_t > 1/b$ since the tangent case has $kb > 1$. Putting the two stretches together:

range of k	roots in I_1	roots in I_2	total
$k \leq 1/b$	1	1	2
$1/b < k < k_t$	1	2	3
$k = k_t$	1	1	2
$k_t < k < 1/a$	1	0	1
$k \geq 1/a$	0	0	0

So the equation may have 0, 1, 2 or 3 roots in the interval, as required.

Exactly one root

From the table, there is exactly one root precisely when $k_t < k < \dfrac{1}{a} = \dfrac{2}{(4n+1)\pi}$. It remains to describe k_t.

Ask yourself How do we turn "the line touches the curve" into equations?

The line $y = kx$ touches $y = \sin x$ at $x = t$ when the two graphs meet there *and* have

equal slopes:

$$\sin t = k_t t, \qquad \cos t = k_t.$$

The point t lies in I_2, so write $t = b - \delta$ with $0 < \delta < \frac{\pi}{2}$. Since $\sin b = 1$ and $\cos b = 0$,

$$\sin t = \sin(b - \delta) = \cos\delta, \qquad \cos t = \cos(b - \delta) = \sin\delta.$$

So the slope condition gives $k_t = \sin\delta$, and the meeting condition becomes

$$\cos\delta = \left((4n+5)\frac{\pi}{2} - \delta\right)\sin\delta.$$

Hence exactly one root occurs when

$$\sin\delta < k < \frac{2}{(4n+1)\pi},$$

with δ as described, which is what we had to show.

Approximations for large n

Divide the equation for δ by $\sin\delta$ (non-zero since $0 < \delta < \frac{\pi}{2}$):

$$\cot\delta = b - \delta, \qquad \text{that is} \qquad \tan\delta = \frac{1}{b-\delta}.$$

Since $\delta < \frac{\pi}{2}$, the right-hand side is less than $1/(b - \frac{\pi}{2})$, which is small when n is large. So δ is small, $\tan\delta \approx \delta$, and $b - \delta \approx b$, giving

$$\delta \approx \frac{1}{b} = \frac{2}{(4n+5)\pi}.$$

For the next approximation, write $\delta = \dfrac{1}{b} + \dfrac{c}{b^3}$ and keep terms up to b^{-3} on each side. On the left, $\tan\delta = \delta + \frac{1}{3}\delta^3 + \cdots = \dfrac{1}{b} + \dfrac{c + \frac{1}{3}}{b^3} + \cdots$. On the right,

$$\frac{1}{b-\delta} = \frac{1}{b}\left(1 - \frac{\delta}{b}\right)^{-1} = \frac{1}{b}\left(1 + \frac{\delta}{b} + \cdots\right) = \frac{1}{b} + \frac{1}{b^3} + \cdots,$$

using $\delta \approx 1/b$ in the correction term. Comparing the b^{-3} terms gives $c + \frac{1}{3} = 1$, so $c = \frac{2}{3}$ and

$$\delta \approx \frac{1}{b} + \frac{2}{3b^3} = \frac{2}{(4n+5)\pi} + \frac{16}{3(4n+5)^3\pi^3}.$$

(Equivalently, one step of the iteration $\delta \mapsto \arctan\dfrac{1}{b-\delta}$ starting from $\delta = 1/b$ gives $\arctan\dfrac{b}{b^2-1}$, which has the same expansion.) It follows that the lower end of the interval for k is $\sin\delta \approx \dfrac{1}{b} + \dfrac{1}{2b^3}$.

Checks

- For $n = 1$, solving $\cos\delta = (\frac{9\pi}{2} - \delta)\sin\delta$ numerically gives $\delta = 0.070973$, while $1/b = 0.070736$ and $1/b + 2/(3b^3) = 0.070971$: the improved approximation is about 150 times closer. For $n = 10$ the error falls to about 5×10^{-10}.
- Counting sign changes of $kx - \sin x$ on a fine grid for $n = 1, 2, 3, 5$ and a value of k from each row of the table gives $2, 3, 1, 0$ roots exactly as predicted, and random k between $\sin\delta$ and $2/((4n+1)\pi)$ always gives one root.

Common slips

- Forgetting the tangent case, or getting the order of $1/b$, k_t and $1/a$ wrong, which garbles the count.
- Using only $\tan\delta \approx \delta$ in the second approximation, which gives $\delta \approx b/(b^2 - 1) \approx 1/b + 1/b^3$. The $\frac{1}{3}\delta^3$ term in $\tan\delta$ is of the same order as the correction and must be kept.

Beyond STEP

Getting successively better approximations to the root of an equation with a small or large parameter is *asymptotic analysis*: you substitute a series in powers of $1/b$ and match coefficients, exactly as here. The same idea, applied to $\tan x = x$, describes the large roots of that equation, which turn up in heat-flow and vibration problems. For more on the analysis behind such expansions, see Bartle and Sherbert, *Introduction to Real Analysis*; Siklos, *Advanced Problems in Mathematics*, has more STEP-style practice.

WORKED SOLUTION

STEP III 1990, Question 11

Mechanics • Rigid Body Rotation

STEP III 1990, Question 11

Read the question before you go on. It is in the STEP Support Programme's question database:

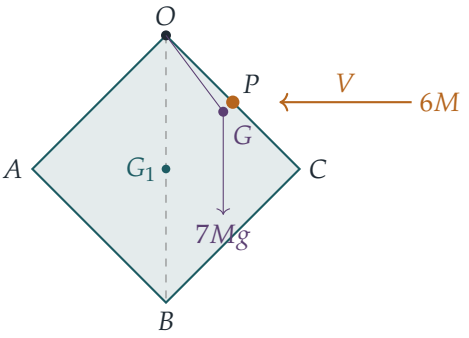
step.maths.org/questions/90-s3-q11

The key idea

The impact is sudden, and the axis at O exerts a large impulsive force during it, so linear momentum is *not* conserved. But that impulse acts at O and so has no moment about O: angular momentum about the axis is conserved through the collision. After that the system is a rigid body on a fixed axis, gravity is the only force doing work, and conservation of energy tells us whether it gets over the top.

Solution

We model the lamina as a uniform rigid square, the particle as a point mass that sticks to the lamina at the instant of impact, and the axis as smooth. Take O as origin. With OB vertically downwards, the centre G_1 of the square is at depth $\sqrt{2}\,a$ (half the diagonal $2\sqrt{2}\,a$), and A and C are at horizontal distance $\sqrt{2}\,a$ either side of OB, also at depth $\sqrt{2}\,a$. The midpoint P of OC is therefore at horizontal distance $a/\sqrt{2}$ from O and depth $a/\sqrt{2}$, with $OP = a$.



(The diagram shows the particle arriving from the right; arriving from the left gives the same answer, as we shall see.)

Moment of inertia about the axis

For a uniform square lamina of side $2a$, the moment of inertia about the perpendicular axis through its centre is $\frac{1}{12}M\big((2a)^2+(2a)^2\big) = \frac{2}{3}Ma^2$ (by the perpendicular axis theorem, adding the two in-plane axes, each $\frac{1}{3}Ma^2$). By the parallel axis theorem, with $OG_1 = \sqrt{2}\,a$,

$$I_{\text{lamina}} = \tfrac{2}{3}Ma^2 + M\big(\sqrt{2}\,a\big)^2 = \tfrac{8}{3}Ma^2.$$

The particle ends up at P, a distance a from O, so the combined body has

$$I = \tfrac{8}{3}Ma^2 + 6Ma^2 = \tfrac{26}{3}Ma^2.$$

The impact

> **Ask yourself** Why can't we use conservation of linear momentum for the collision?

Because the axis is fixed: to stop O moving, it must deliver an impulse to the lamina during the impact, and that impulse is unknown. Its line of action passes through O, though, so it has zero moment about O, and gravity gives no impulse in an instant. So angular momentum about the axis is conserved.

Just before impact, the particle's angular momentum about O is its momentum $6MV$ times the perpendicular distance from O to its (horizontal) line of motion, which is the depth of P, namely $a/\sqrt{2}$. The lamina is at rest. Just afterwards the whole body turns with angular speed ω, so

$$\tfrac{26}{3}Ma^2\,\omega = 6MV\cdot\frac{a}{\sqrt{2}} \qquad\Longrightarrow\qquad \omega = \frac{9V}{13\sqrt{2}\,a}.$$

The kinetic energy immediately after impact is then

$$\tfrac{1}{2}I\omega^2 = \tfrac{1}{2}\cdot\tfrac{26}{3}Ma^2\cdot\frac{81V^2}{338a^2} = \frac{27}{26}MV^2.$$

Making complete revolutions

The combined centre of mass G (mass $7M$) has horizontal displacement and depth from O

$$\frac{M\cdot 0 + 6M\cdot\frac{a}{\sqrt{2}}}{7M} = \frac{3\sqrt{2}}{7}a, \qquad \frac{M\cdot\sqrt{2}\,a + 6M\cdot\frac{a}{\sqrt{2}}}{7M} = \frac{4\sqrt{2}}{7}a,$$

so $OG = \frac{\sqrt{2}}{7}a\sqrt{3^2+4^2} = \frac{5\sqrt{2}}{7}a$.

The body is rigid and pinned at O, so (unlike a particle on a string) it cannot "fall off" its circle; it makes complete revolutions exactly when it still has positive kinetic energy at the position of highest potential energy, which is when G is vertically above O. From its starting depth $\frac{4\sqrt{2}}{7}a$ below O to a height $\frac{5\sqrt{2}}{7}a$ above O, G rises

$$\frac{4\sqrt{2}}{7}a + \frac{5\sqrt{2}}{7}a = \frac{9\sqrt{2}}{7}a,$$

whichever way the body turns, so the direction from which the particle came doesn't matter. Only gravity does work after the impact (the smooth axis does none), so by conservation of energy we need

$$\frac{27}{26}MV^2 > 7Mg\cdot\frac{9\sqrt{2}}{7}a = 9\sqrt{2}\,Mga,$$

that is,

$$\boxed{V^2 > \frac{26\sqrt{2}}{3}ga.}$$

The mass M cancels, as it must, since every mass in the problem is a multiple of M. (At exactly equality the body would only creep towards the top position, never passing it, so the inequality is strict.)

Checks

- The moment of inertia, centre of mass (tan of its angle from the downward vertical is $\frac{3}{4}$) and kinetic energy factor $\frac{27}{26}$ were all confirmed numerically, the first by integrating over a grid on the tilted square.
- Stepping the equation of motion $I\ddot{\theta} = -7Mg\,x_G$ numerically with $a = 1.3$, $g = 9.81$: with V^2 at 98% of $\frac{26\sqrt{2}}{3}ga \approx 15.6$ the body swings back, while at 102% it goes right round, for either direction of approach.
- Sanity check on the energy: only $\frac{27}{26}MV^2$ out of the particle's $3MV^2$ survives the impact, as expected for a perfectly inelastic collision.

Common slips

- Conserving linear momentum through the impact: the axis gives an impulse, so it isn't conserved.
- Using the distance $OP = a$ as the moment arm of the particle's momentum, instead of the perpendicular distance $a/\sqrt{2}$ to its line of motion.
- Forgetting that the centre of mass has moved off OB once the particle sticks, so the rise to the top is not simply twice the original depth.

Beyond STEP

This is the mechanics of rigid bodies rotating about a fixed axis, where impulsive reactions at the axis make angular momentum, rather than linear momentum, the conserved quantity in collisions. A neat follow-up is the *centre of percussion*: the point where a blow produces no impulsive reaction at the axis at all, which is why a cricket bat has a "sweet spot". Kleppner and Kolenkow, *An Introduction to Mechanics*, treats rigid-body rotation and angular impulse carefully.

WORKED SOLUTION

STEP III 1990, Question 12

Mechanics • Particle on Moving Wedge

STEP III 1990, Question 12

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q12

The key idea

Two things are conserved while the particle slides: horizontal momentum (the table is smooth, so there are no external horizontal forces) and energy (every surface is smooth). These two laws, together with the fact that the particle always moves along the face AC *relative to the wedge*, fix every velocity. After that, the bounce and the flight are just projectile motion while the wedge drifts at constant speed.

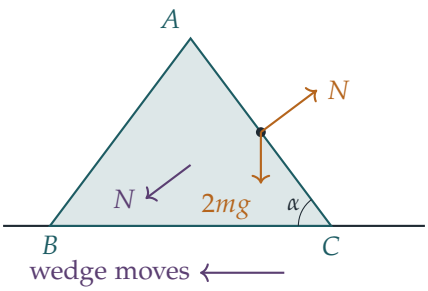
Solution

Setting up

Model the particle as a point mass $2m$ and treat the wedge as a rigid body of mass m. All contacts are smooth. Work in the vertical cross-section through A, B, C. Put B at the origin, with $C = (6a, 0)$ and $A = (3a, 4a)$. The height of A is $4a$ because the triangle is isosceles with $AB = AC = 5a$ and half-base $3a$. Let α be the angle ACB. Then

$$\sin\alpha = \tfrac{4}{5}, \qquad \cos\alpha = \tfrac{3}{5}.$$

The particle slides down AC, moving in the positive x-direction, and the wedge recoils in the negative x-direction.



Let u be the speed of the particle *relative to the wedge*, directed down AC. Let V be the

wedge's velocity, which is horizontal. The particle's velocity is the wedge's velocity plus its velocity relative to the wedge:

$$\mathbf{v}_P = \big(V + u\cos\alpha,\ -u\sin\alpha\big) = \big(V + \tfrac{3}{5}u,\ -\tfrac{4}{5}u\big).$$

Momentum and energy during the slide

The only external forces are gravity and the vertical reactions of the table, so horizontal momentum is conserved. It starts at zero, so

$$2m\big(V + \tfrac{3}{5}u\big) + mV = 0 \quad\Longrightarrow\quad V = -\tfrac{2}{5}u.$$

The particle's velocity is therefore $\big(\tfrac{1}{5}u,\ -\tfrac{4}{5}u\big)$ and the wedge's is $-\tfrac{2}{5}u$.

? **Ask yourself** Why can't we just conserve energy for the particle on its own?

The normal reaction on the particle is perpendicular to the face AC, but the face is moving. So the particle's actual velocity is not along AC, and N does work on it. Only the *system* conserves energy, because the pair of normal forces does zero total work.

The particle falls a height $4a$ by the time it reaches the bottom of AC. Energy for the whole system gives

$$\begin{aligned}
\tfrac{1}{2}(2m)\Big(\tfrac{1}{25} + \tfrac{16}{25}\Big)u^2 + \tfrac{1}{2}m\cdot\tfrac{4}{25}u^2 &= 2mg\cdot 4a\\
\tfrac{19}{25}u^2 &= 8ga\\
u^2 &= \frac{200ga}{19}.
\end{aligned}$$

The particle stays on the face throughout. Resolving with accelerations shows that the normal reaction is the constant $N = \tfrac{10}{19}mg > 0$.

Where everything is at the impact

Momentum is zero throughout, so the centre of mass of the system has no horizontal motion. Let the particle move x_P and the wedge x_W horizontally by the time the particle reaches the table. Then $2m\,x_P + m\,x_W = 0$. Relative to the wedge, the particle has gone from A to C, a horizontal distance of $3a$, so $x_P - x_W = 3a$. Hence

$$x_P = a, \qquad x_W = -2a.$$

At that instant the particle is at $x = 3a + a = 4a$ and the corner C is at $6a - 2a = 4a$. They are at the same place, as they must be, since the particle has just reached C.

The bounce and the flight

The table is fixed and smooth, and the impact is perfectly elastic. So the particle's horizontal velocity is unchanged and its vertical velocity is reversed. It leaves the table

with velocity $\left(\frac{1}{5}u,\ \frac{4}{5}u\right)$. Its time of flight back to the table is

$$T = \frac{2 \cdot \frac{4}{5}u}{g} = \frac{8u}{5g}.$$

During the flight it moves horizontally a distance $\frac{1}{5}u \cdot T = \dfrac{8u^2}{25g}$ to the right.

Once the particle has left it, the wedge has no horizontal force on it, so it carries on at speed $\frac{2}{5}u$ to the left. In time T it moves $\frac{2}{5}u \cdot T = \dfrac{16u^2}{25g}$. The particle is moving right and the wedge left, so the particle cannot meet the wedge again.

The distance between D and E is the sum of these two distances, because both started from $x = 4a$ and they moved in opposite directions:

$$DE = \frac{8u^2}{25g} + \frac{16u^2}{25g} = \frac{24u^2}{25g} = \frac{24}{25} \cdot \frac{200a}{19} = \frac{192}{19}a,$$

as required.

Checks

- Total kinetic energy at the impact: $\frac{19}{25}mu^2 = \frac{19}{25} \cdot \frac{200}{19}mga = 8mga$. This equals the potential energy lost.
- Momentum: $2m \cdot \frac{1}{5}u + m \cdot (-\frac{2}{5}u) = 0$.
- Stepping the motion numerically with $a = g = m = 1$ gives $DE \approx 10.1053$, and $\frac{192}{19} = 10.10526\ldots$.

Common slips

- Taking the particle's velocity to be along AC. That is only its velocity *relative to* the moving wedge.
- Writing $\frac{1}{2}(2m)u^2 = 2mg \cdot 4a$ for the particle on its own. This ignores the wedge's kinetic energy and the work done by N.
- Measuring DE from the original position of C rather than from the impact point $x = 4a$, where both the particle and C are at that moment.

Beyond STEP

A wedge sliding on a smooth table is a classic example of a system with *constraints*. Lagrangian mechanics handles such systems by choosing coordinates (here, the wedge's position and the particle's distance down the slope) so that the constraint forces never need to appear. The conserved horizontal momentum then comes from the symmetry

of the problem under horizontal translation, which is an example of Noether's theorem. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers moving frames and momentum conservation in multi-body systems.

WORKED SOLUTION

STEP III 1990, Question 13

Mechanics • Motion Inside a Sphere

STEP III 1990, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q13

The key idea

There are two stages: circular motion on the sphere until the normal reaction vanishes, then free flight as a projectile. The condition for leaving fixes the speed at the moment of leaving in terms of θ, and then the hint does the rest: write down the condition for the projectile to be back on the sphere, $x^2 + y^2 = a^2$, as an equation in t. It collapses beautifully, because the leaving condition makes several terms cancel.

Solution

The model

We treat P as a particle and the sphere (radius a) as smooth, so the only forces on P while it is in contact are its weight mg and the normal reaction R, directed towards O. The motion takes place in a vertical plane through O; we take axes with origin O, x horizontal and y vertically upwards, and let P climb the right-hand side of the sphere.

For P to leave the surface at all, it must do so above the level of O: below that level the component of gravity along PO points away from O, so R cannot be zero there while P moves on the circle. Hence $0 < \theta < \frac{\pi}{2}$.

Leaving the surface

Let P have speed v when OP makes angle θ with the upward vertical, so P is at height $a\cos\theta$ above O. Resolving towards O, the weight has component $mg\cos\theta$ in that direction, so

$$R + mg\cos\theta = \frac{mv^2}{a}.$$

P leaves when $R = 0$, so at that moment

$$v^2 = ag\cos\theta.$$

(Conservation of energy shows that the speed of projection must be u with $u^2 = v^2 + 2ga(1+\cos\theta) = ag(2+3\cos\theta)$, so every θ in $(0, \frac{\pi}{2})$ is achievable by choosing u suitably. As P climbs, v falls and R falls, so R first vanishes exactly at this point.)

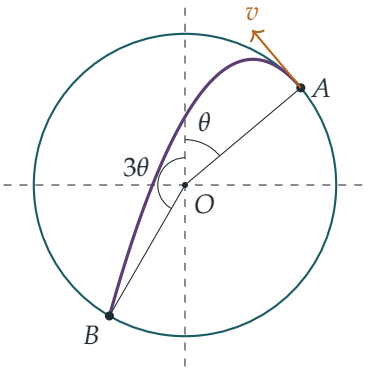
The free flight

At the moment of leaving, call it $t = 0$, P is at $(a\sin\theta, a\cos\theta)$, and its velocity is along the tangent, perpendicular to OP and pointing up and over the top:

$$\text{velocity} = v(-\cos\theta,\ \sin\theta).$$

After that only gravity acts, so

$$x = a\sin\theta - vt\cos\theta, \qquad y = a\cos\theta + vt\sin\theta - \tfrac{1}{2}gt^2.$$



P leaves at A and, as we now show, strikes the sphere again at B.

P is on the sphere when $x^2 + y^2 = a^2$. Expanding, the terms that are constant or linear in t are

$$a^2\sin^2\theta + a^2\cos^2\theta = a^2 \qquad \text{and} \qquad -2avt\sin\theta\cos\theta + 2avt\cos\theta\sin\theta = 0,$$

so

$$\begin{aligned} x^2 + y^2 - a^2 &= v^2t^2\cos^2\theta + v^2t^2\sin^2\theta - gt^2\left(a\cos\theta + vt\sin\theta\right) + \tfrac{1}{4}g^2t^4 \\ &= \left(v^2 - ag\cos\theta\right)t^2 - gv\sin\theta\, t^3 + \tfrac{1}{4}g^2t^4 \\ &= gt^3\left(\tfrac{1}{4}gt - v\sin\theta\right), \end{aligned}$$

using $v^2 = ag\cos\theta$ in the last step.

Ask yourself Why does $t = 0$ come out as a triple root, and not just a single one?

At the moment of leaving, gravity alone is providing exactly the acceleration v^2/a towards O needed to move on the circle. So the parabola starts off with the same

direction *and* the same curvature as the sphere: it hugs the surface very closely before peeling away inside it. That is why the t^2 term cancels as well as the t term.

For $0 < t < 4v\sin\theta/g$ the expression is negative, so P is strictly inside the sphere, and it next meets the surface at

$$T = \frac{4v\sin\theta}{g}.$$

Where it lands

At $t = T$, using $vT\sin\theta = 4v^2\sin^2\theta/g = 4a\cos\theta\sin^2\theta$ and $\frac{1}{2}gT^2 = 8v^2\sin^2\theta/g = 8a\cos\theta\sin^2\theta$,

$$y = a\cos\theta + 4a\cos\theta\sin^2\theta - 8a\cos\theta\sin^2\theta = a\cos\theta\left(1 - 4\sin^2\theta\right).$$

(This is $a\cos 3\theta$, since $\cos 3\theta = 4\cos^3\theta - 3\cos\theta = \cos\theta(1 - 4\sin^2\theta)$; similarly $x = -a\sin 3\theta$, so the impact point B is at angle 3θ from the upward vertical on the far side, as in the diagram.)

P strikes below the level of O exactly when $y < 0$. Since $0 < \theta < \frac{\pi}{2}$, $\cos\theta > 0$, so

$$y < 0 \iff 1 - 4\sin^2\theta < 0 \iff \sin\theta > \tfrac{1}{2} \iff \theta > \tfrac{\pi}{6},$$

the last step because sin is increasing on $(0, \frac{\pi}{2})$. When $\theta = \frac{\pi}{6}$ exactly, P strikes the sphere exactly level with O. So the angle that θ must exceed is

$$\boxed{\theta = \frac{\pi}{6} = 30^\circ}$$

and every θ with $30^\circ < \theta < 90^\circ$ does give an impact below the level of O.

Checks

- $\theta = 60^\circ$: the formula gives $y = a\cos 180^\circ = -a$, so P lands back at the lowest point, the point of projection. With $\theta = 45^\circ$ it lands at 135° from the upward vertical; both are $y < 0$, as they should be.
- As $\theta \to 0$, $T \to 0$ and $B \to A$: a particle that only just leaves at the top falls straight back onto the surface.
- A numerical simulation (stepping the motion on the sphere until $R = 0$, then the projectile until it meets the sphere) for $\theta = 10^\circ, 25^\circ, 29^\circ, 30.5^\circ, 31^\circ, 45^\circ, 60^\circ, 80^\circ$ matched $T = 4v\sin\theta/g$ and $y = a\cos 3\theta$ to four decimal places; for example $\theta = 31^\circ$ gives $y/a = -0.0523$.

Common slips

- Getting the direction of the velocity wrong: it is perpendicular to OP and still heading up and over, with components $(-v\cos\theta,\ v\sin\theta)$, not $(v\sin\theta,\ v\cos\theta)$.

- Forgetting to use $v^2 = ag\cos\theta$ before solving $x^2 + y^2 = a^2$, which leaves an awkward quartic instead of a clean factorisation.
- Dividing $1 - 4\sin^2\theta < 0$ through without noting that $\cos\theta > 0$ because P leaves above the level of O.

Beyond STEP

The triple root at $t = 0$ is a statement about *osculation*: the parabola and the circle share their tangent and their curvature at the leaving point, so they have contact of second order. Curvature and the osculating circle are central ideas in the differential geometry of curves, and they reappear in mechanics whenever we split acceleration into tangential and normal parts. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats motion in polar coordinates and constrained motion of this kind carefully.

WORKED SOLUTION

STEP III 1990, Question 14

Mechanics • Systems of Forces

STEP III 1990, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

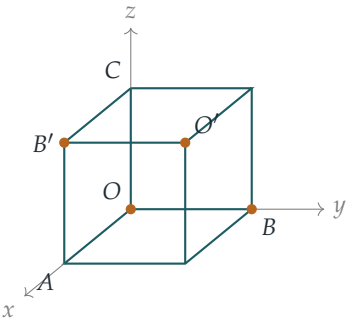
step.maths.org/questions/90-s3-q14

The key idea

A system of forces on a rigid body comes down to two vectors: the resultant $\mathbf{R} = \sum \mathbf{F}_i$ and the moment about a chosen point, $\mathbf{G}_P = \sum(\mathbf{r}_i - \mathbf{p}) \times \mathbf{F}_i$. Moving the base point from O to P changes the moment by a simple rule, $\mathbf{G}_P = \mathbf{G}_O - \mathbf{p} \times \mathbf{R}$. Every part of the question is this rule used with a different P.

Solution

Model the cube as a rigid body with edge a, so $A = (a, 0, 0)$, $B = (0, a, 0)$, $C = (0, 0, a)$, and the vertex opposite B is $B' = (a, 0, a)$.



Finding λ, μ, ν

The force at O has no moment about O, so only the other three count. Using $(x,y,z)\times(p,q,r)=(yr-zq,\ zp-xr,\ xq-yp)$:

$$\overrightarrow{OO'}\times\begin{pmatrix}\lambda\\0\\1\end{pmatrix}=a\begin{pmatrix}1\\1\\1\end{pmatrix}\times\begin{pmatrix}\lambda\\0\\1\end{pmatrix}=a\begin{pmatrix}1\\\lambda-1\\-\lambda\end{pmatrix},$$

$$\overrightarrow{OB}\times\begin{pmatrix}-1\\0\\2\end{pmatrix}=a\begin{pmatrix}0\\1\\0\end{pmatrix}\times\begin{pmatrix}-1\\0\\2\end{pmatrix}=a\begin{pmatrix}2\\0\\1\end{pmatrix},$$

$$\overrightarrow{OB'}\times\begin{pmatrix}1\\\mu\\\nu\end{pmatrix}=a\begin{pmatrix}1\\0\\1\end{pmatrix}\times\begin{pmatrix}1\\\mu\\\nu\end{pmatrix}=a\begin{pmatrix}-\mu\\1-\nu\\\mu\end{pmatrix}.$$

Adding, the total moment about O is $a\,(3-\mu,\ \lambda-\nu,\ 1-\lambda+\mu)$. Setting each component to zero gives $\mu=3$, then $\lambda=1+\mu=4$, then $\nu=\lambda=4$:

$$\lambda=4,\qquad \mu=3,\qquad \nu=4.$$

None of this depends on α,β,γ, and the resultant is

$$\mathbf{R}=\begin{pmatrix}\alpha+\lambda-1+1\\\beta+\mu\\\gamma+1+2+\nu\end{pmatrix}=\begin{pmatrix}\alpha+4\\\beta+3\\\gamma+7\end{pmatrix},\qquad \mathbf{G}_O=\mathbf{0}.$$

Part (i)

With $\alpha=\beta=\gamma=0$ we have $\mathbf{R}=(4,3,7)$. Two systems are equivalent when they have the same resultant and the same moment about one point (and hence about every point). So the single force at B must be $\mathbf{R}$ itself, and the couple must equal the moment of the original system about B:

$$\mathbf{G}_B=\mathbf{G}_O-\overrightarrow{OB}\times\mathbf{R}=\mathbf{0}-a\begin{pmatrix}0\\1\\0\end{pmatrix}\times\begin{pmatrix}4\\3\\7\end{pmatrix}=-a\begin{pmatrix}7\\0\\-4\end{pmatrix}=a\begin{pmatrix}-7\\0\\4\end{pmatrix}.$$

? **Ask yourself** Why is the moment about B equal to $\mathbf{G}_O-\overrightarrow{OB}\times\mathbf{R}$?

For each force, $(\mathbf{r}_i-\mathbf{b})\times\mathbf{F}_i=\mathbf{r}_i\times\mathbf{F}_i-\mathbf{b}\times\mathbf{F}_i$; summing over i gives $\mathbf{G}_O-\mathbf{b}\times\mathbf{R}$.

So the equivalent system is a force $(4,3,7)$ acting at B together with a couple of moment $a(-7,0,4)$. As a check, the force at B plus this couple has moment about O equal to $a(7,0,-4)+a(-7,0,4)=\mathbf{0}$, as it should.

Part (ii)

Now $\mathbf{R} = (2+4,\ 3+3,\ 2+7) = (6,6,9) = 3(2,2,3)$, and still $\mathbf{G}_O = \mathbf{0}$. At a point P with position vector $\mathbf{r} = (x,y,z)$,

$$\mathbf{G}_P = \mathbf{G}_O - \mathbf{r} \times \mathbf{R} = -\mathbf{r} \times \mathbf{R}.$$

This is zero exactly when $\mathbf{r}$ is parallel to $\mathbf{R}$ (or $\mathbf{r} = \mathbf{0}$), because $\mathbf{R} \neq \mathbf{0}$. So the locus is the line through O in the direction $(2,2,3)$:

$$\frac{x}{2} = \frac{y}{2} = \frac{z}{3}.$$

Physically, the whole system is equivalent to the single force $(6,6,9)$ acting along this line.

When the cube translates (no rotation), every point moves by the same displacement, so the work done by the forces is $\mathbf{R} \cdot \mathbf{s}$, where $\mathbf{s}$ is the displacement. For a distance d along the line, in the direction of $(2,2,3)$, $\mathbf{s} = \dfrac{d}{\sqrt{17}}(2,2,3)$ and

$$W = (6,6,9) \cdot \frac{d}{\sqrt{17}}(2,2,3) = \frac{51d}{\sqrt{17}} = 3\sqrt{17}\,d.$$

The question leaves the distance unnamed; per unit distance the work is $3\sqrt{17} \approx 12.37$ units. Since $\mathbf{R}$ lies along the line, this is just $|\mathbf{R}|\,d = \sqrt{36+36+81}\,d = \sqrt{153}\,d$, the same number. (Moving the other way along the line gives $-3\sqrt{17}\,d$.)

Checks

- With $\lambda = 4$, $\mu = 3$, $\nu = 4$, the three moments about O are $a(1,3,-4)$, $a(2,0,1)$ and $a(-3,-3,3)$, which sum to zero.
- In (ii), take $P = (2,2,3)$: $\mathbf{r} \times \mathbf{R} = (2,2,3) \times (6,6,9) = \mathbf{0}$. At $P = (1,0,0)$ the moment is $-(0,-9,6) \neq \mathbf{0}$.
- A computer with exact fractions confirmed every moment, checked that the equivalent system in (i) has the same moment as the original about 20 random points, and confirmed the work per unit distance is $3\sqrt{17} \approx 12.369$.

Common slips

- Getting the order wrong in $\mathbf{r} \times \mathbf{F}$: reversing it flips every sign.
- Taking the couple in (i) to be $+\overrightarrow{OB} \times \mathbf{R}$. Moving the force from O to B needs the compensating couple with the opposite sign.
- Finding the work by summing over the forces at different vertices. In a pure translation only the resultant matters.

Beyond STEP

Any system of forces on a rigid body reduces to a single force along a line together with a couple about that same line. This is a *wrench*, and its pitch is $\mathbf{R} \cdot \mathbf{G} / |\mathbf{R}|^2$. In (ii) the pitch is zero, so the wrench is just a single force, and the zero-moment locus is its line of action, the central axis. These ideas lead to rigid-body dynamics and screw theory in robotics. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers torque and rigid bodies at the right level.

WORKED SOLUTION

STEP III 1990, Question 15

Probability and Statistics • Geometric Distributions

STEP III 1990, Question 15

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q15

The key idea

Each throw shows A with probability $\frac{1}{4}$ and B with probability $\frac{3}{4}$, independently. So the event $\{M = m,\ N = n\}$ is one specific string of throws, and its probability is a product. That product splits into a factor depending only on m times a factor depending only on n, and this does almost all the work: the marginals are geometric, the variables are independent, and every later probability is a geometric series.

Solution

Write $p = \mathrm{P}(A) = \frac{3}{12} = \frac{1}{4}$ and $q = \mathrm{P}(B) = \frac{9}{12} = \frac{3}{4}$ for a single throw. Throws are independent.

The event $\{M = m,\ N = n\}$ happens exactly when the first $m + n + 1$ throws are

$$\underbrace{A\,A\cdots A}_{m}\ \underbrace{B\,B\cdots B}_{n}\ A,$$

so by independence, for $m \geq 0$ and $n \geq 1$,

$$\mathrm{P}(M = m,\ N = n) = \left(\tfrac{1}{4}\right)^m \left(\tfrac{3}{4}\right)^n \tfrac{1}{4} = \left(\tfrac{1}{4}\right)^{m+1} \left(\tfrac{3}{4}\right)^n.$$

(With probability 1 the sequence does eventually show a B and then an A, so these probabilities sum to 1; the marginals below confirm this.)

Part (i)

For the marginal of M, sum over $n \geq 1$, using $\sum_{n\geq 1} \left(\frac{3}{4}\right)^n = \dfrac{3/4}{1 - 3/4} = 3$:

$$\mathrm{P}(M = m) = \left(\tfrac{1}{4}\right)^{m+1} \cdot 3 = \tfrac{3}{4}\left(\tfrac{1}{4}\right)^m, \qquad m = 0, 1, 2, \ldots$$

For the marginal of N, sum over $m \geq 0$, using $\sum_{m\geq 0}\left(\frac{1}{4}\right)^{m+1} = \dfrac{1/4}{1-1/4} = \frac{1}{3}$:

$$\mathrm{P}(N=n) = \tfrac{1}{3}\left(\tfrac{3}{4}\right)^{n} = \tfrac{1}{4}\left(\tfrac{3}{4}\right)^{n-1}, \qquad n = 1,2,3,\ldots$$

Both make sense directly: M counts As before the first B, and N is the waiting time for the first A once the B-run has started. Each sums to 1.

Part (ii)

We use $\sum_{k\geq 1} kx^{k-1} = \dfrac{1}{(1-x)^2}$ for $|x|<1$ (differentiate the geometric series). Then

$$\mathrm{E}(M) = \tfrac{3}{4}\sum_{m\geq 1} m\left(\tfrac{1}{4}\right)^{m} = \tfrac{3}{4}\cdot\tfrac{1}{4}\cdot\frac{1}{(3/4)^2} = \tfrac{1}{3},$$

$$\mathrm{E}(N) = \tfrac{1}{4}\sum_{n\geq 1} n\left(\tfrac{3}{4}\right)^{n-1} = \tfrac{1}{4}\cdot\frac{1}{(1/4)^2} = 4.$$

Independence

Multiplying the marginals,

$$\mathrm{P}(M=m)\,\mathrm{P}(N=n) = \tfrac{3}{4}\left(\tfrac{1}{4}\right)^{m}\cdot\tfrac{1}{4}\left(\tfrac{3}{4}\right)^{n-1} = \left(\tfrac{1}{4}\right)^{m+1}\left(\tfrac{3}{4}\right)^{n} = \mathrm{P}(M=m,\ N=n)$$

for every $m \geq 0$, $n \geq 1$. So M and N *are* independent.

Ask yourself Surely the length of the A-run affects the B-run? Why doesn't it?

The A-run ends at the first B, and from that throw on the process starts afresh: the remaining throws are independent of everything before. The length of the B-run depends only on those later throws, so knowing M tells us nothing about N. The factorisation of the joint probability is the algebraic form of this.

The probabilities involving N and M

For an integer $k \geq 1$, sum the tail of the geometric distribution:

$$\mathrm{P}(N>k) = \sum_{n\geq k+1}\tfrac{1}{4}\left(\tfrac{3}{4}\right)^{n-1} = \tfrac{1}{4}\cdot\frac{(3/4)^k}{1-3/4} = \left(\tfrac{3}{4}\right)^{k}.$$

This formula also gives $\mathrm{P}(N>0)=1$, which is correct since $N \geq 1$.

For $\mathrm{P}(N>M)$, condition on M and use independence:

$$\begin{aligned}\mathrm{P}(N>M) &= \sum_{m\geq 0}\mathrm{P}(M=m)\,\mathrm{P}(N>m) = \sum_{m\geq 0}\tfrac{3}{4}\left(\tfrac{1}{4}\right)^{m}\left(\tfrac{3}{4}\right)^{m}\\ &= \tfrac{3}{4}\sum_{m\geq 0}\left(\tfrac{3}{16}\right)^{m} = \tfrac{3}{4}\cdot\tfrac{16}{13} = \tfrac{12}{13}.\end{aligned}$$

For $\mathrm{P}(N = M)$ the common value must be at least 1, since $N \geq 1$:

$$\mathrm{P}(N = M) = \sum_{m\geq 1} \tfrac{3}{4}\left(\tfrac{1}{4}\right)^m \cdot \tfrac{1}{4}\left(\tfrac{3}{4}\right)^{m-1} = \tfrac{3}{64}\sum_{m\geq 1}\left(\tfrac{3}{16}\right)^{m-1}$$
$$= \tfrac{3}{64}\cdot\tfrac{16}{13} = \tfrac{3}{52}.$$

Finally, the three events $N > M$, $N = M$, $N < M$ are exclusive and exhaustive, so

$$\mathrm{P}(N < M) = 1 - \tfrac{12}{13} - \tfrac{3}{52} = \tfrac{52-48-3}{52} = \tfrac{1}{52},$$

as required.

Checks

- The means fit intuition: A is rare, so the opening A-run is short ($\tfrac{1}{3}$ on average), while we wait on average $1/p = 4$ throws for an A to end the B-run.
- Directly, $\mathrm{P}(M > n) = \left(\tfrac{1}{4}\right)^{n+1}$, so $\mathrm{P}(N < M) = \sum_{n\geq 1}\tfrac{1}{4}\left(\tfrac{3}{4}\right)^{n-1}\left(\tfrac{1}{4}\right)^{n+1} = \tfrac{1}{64}\sum_{n\geq 1}\left(\tfrac{3}{16}\right)^{n-1} = \tfrac{1}{64}\cdot\tfrac{16}{13} = \tfrac{1}{52}$, agreeing with the subtraction. The direct reason for $\mathrm{P}(N > k) = \left(\tfrac{3}{4}\right)^k$: after the first B, the next k throws must all be B.
- Exact summation of the joint distribution gives $\mathrm{P}(N > M) = 0.92308$, $\mathrm{P}(N = M) = 0.05769$, $\mathrm{P}(N < M) = 0.01923$; a simulation of 200 000 series of throws gave 0.923, 0.058, 0.019 and sample means 0.334 and 3.99.

Common slips

- Including $m = 0$ in the sum for $\mathrm{P}(N = M)$: N can never be 0.
- Forgetting that M can be 0 (the first throw may be B), which shifts the marginal of M and gives the wrong mean.

Beyond STEP

The run lengths here are independent because a sequence of independent trials "restarts" at any time determined only by the past, such as the first B. This is the *strong Markov property*, and it underlies renewal theory and the analysis of Markov chains, which are first- and second-year undergraduate topics. Grimmett and Welsh, *Probability: An Introduction*, treats geometric distributions, independence and Markov chains at the right level.

WORKED SOLUTION

STEP III 1990, Question 16

Probability and Statistics • Geometric Probability

STEP III 1990, Question 16

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/90-s3-q16

This question has a full guided treatment in *Beyond STEP*, Section 12.21, *Breaking a Stick*: https://beyondstep.keithmcnulty.org/ch12/broken-stick.html

The key idea

Three lengths with a fixed total make a triangle exactly when each one is less than half the total. Once a triangle exists, it's obtuse exactly when the square of the longest side exceeds the sum of the squares of the other two (the cosine rule). Both parts turn into conditions on the random variables, and the probabilities become lengths of intervals in (i) and areas of regions in (ii).

Solution

Part (i)

The three pieces have lengths X, $\frac{1}{2}(1-X)$ and $\frac{1}{2}(1-X)$, with total 1. Three positive lengths form a triangle if and only if each is less than the sum of the other two, that is, less than half the perimeter, here $\frac{1}{2}$. The two equal pieces satisfy $\frac{1}{2}(1-X) < \frac{1}{2}$ whenever $X > 0$, so the only condition is $X < \frac{1}{2}$. Let T be the event that a triangle can be formed. Then

$$\mathrm{P}(T) = \mathrm{P}\left(0 < X < \tfrac{1}{2}\right) = \tfrac{1}{2}.$$

A triangle is obtuse if and only if the square of its longest side is greater than the sum of the squares of the other two. Which side is longest depends on X.

- If $X \leq \frac{1}{3}$, then $X \leq \frac{1}{2}(1-X)$, so one of the equal pieces is a longest side. The test would be $\frac{1}{4}(1-X)^2 > \frac{1}{4}(1-X)^2 + X^2$, i.e. $X^2 < 0$, which never happens. The triangle is isosceles with its base no longer than the equal sides, so it's never obtuse.

- If $\frac{1}{3} < X < \frac{1}{2}$, the longest side is X, and the triangle is obtuse if and only if

$$X^2 > 2 \cdot \tfrac{1}{4}(1-X)^2 \iff X > \frac{1-X}{\sqrt{2}},$$

taking positive square roots, since both sides are positive. This rearranges to $X(\sqrt{2}+1) > 1$, i.e. $X > \dfrac{1}{\sqrt{2}+1} = \sqrt{2}-1 \approx 0.414$, which does lie in $\left(\frac{1}{3}, \frac{1}{2}\right)$.

Let O be the event that the triangle is obtuse. Then $O \cap T$ is the event $\sqrt{2}-1 < X < \frac{1}{2}$, and since X is uniform,

$$\mathrm{P}(O \mid T) = \frac{\mathrm{P}(\sqrt{2}-1 < X < \frac{1}{2})}{\mathrm{P}(T)} = \frac{\frac{1}{2} - (\sqrt{2}-1)}{\frac{1}{2}} = 1 - 2\sqrt{2} + 2 = 3 - 2\sqrt{2},$$

as required.

Part (ii)

The construction is always possible. The triangle R with vertices $(1,0)$, $(1,1)$, $(0,1)$ is the set of points with $x \le 1$, $y \le 1$ and $x + y \ge 1$. The proposed sides x, y, $2-x-y$ have total 2, and they form a (possibly degenerate) triangle if and only if all three are non-negative and none exceeds half the perimeter, 1:

- $0 \le x \le 1$ and $0 \le y \le 1$ hold on R;
- $2 - x - y \le 1$ is just $x + y \ge 1$, and $2 - x - y \ge 0$ follows from $x \le 1$, $y \le 1$.

So every point of R gives a triangle. It is degenerate only on the edges of R, which have probability zero.

When is $\angle ABC$ obtuse? The angle at B lies between BA and BC, so the side opposite it is $CA = y$. By the cosine rule, $\angle ABC$ is obtuse if and only if $CA^2 > AB^2 + BC^2$:

$$y^2 > x^2 + (2-x-y)^2.$$

Expanding $(2-x-y)^2 = (2-x)^2 - 2(2-x)y + y^2$, the y^2 terms cancel, and the condition becomes

$$2(2-x)y > x^2 + (2-x)^2 = 2x^2 - 4x + 4.$$

Since $x \le 1$, we have $2 - x > 0$, so we can divide by $2(2-x)$ without reversing the inequality:

$$y > \frac{x^2 - 2x + 2}{2 - x},$$

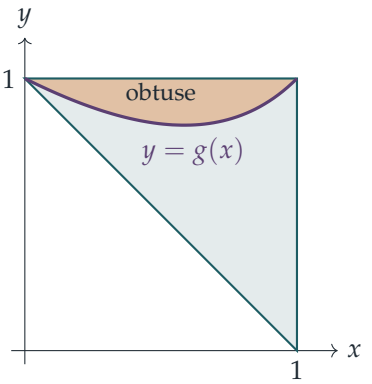
as required. Call the right-hand side $g(x)$.

Ask yourself Does the curve $y = g(x)$ actually stay inside the triangle R?

We need it to lie between the lines $y = 1 - x$ and $y = 1$ for $0 \le x \le 1$, so that the obtuse region is simply the part of R above the curve. Indeed,

$$1 - g(x) = \frac{(2-x) - (x^2 - 2x + 2)}{2 - x} = \frac{x(1-x)}{2-x} \ge 0, \qquad g(x) - (1-x) = \frac{x}{2-x} \ge 0,$$

with $g(0) = g(1) = 1$. So the curve runs from $(0,1)$ to $(1,1)$, dipping below $y = 1$ but never below the hypotenuse $x + y = 1$.



The probability. The triangle R has area $\frac{1}{2}$, so the joint density is 2 on R, and

$$\mathrm{P}(\angle ABC \text{ obtuse}) = 2 \times \text{area}\{(x,y) : g(x) < y \le 1,\ 0 \le x \le 1\} = 2\int_0^1 (1 - g(x))\,\mathrm{d}x.$$

To integrate g, substitute $u = 2 - x$, so $x = 2 - u$ and

$$x^2 - 2x + 2 = (2-u)^2 - 2(2-u) + 2 = u^2 - 2u + 2, \qquad g = \frac{u^2 - 2u + 2}{u} = u - 2 + \frac{2}{u}.$$

As x runs from 0 to 1, u runs from 2 to 1, and $\mathrm{d}x = -\mathrm{d}u$, so

$$\begin{aligned}\int_0^1 g(x)\,\mathrm{d}x &= \int_1^2 \left(u - 2 + \frac{2}{u}\right)\mathrm{d}u = \left[\tfrac{1}{2}u^2 - 2u + 2\ln u\right]_1^2 \\ &= (2 - 4 + 2\ln 2) - \left(\tfrac{1}{2} - 2\right) = 2\ln 2 - \tfrac{1}{2}.\end{aligned}$$

Hence

$$\mathrm{P}(\angle ABC \text{ obtuse}) = 2\left(1 - 2\ln 2 + \tfrac{1}{2}\right) = 3 - 4\ln 2,$$

as required.

Checks

- Numerically, $3 - 2\sqrt{2} \approx 0.1716$ and $3 - 4\ln 2 \approx 0.2274$. A Monte Carlo simulation with 10^6 trials gave about 0.1719 and 0.2278.

- Both answers lie between 0 and 1. In (ii), the curve touches $y = 1$ at both ends: at $x = 0$ the side BC vanishes, and at $x = 1$ the side AB vanishes, so both are degenerate cases, as you'd expect.
- A Simpson's-rule evaluation of $\int_0^1 g(x)\,\mathrm{d}x$ gives 0.886294, matching $2\ln 2 - \frac{1}{2}$.

Common slips

- In (i), forgetting to condition: the unconditional probability of an obtuse triangle is $\frac{3}{2} - \sqrt{2}$, half the required answer.
- In (ii), using the wrong side in the cosine rule. The angle at B is opposite $CA = y$, not AB.
- Forgetting that the density on R is 2, not 1, which gives an answer half the size.

Beyond STEP

Turning a probability into a length or an area is *geometric probability*, and breaking a stick at random is one of its classic problems: break a stick at two independent uniform points and the chance of a triangle is $\frac{1}{4}$. Part (ii) is a first look at continuous joint distributions, where probabilities are double integrals of a joint density over a region. Grimmett and Welsh, *Probability: An Introduction*, treats joint densities and has several problems of this kind.

Beyond STEP

STEP is the hardest mathematics examination most school students ever sit. It is also one of the best: its questions are short, unfamiliar and deep, and the best of them are small pieces of university mathematics in disguise.

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1990 Worked Solutions. The whole book, every chapter and all the worked solutions are free to read, download and share at **beyondstep.keithmcnulty.org**

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